

Teaching translation in the age of generative AI

New paradigm, new learning?

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Translation and Multilingual Natural
Language Processing

Translation and Multilingual Natural Language Processing

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Introduction

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1 Why this book?

Since OpenAI launched ChatGPT back in November 2022, generative artificial intelligence (GenAI) has acted as a major disruptor in the translation industry and beyond. Of course, in any given industry disruptors are not intrinsically good or bad. They merely act as game changers, good and bad. Also commonly associated with “chatbots”, GenAI tools certainly are a game changer for the translation industry and, consequently, for translator education too. Using deep learning to train on vast corpora, not only do LLMs have the ability to “generate text which is often indistinguishable from text written by a human” (Moorkens et al. 2025: 188), but they can do much more than just translate texts. In this book, we shall therefore make a distinction between artificial intelligence (AI) and GenAI. In our context, AI refers broadly to all AI technologies, including neural machine translation (NMT). GenAI, however, refers to machine learning tools that generate media, including chat-based Large Language Models (LLMs) like GPT-4. Since this book deals mostly with text, the terms GenAI and LLMs tend to appear synonymously.

Unlike freely available machine translation (MT) tools like Google Translate or DeepL, LLMs can also be prompted to adapt their translations to take the source text and/or the target text context into consideration. For free (initially). In a matter of seconds. It is easy to see how GenAI can potentially contribute to the further democratisation of translation by making it freely accessible in contexts where translation would have been neither practical nor affordable until now. This disruptor also provides translators with new resources that can potentially help them not just to increase the speed and quality of the translation work they deliver to clients, but also with the way they manage their translation projects or

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go about finding new clients. In the 2024 edition of the annual ELIS (European Language Industry Survey) report, for instance, the independent translators who saw GenAI as a positive trend explained that they leveraged its possibilities “as a tool (e.g. terminology extraction), as a source for editing work, and as a motivator for clients to choose human translation due to bad AI experiences” (ELIS 2024: 24).

Of course, the use of AI chatbots for translation work also comes with some important limitations that need our attention. One of them is that they need vast amounts of data to learn, meaning that their ability to translate depends largely on the amount of language data available to train on for any given language. Even though LLMs differ from all other machine learning-based MT systems in that they can be trained on monolingual data instead of bilingual corpora, thus increasing the amount of available data they can train on, they remain mostly trained on English data. Because of this, their translation performance is likely to be much less convincing for so-called low-resource languages compared with high-resource languages. In addition, AI chatbots are trained to identify and replicate language patterns. A natural corollary of this is that they lack “any form of genuine comprehension or the cognitive processes that humans use to understand language and context” (Moorkens et al. 2025: 189). Finally, when GenAI lacks the data it needs to produce part of a translation, it can “hallucinate”, i.e. make up content in the target language.

These limitations notwithstanding, GenAI has started reshaping what it means to work as a professional translator in an industry that was already largely technology-driven, and where arguably automation has long been used as a way to increase the speed and reach of translation while reducing its cost. This has sometimes been to the detriment of working conditions and job satisfaction for human translators (see, for instance, Lambert & Walker 2024). This, in fact, makes it even more urgent for all of us translator educators to engage with GenAI so that we can, in the process, re-examine the role and agency of human translators within the translation process through that lens. In other words, GenAI should encourage us to rethink critically the value and values that humans bring to the translation process. As part of this, we must (re-)interrogate what we do on our translation programmes so that we empower the next generation of professional translators to achieve the kind of “human-centred augmented translation” that will benefit both individuals and society (O’Brien 2024: 391).

Admittedly, however, for some of us translator educators the prospect of engaging with GenAI may initially have felt somewhat overwhelming. First of all, the timing may not have been ideal. OpenAI’s ChatGPT was released shortly after we came out blinking from a series of pandemic-related lockdowns, during

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which teaching happened mostly online. For some of us, this meant having to interact with (new) technologies that we may have felt were not always tailored to our own needs and/or that we weren't always comfortable using. As a result, most of us got to experience first-hand the “technostress” many professional translators experience when they are asked to use translation tools and technologies they are not entirely comfortable with (Penet 2024). This is something we should take seriously, as technostress can “reduc[e] performance and har[m] individual wellbeing” (Koskinen 2020: 146). With its pedagogical approach, this edited volume therefore comes as an attempt to help alleviate feelings of stress among some of us as translator trainers.

Ideally, we should all feel empowered to engage with GenAI on our programmes, where it also acts as a disruptor. Again, this was reflected in the findings of the latest ELIS reports. In its 2024 survey, university staff ranked GenAI implementation as the most widely shared challenge, with close to 90% of them seeing it as an issue. This prompted the report's authors to comment that: “Generative AI and how to implement it in the universities' programmes has taken the challenge chart of university staff by storm, topping even their concerns about the visibility of the profession and the eternal lack of time” (ELIS 2024: 27). If, the following year, visibility of the profession had reclaimed top spot in the list of challenges, it was still closely followed by GenAI implementation, which remained a concern for over 80% of university staff (ELIS 2025: 25). Yet, adapt and implement we must! This is because, still according to the report, in 2025 “[a]ctual MT use by language companies has increased [...] and reaches now the magic mark of 50% of handled projects. AI makes its entry with an impressive 34%” (ELIS 2025: 35). In a similar vein, Slator, an online portal for language industry news and research, recently made the following point:

‘Skate to where the puck is going, not to where it is’ has always been good advice for ice hockey. But it is remarkably prescient for the language services industry as we prepare for the tsunami of disruption brought about by generative artificial intelligence (AI) and large language models, or LLMs. The speed of development has been frenetic since the release of OpenAI's ChatGPT in November 2022. [...] It would be a brave forecaster to predict how the corporate landscape will look five years from now—but operating from where the puck is today risks asking the wrong questions and missing the boat (Welocalize 2023).

Even though this message was clearly intended for language service providers, it certainly holds some relevance for us educators. Whatever we may think or feel

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about GenAI, we owe it to both our students and to the translation industry to engage with it, and to do so in a way that will help them anticipate where the puck is going. However, we must do so in a way that is ethical and sustainable as, arguably, GenAI is much more than just another tool in the translator's toolkit. This is what this book intends to help you with.

We have called this edited volume *Teaching Translation in the Age of Generative AI: New Paradigm, New Learning*. Given the very nature of GenAI, the word paradigm, with its historical meaning of "pattern", seems rather appropriate. Beyond this, the word captures GenAI's potential to be a game changer not just in the translation industry, but also in translator education. With this new paradigm, which is characterised by the emergence of new translation practices, comes new learning for both students and educators. This is what we have attempted to capture here with the help of experts from across the globe. In fact, if this edited volume was a film, reviewers would no doubt talk about a "star-studded cast". As editors, we are truly grateful to all of our colleagues who have so generously accepted the invitation to contribute their expertise, making it available via open access. Like us, they share our ambition to make this book as accessible as possible so it can be of help to as many colleagues in translation education as possible as they learn to negotiate the uncharted territory of GenAI. We are also grateful to Language Science Press and the editors of the "Translation and Multilingual Natural Language Processing" for giving our book a good home. This open access volume builds on outstanding past publications in the series, not least Kenny's *Machine Translation for Everyone: Empowering users in the age of artificial intelligence* 2022. This volume also complements nicely Pym and Hao's *How to Augment Language Skills: Generative AI and Machine Translation in Language Learning and Translator Training* 2025 as it focuses more specifically on GenAI in translator education.

2 Contents of this book

This book is articulated around three main parts. Part 1 explores the new skills and competences we need to help our students develop in the age of GenAI. It opens with Gary Massey and Maureen Ehrensberger-Dow's detailed discussion in Chapter 1 of translation competence in the age of GenAI. They make the case that GenAI is a collaborative technology that can be complemented by human agency if human translators bring their own, complementary skills to the table. To that aim, translation programmes should foreground textual, digital skills and interlingual skills when training students to work with GenAI. In Chapter 2, Erik

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Angelone discusses the importance of deliberate practice in translator training to help students develop the competence and expertise they will need in their future careers. He then shows how immediate, informative feedback is a key condition for the development of deliberate practice and how we can use AI chatbots in our teaching as More Knowledgeable Others (MKO) that provide students with the required feedback to foster translational expertise through deliberate practice. Chapter 3, by Ramón Inglada, argues that, in the age of automated translation, the concept suitability should take precedence over that of correctness. To make sure our students are in a position to evaluate the suitability of the translation solutions generated by (Gen)AI, he argues, we should help them develop three new skills, namely the skills of selection, assessment and prompting, alongside the traditional core skills for translation competence. He then gives concrete examples of how GenAI can be used in the classroom to foster these new skills among trainee translators.

In Part 2, the focus shifts to the new knowledge we and our students need to delve into in the age of GenAI. This knowledge often underpins the new skills and competences mentioned in Part 1. In Chapter 4, for instance, Lynne Bowker argues the importance of teaching our translation students about data (and, therefore, corpora) as it can help them make more informed and responsible use of the technology. She recognises, however, that doing so from an expert-to-expert position may not always be most effective, as it is easy to get lost in a rabbit-hole of intricate details when talking about AI networks. Instead, she shows through concrete and compelling examples that borrowing techniques from science communication – whose goal is to make expert communication available to non-experts – can make our teaching about data much more impactful for students. This is followed by Chapter 5, in which Masaru Yamada discusses prompting for GenAI. More specifically, he shows through detailed examples the usefulness of traditional translation studies concepts to prompt AI chatbots, thus opening new avenues for translator education. In an echo to Bowker's chapter, Joss Moorkens and Gökhan Doğru tackle the all-important topic of AI ethics in Chapter 6. They consider ways to introduce discussion and reflection in the classroom that maximise the impact of this teaching. To that end, they focus in particular on case studies and the use of mapping and images to foster systemic thinking around translation and GenAI.

Finally, Part 3 puts some flesh on the bones of Parts 1 and 2 as it reviews some of the new teaching approaches adopted by colleagues since the advent of GenAI. It does so by introducing the reader to a series of vignettes taken from a variety of translation-related disciplines and contexts. Chapter 7 opens with Senem Öner

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Bulut presenting the results of a research project in which staff and students collaboratively and experimentally explored the dynamics of prompt engineering for translation, as well as the new skills human translators may need to develop to prompt effectively. In Chapter 8, Maria Zimina-Poirot gives concrete examples of activities to develop students' critical and ethical use of GenAI as part of the translation workflow. Through these, she convincingly shows how such a use of GenAI can help augment human contributions. This is followed by Chapter 9, in which David Orrego-Carmona moves the lens to subtitler training as another area of translator education where "the teaching must shift from teaching specific technical and translation skills to developing adaptable professionals who can critically engage with evolving technologies while maintaining high-quality standards". To that aim, he presents us with series of innovative classroom activities that can be replicated to help our students' "digital reflexivity" in subtitling. In Chapter 10, Sonia Vandepitte focuses on another specific area of translator education, namely translation into one's second language (L2 translation). Reporting on the findings of an exploratory research project involving students of an L2 translation class on a postgraduate programme, she shows how integrating GenAI in L2 translation at the editing stage of the process not only enhances students' reflexivity, but it can also increase their intrinsic motivation, which is crucial in L2 translation training. In Chapter 11, we move to interpreting as Sahar Othomani and Nermin al Sharman introduce us to the [AI]phra project, a simulation software created to enhance interpreter training in the MENA (Middle East and North Africa) region through gamification. Piloted in 2023, the key motivator behind [AI]phra is to unleash the potential of AI to address the at-times limited access to interpreter education in the MENA region. Finally, in Chapter 12 we move slightly away from translator education to explore the potential of embracing MT in second language (L2) education. In this final chapter, Atsushi Mizumoto thus introduces us to the Metacognitive Resource Use Framework, which "positions learners as metacognitive agents capable of strategically utilizing a wide range of language resources, including MT and generative AI tools".

3 One final thought

As GenAI continues to reshape what it means to translate and, therefore, what it means to be a translator (educator), our main aim with this volume is to help fellow translator educators better understand the new skills and competences that we need to foster in our teaching, scaffolded by new knowledge. We also hope that, taking a leaf from some of the case studies in Part 3, many colleagues

will be spurred on to further explore what it means to (learn to) translate in the age of GenAI in partnership with students. This, in turn, could lead to a welcome increase in the number of research publications on the topic at a time where we all need to learn to navigate our way through this new paradigm.

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Acknowledgments

Part I

New paradigm: New skills & competences

Chapter 1

Translation competence in the age of generative AI: Debates, dilemmas, directions

Gary Massey^a & Maureen Ehrensberger-Dow^a

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Until recently, the description and modelling of the competences and skills needed to translate successfully, and the ways in which they develop, have seen a steady evolution and predictable expansion, largely to accommodate technological advances and an increasing awareness of situatedness. However, the impact of neural MT and, now, generative AI (GenAI) has been unprecedented in rapidly transforming the core tasks of translators. Together with a proliferation of creative roles in a diversifying language industry, the volatility indicates a paradigm shift that is beginning to supplant even the once stable epithet “translator”. It also questions the efficacy of current descriptions of skills and confronts educators with dilemmas of balancing specialisation and generalisation, routinisation and adaptivity, core and transferable skills. This chapter considers relevant aspects of modelling competences and their development and examines related debates and dilemmas engaging educators and employers in the current and foreseen language-industry climate. It outlines directions for training in the age of translating with(out) GenAI, proposing an approach that, alongside core textual, interlingual translation and digital skills, combines transferable skills with human-machine/human-agent interaction (HMI/HAI) competence.

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1 Introduction

1.1 General

What elements constitute translation competence (TC)¹ and how they can be developed have represented pivotal issues in applied Translation Studies since its inception (Holmes 1988: 77). They continue to trigger considerable debate amongst researchers, educators and practitioners, and they present educators and their organisations with dilemmas about how and where to target resources. In this chapter, we explore some debates and dilemmas triggered by changes to professional and educational practices brought about by generative artificial intelligence (GenAI). We also outline potential directions for human agents in the age of translating with and without GenAI. In our view, this necessitates an approach combining transferable skills with human-machine/human-agent interaction (HMI/HAI) competence, alongside “old-school” translation skills. Before proceeding to the debates, dilemmas and directions, however, we should define, describe and demarcate what exactly we mean by professional interlingual TC in the age of GenAI.

1.2 Definitions, descriptions and demarcation

The word interlingual is used in Translation Studies and the language industry to designate a particular type of language mediation activity, namely that which takes place between natural languages and the particular cultures they represent. Alongside interpreting, translation represents a prototypical form of professional interlingual language mediation. In a language industry characterised by rapidly diversifying job titles and tasks (Bond 2018, Slator 2020: 11–17), the prototypical conceptualisation of what so-called (professional) translators do, i.e. translate content written in a source-language (SL) document into a target-language (TL) document, appears increasingly outmoded. Language professionals adaptively engage in a whole range of service provision as the lines between core and adjacent services blur (Angelone 2023, Angelone et al. 2024: 3–5).

The major driver behind this state of affairs has been the technologisation of the industry as language service providers (LSPs) operate more and more downstream and upstream of previously core translation, localisation and interpret-

¹This chapter applies the definitions of “skill” and “competence” from the latest EMT (2022: 3) competence framework. A skill is the “ability to apply knowledge and use know-how to complete tasks and solve problems”, while competence refers to “the proven ability to use knowledge, social and/or methodological abilities, in work or study situations and in professional and personal development”.

1 Translation competence in the age of generative AI

ing activities. Indeed, the industry itself prefers the generic term linguist, and with good reason. Since 2016, the higher accuracy and fluency of neural machine translation (NMT) engines have led to more extensive automation of the translation process. Linguists have thus seen more deployment downstream of language transfer (or conversion) per se in post-editing (PE), output evaluation and quality assurance tasks. Most recently, the introduction of automated MT quality estimation (QE) to determine how much human PE is required is itself replacing more costly human MT quality evaluation (Slator 2024: 50–51). Upstream of transfer proper, linguists have been expanding their repertoire to perform tasks that include linguistic consultancy, MT pre-editing and multilingual content creation, for example in marketing, PR and corporate communications (often referred to as transcreation). Nevertheless, we shall retain the term translation, both in line with the title of the present volume and in recognition of the convincing argument made by do Carmo & Moorkens (2021) that the ultimate ethos of translation is to help communication flow between humans, regardless of the new technologies deployed and of the roles and activities these necessitate.

GenAI has now entered the technological mix, and its applications clearly go far beyond interlingual mediation. Opportunities and risks have been identified in higher education in general (Atlas 2023, Gimpel et al. 2023, OECD 2023), where the potential of GenAI as an educational tool extends to teachers and students alike. Properly used, it can support course design, materials development, assessment, text and image production, coding, critical thinking and individualised learning. Improperly used, it can facilitate academic dishonesty, increase technological dependency, atrophy human skills and agency, undermine data protection and intellectual property rights, reinforce discrimination (due to data bias) and increase inequality (due to unequal access).

Within the narrower confines of interlingual translation, GenAI differs decisively from AI-based language technology such as NMT in being capable of generating multilingual text, images and other content in response to prompts, including the prompt to translate. These prompts elicit statistically probable output generated by deep-learning models able to process and generate natural languages, known as large language models (LLMs). The prompt-response cycle can be iterative, allowing users to ask for more information or to provide more detailed prompts should the results require improvement (Gimpel et al. 2023: 22). Adequately structuring the instructions that prompt a GenAI model – also known as prompt design or, at the more technical level, prompt engineering – thus becomes central to its effective use. The skills and competences (see Footnote 1) necessary to embrace GenAI in the education and practice of professional

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translators thus differ in major respects from those needed to master CAT tools and other translation technologies.

It is indisputable that such skills and competences are essential to a language industry already transitioning from NMT to GenAI. Unbabel co-founder and CEO João Graça points out that LLMs are attracting more research and development, and can handle more complex tasks than NMT, such as automatic PE, source correction and cultural adaptation.² Lionbridge, one of the world's leading LSPs, claims GenAI is "taking automated translation and localisation to new heights".³ *Slator* (2024: 52) reports that translation management systems (TMS) providers are already integrating GPT models to leverage LLMs' capacity to rephrase, summarise and suggest new translations. LLMs can be used to incorporate QE into workflows and gauge PE effort, do PE or even provide linguistic insights throughout the translation workflow. *Slator's* results also show two-thirds of MT providers offering fine-tuned LLMs and over 80% offering MT-LLM hybrid solutions. LSPs have the potential to become major providers and enablers of multilingual, multimodal GenAI content through content creation, validation, localisation, transmission and management. In keeping with the broadening portfolios of LSPs, multilingual text generation is currently "the most in-demand language AI application, after machine translation" (*Slator* 2024: 74). Fine-tuning and prompting LLMs to create and translate/localise content are emerging activities for language professionals to complement longer-established PE and quality assurance roles.

Competent – or ideally expert – leverage of GenAI is a must for all those aiming to work in the language industry. But which skills and competences does and will this require? And in a dawning age of (language) work increasingly dominated by AI, what value can human intelligence and skills bring to bear? It is time to consider the debates around professional interlingual translation and the dilemmas that educators are facing.

2 Debates and dilemmas

2.1 Agency

Asscher (2023) raised the issue of whether MT can actually be regarded as a proper point of interest in Translation Studies from a definitional perspective. He concludes that it is indeed compatible with both prescriptive, equivalence-oriented definitions of what translation is as well as descriptive definitions based

²*Slatorpod* #216, 12 July 2024 (<https://www.youtube.com/watch?v=6smTEp3CXwQ>)

³<https://www.lionbridge.com/generative-ai/>

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largely on how translations are received. But he also points out that the perceived threat of AI to the social and professional status of translators, and of those who educate them (see also [ELIS 2024: 24–25](#)), may well redraw definitional boundaries and distinguish between MT and human translation on the basis of the perceived creative and moral authenticity of conscious agency ([Asscher 2023: 14–16](#)).

Asscher’s argument draws on current positions on AI from the broader humanities and social sciences, such as moral philosophy (e.g. [Sebastián & Rudy-Hiller 2021](#)). However, there is evidence of similar preoccupations in education policy and science, with emphasis falling on the importance of human agency, including teacher agency, and associated transferable skills – responsible action, critical thinking, systems thinking, logical reasoning, cultural agility, problem solving and emotion regulation ([Gimpel et al. 2023](#), [OECD 2023](#)).

Within the narrower confines of Translation Studies, it appears that agency is already emerging as the fulcrum around which the value of professional human translation can be measured. For example, [Vieira \(2020: 327–329\)](#) presents PE as a spectrum ranging from an MT-centred process typified by automatic PE to human-centred PE where humans have full control of output. Human agency moves back and forth along the spectrum as client, commission and employer demands require. The human-centred pole is more palatable to professional translators ([Vieira 2020: 327](#)), whose resistance to PE seems to derive from anxiety over a perceived loss of agency ([Cadwell et al. 2018](#), [Sakamoto 2019](#)). In the discourse surrounding PE, agency has thus become a touchstone of professionalism and professional self-concept.

Similarly, [Rico & González Pastor \(2022: 188\)](#) found in their pre-GenAI study of attitudes to teaching MT that the translation educators made much of the “human factor”. Though the data are unclear on precisely where they believe that agency resides, the participants place the translator at the centre of translation production. Interestingly, this prompts the researchers to posit that “MT has overcome its condition of tool”, and to claim that teaching MT should adopt a holistic approach “beyond an instrumentalist agenda that concentrates on the technical properties of the technologies” and which “evidences how the human factor is key to the translation process” ([Rico & González Pastor 2022: 190, 193](#)). Referring to other research, they also tentatively consider whether complementary effects of cognitive impairment or augmentation could influence competence development ([Rico & González Pastor 2022: 192](#)). One wider implication is that, even before the advent of GenAI, the paradigmatic human agency of the tool-user is being subverted by a more discerning and subtle appreciation of the rapidly evolving interactivity between translation technologies and their users.

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The changed nature of that relationship appears to be reflected in the metaphors used to describe GenAI systems: ChatGPT and similar systems are referred to as “conversational agents” (Moorkens & Guerberof Arenas 2024: 75), and educational sources describe them as a “language partner”, “writing partner”, “writing collaborator”, “co-partner to formulate text”, “learning partner”, “intellectual sparring partner”, or “partner to generate codes” (e.g. Atlas 2023: ii: 64, Gimpel et al. 2023: 19–24). Business consultants and IT case researchers alike refer to human-GenAI interaction as “co-creation” (e.g. Eapen et al. 2023, Nah et al. 2023: 296).⁴

GenAI has now advanced well beyond the material agents in the “dance of agency” so deftly explored by Olohan (2011) in interactions between translators and TM. This is a question not just of degree (speed, accuracy and adequacy of suggestions, etc.) but also, and most pertinently, of quality (reciprocal interactions and learning). The model pivotal to Olohan’s (2011: 344) conceptualisation of agency decisively distinguishes between the capacity of human agents to both resist and accommodate to technologies and of non-human agents to simply resist. But GenAI is patently capable of accommodation in its own right (deep learning). This renders irrelevant the perceived divide between the poles of social and technological determinism that Olohan (2011: 345) observes among translators commenting on TM, and which O’Brien (2024) criticises in her call to overcome the antagonism between technology and translators by focusing on a human-centred AI (HCAI) that amplifies rather than emulates human abilities (cf. Shneiderman 2020). As Risku & Windhager (2013: 36–37) point out, any technology or artefact used by translators in their work aligns with the concept of non-human “actants” in actor-network theory (ANT, e.g. Latour 2005). GenAI, however, quite obviously goes further in fulfilling the conditions of what the human-machine interaction (HMI) sub-field of human-agent interaction (HAI) has called “joint activity” (Bradshaw et al. 2011: 288): “the essence of joint activity is interdependence [...] to produce something that is a genuine joint product”.

Together with other effects of (Gen)AI on the processes and practices of professional translation, the agency issue yields multiple potential ramifications for the development and exercise of professional TC. It is to these that the next sections turn.

⁴The anthropomorphism is more than a conceit. Research on conversational agents used as social companions shows that human-AI friendship is similar to human-human relationships and contributes to social health (Brandtzaeg et al. 2022, Chaturvedi et al. 2023, Guingrich & Graziano 2024). Virtual agents capable of reciprocal adaptive behaviour are being tested for use in cognitive behaviour therapy and social skills training (Woo et al. 2024).

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2.2 Modelling competence

The reciprocity and mutuality of GenAI already go some way towards realising aspects of human-centred augmented translation so engagingly discussed by O'Brien (2024). She argues for HCAI as a framework for amplifying translators' abilities and empowerment while maintaining human control. We would claim that GenAI represents a collaborative technology capable of complementing human agency and supporting empowerment to a degree unseen in dedicated (augmented) translation technologies to date. Even before ChatGPT launched the GenAI era late in 2022, Fügner et al. (2021: 1552) could confidently assert that "human performance can improve individually by receiving AI advice [...] due to complementary knowledge". Although their results also showed "significant downsides" when AI advice is provided as a "one-size-fits-all" solution in group decision-making, the individualisation made possible by appropriate prompting would suggest that this is no longer an issue.

As in all complementary partnerships, the actants (to use the ANT term) must bring their own strengths to the table. Competent translators will for their part still need the transferable human skills that enable them to regulate cognitive and affective performance without AI support, the experience and knowledge on which they are predicated, and the competences necessary to deploy them within the sociotechnical environments where they work. These will have to be aligned with the new key skill of interacting with GenAI by prompting and reprompting LLMs,⁵ based on knowledge and experience of how GenAI works. We further propose that this skill could be embedded in a broader area of HMI/HAI competence (see below) which takes due account of GenAI's technological agency.⁶

It is here that current TC models reveal inadequacies. Although they give due weight to the transferable personal and interpersonal skills that facilitate human agency, they have perpetuated an instrumentalist teaching agenda by focussing almost exclusively on aspects of CAT tool use (including MT) that not only underplay the human factor in HMI/HAI but also assume no technological agency. For example, in the case of the EMT competence frameworks, the only hint of possible reciprocal effects between the technological environment and human performance has been the inclusion of an organisational and physical ergonomic descriptor as an aspect of personal competence (EMT 2022: 10).

⁵See Chapter 5 by Yamada in the present volume.

⁶This is not to say that there is equivalence between conscious human agency and AI agency. However, they share features that suggest "family resemblances" in the sense developed by Wittgenstein (e.g. 1958: 32).

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Tellingly, it makes no mention of ergonomic factors of technology design and use that play into cognitive performance, despite empirical evidence that they can be major enhancers, distractors or stressors (Ehrensberger-Dow et al. 2016). Otherwise, the EMT technology competence category focuses on non-interactive aspects of technology use: critically assessing, adapting to and effectively deploying available resources, including appropriate use of MT, file management and TMS, and demonstrating data literacy. Interactivity is visible only in interpersonal relations: checking, reviewing, revising and evaluating the work of others, working in multicultural and multilingual teams, interacting with clients and networking with language professionals and LSPs (EMT 2022: 8–11).

The older but no less influential PACTE Group’s competence model (Hurtado Albir 2017: 35–41) is similar in this respect. It includes technology under what is called instrumental sub-competence, “related to the use of documentation resources and information and communication technologies applied to translation” (Hurtado Albir 2017: 40). The descriptor again demonstrates how competence models have fuelled the instrumentalist agenda criticised above.⁷

Nevertheless, the PACTE model has been a major source for others, who have supplemented it with additional components where appropriate. Those additions reflect a growing awareness that translation is far less a discrete cognitive act performed solely by the mind of an individual than an event of embodied cognition performed by complex systems involving humans, their social, technical and physical environments together with their cultural artefacts (Risku 2010: 103). For example, an interpersonal component is added to the most recent PACTE-informed model: Prieto Ramos’s (2024) adaptation of his previous model of legal TC (Prieto Ramos 2011) to institutional translation. The model’s “instrumental competence” category goes beyond the PACTE model by specifying the use of reliable resources for information mining, of computer tools for translation and revision tasks, and of MT (Prieto Ramos 2024: 155). But it does not speak to possible implications of human interactivity with agentic AI systems.

Despite some overt (and acknowledged) similarities, Prieto Ramos (2024: 153) is highly critical of the latest EMT framework (2022) for failing to redress what he sees as a downgrading of thematic competence (i.e. domain specialisation, e.g. in law) in its previous iteration (EMT 2017). It is hard to argue against him, since thematic competence figures high on the list of professional expectations uncovered by his own research and that of others, especially with respect to

⁷The relevant descriptors in the highest level of the more recent competence framework (EFFORT) are similar, in that their focus is on non-interactive aspects of technology. <https://www.effortproject.eu/wp-content/uploads/Level-C-en.pdf>

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legal and institutional legislative translation (Esfandiari et al. 2019, Lafeber 2023, Prieto Ramos & Guzmán 2023). The research receives equal credence from emic (insider) language industry heuristics. Describing the Slator (2022) “expert in the loop” model, for example, industry commentator Florian Faes responds to a question by the first author that only language professionals with high levels of domain and language expertise, combined with appropriate technological and prompt design/engineering competence, will provide the added human value needed to work with (Gen)AI and that such experts will stay in high demand (Faes & Massey 2024: 27–29).

In other respects, however, the EMT framework (2022: 10) falls in with the broad consensus, already articulated above, that transferable personal and interpersonal skills (called generic or soft skills) are central to employability and to the adaptability needed in today’s work environment. The point is echoed by a recent PE model (Nitzke & Hansen-Schirra 2021: 69–79), visualised as a house, where the basic bilingual, extra-linguistic and information research competences that professional translators possess form the foundations. The roof is made of soft skills such as risk assessment, service competences, self-efficacy, a professional self-concept, an ethical attitude, concentration, stress-resistance, logical reasoning, analytical thinking and an affinity with technology. The model is one of several developed for tasks such as revision and PE (Robert et al. 2023), which were once considered integral aspects of TC. These reflect the growing diversification of language-industry tasks and the concomitant need for language professionals to have transferable skills enabling them to adapt to new activities.

2.3 Developing competence

Industry diversification has prompted Angelone (2023) to call for fostering adaptive (as opposed to routinised) expertise across translator training curricula by consistently exposing learners to challenging situations where they need to apply their knowledge flexibly – itself a transferable skill set. But what balance should be struck between routine and adaptivity, between core and transferable skills? These are two dilemmas confronting translation teachers and their institutions with limited material and temporal resources at their disposal. Related to them is a third, namely employability tensions between domain or task specialisation (Prieto Ramos 2024) and a generalist approach more congruent with adaptivity training (EMT 2022). In all these cases, choices have to be made in specific educational contexts as to the most likely prospects for graduate employment, with curriculum developers nimble enough to adapt quickly to market changes.

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The most widespread issue, however, is not unique to translator education. In the context of GenAI use in society in general, Teubner et al. (2023: 98) refer to the “educators’ dilemma” – between banning GenAI use, as many translation teachers might have been inclined to do with freely available, high quality NMT post-2016,⁸ and educating students about how to use it well. Their prediction (Teubner et al. 2023: 97) about “a shift of focus away from the act of writing itself towards the actual content and the ideas being communicated” is very pertinent to the context of translation training. They suggest that this could lead to more importance being placed on reading and evaluating options, recalling Pym’s (2003) minimalist proposal to describe TC.

Viewed through the prism of HMI/HAI, GenAI can be regarded as an interactive, agentic partner in translation production and learning processes. Conceptualising the role of GenAI in this way seamlessly integrates into social constructivist and emergentist experiential learning (Kiraly 2000, Kiraly 2013). Here, learning is understood as a collaborative process of work on authentic or simulated translation projects, supplemented with appropriate scaffolding. Teachers essentially partner with their students as learners in proactively seeking knowledge (Kiraly 2013: 214–215). Various qualitative studies have validated the approach, with reported outcomes including increased student responsibility, learner autonomy, critical reflection, self-regulation, motivation and self-efficacy (Kiraly 2013) – those very transferable skills that educators see as a necessary complement to AI and which O’Brien (2024) would want AI to support.

Alongside such transferable skills, the language industry continues to expect graduates with skills conventionally associated with translation, such as high reading proficiency in the SL and writing/editing proficiency in the TL. The expert in the loop model (Slator 2022) assumes that the translator in that position also has a solid understanding of the target audience, awareness of cultural differences, sociolinguistic competence and media competence. These “old school” skills have many parallels to what the EMT (2022) subsumes under language, culture and translation competence. In the next section, we discuss some possible directions for translator training that include GenAI and that had not been foreseen in the competence models proposed to date.

⁸But see Mellinger (2017) for a well-timed argument for not doing so.

3 Directions: Skills and competences

3.1 Translating with(out) GenAI

Before discussing the skills and competences for translation in an age of GenAI, it makes sense to reflect briefly on what the activity of producing professional translations involves. This can be done by translating from scratch, or fully human translation,⁹ working with CAT tools or using MT or GenAI to produce a whole target text (TT).

3.1.1 Translating from scratch

Translating from scratch refers to translators coming up with their own version of a TT on the basis of a source text (ST) without having matches or suggestions presented to them automatically by a CAT tool or an MT/GenAI engine, respectively. To do so, they rely on their SL competence to understand the ST, their translation skills to transfer the meaning and message into the TL, and their TL competence to produce a TT appropriate to the needs of the target audience. Appropriateness includes being aware of and observing genre conventions, language register, terminology requirements and client style guidelines. Translation skills include following a translation brief, assessing the target audience needs, relying on internal resources and common sense, acquiring domain knowledge as needed, accessing terminological resources and/or parallel texts, recognising cultural references and/or metaphors, searching for and adapting solutions, etc.

Translating from scratch does not exclude the option of accessing digital resources such as online dictionaries, concordances in CAT tools, MT engines and even GenAI, but the translator is in full control of when and how those resources are used for which parts of the TT. While drafting the TT and almost always also after producing a complete draft, professional translation involves carefully rereading the TT and comparing it with the ST in a bilingual review to ensure completeness, accuracy and appropriateness.

3.1.2 Producing translations with a CAT tool

Working with a CAT tool to produce professional translations differs from translating from scratch in that the ST is segmented by the tool, and previously translated segments that are perfect matches or similar to a minimum match level are

⁹Simply referred to as “translation” before the widespread introduction of language technology (Christensen et al. 2017)

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offered for the translator's approval or revision. Sometimes even perfect matches are not appropriate for the particular TT being produced (because of terminology, register, client style guidelines, etc.), so the translator has the choice of editing them or deleting them to translate from scratch. Many CAT tools now also have the option of accessing MT suggestions for segments or parts of segments that do not have good matches in the TM, either automatically or at the command of the translator.¹⁰

Usually much less typing is involved than in translating from scratch, but otherwise using a CAT tool to produce a TT requires a very similar set of skills. However, there are additional challenges in using a CAT tool to produce a TT. One is the focus on text segments (e.g. sentences or parts of sentences), which can make it more difficult for the translator to retain a sense of the whole text. Although CAT tools can provide cognitive relief by making it easier to be consistent, they can also increase the load by making it harder for the translator to maintain cohesion and coherence (Krüger 2016). Another challenge relates to the phenomenon of priming, in which "cognitive facilitation [...] is triggered by linguistic overlap between earlier and current processing" (Vandepitte et al. 2018: 362). For example, suggestions that are not suitable can serve as inspirations for translators to produce versions that are (Farrell 2023). However, the sheer presence of a TM match or MT suggestion can also block cognitive processing and force the translator to exert extra effort to mitigate the effects of priming (Kolb 2024).

3.1.3 Using MT or GenAI for first drafts

A translation process that includes producing a draft with an MT engine or GenAI is different from translating with a CAT tool or from scratch in certain ways, yet quite similar in others. The similarities lie in evaluating the translation solutions through a careful bilingual review and adapting the TT to be more appropriate. In all types of translation processes, thoroughly understanding the ST and recognising errors, misrepresentations and unmotivated omissions or additions in the draft are just as important as being able to edit the latter to produce a fluent, cohesive and coherent TT.

Translation processes that incorporate GenAI shift the focus from writing to reading, prompting, evaluating and editing. Using a GenAI system with a simple prompt to translate the ST can be expected to produce a draft with features typical of MT output, the quality of which will depend on the particular engine being

¹⁰See Kappus (2024) for an overview of the development and convergence of translation technology.

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accessed. The process of improving such output through PE differs in many respects from editing or revision (do Carmo & Moorkens 2021). As well as a careful monolingual reading of the output, PE should always involve a bilingual review to eliminate any bias and inappropriate additions, omissions or “hallucinations” randomly introduced by the engine (see Dale et al. 2023).

Working with GenAI in a more informed way necessitates a prompting process that provides detailed information about the purpose of the translation, client guidelines, target genre and audience. An iterative process of prompting, evaluation and reprompting to obtain suitable output could precede a prompting sequence to eliminate any problems such as terminological inconsistencies and unnecessary repetition. Similar issues to those mentioned above for MT output (i.e. bias, inappropriate additions and omissions, hallucinations) also need to be rectified.

3.2 Describing translation skills in the GenAI era

It has been convincingly argued elsewhere (e.g. Nitzke & Hansen-Schirra 2021, O’Brien 2021) that using MT to produce high-quality TTs requires a special set of skills, and those arguments also hold when GenAI is simply used to produce a translation of the ST. However, informed use of GenAI includes the awareness that the pre- and drafting phase of creating a TT requires well-considered prompting, and post-drafting demands careful editing. In the following, we consider textual and digital skills that should be foregrounded when training students to work with GenAI to produce TTs as well as the interlingual competence that needs to be developed for this.

3.2.1 Textual skills

As Lafeber (2023: 43) has reported, professional translators should be able to “acquire subject-matter knowledge quickly [...] understand complex topics, figure out obscure meaning, appreciate the authors’ intentions and the readers’ needs”. This high level of literacy suggests that a solid foundation in text analysis for translation (e.g. Nord 2005) is needed as well as the ability to recognise one’s own gaps in knowledge. Rather than hoping that an MT engine or GenAI will produce output that is easier to understand, students should be trained to identify difficult ST passages and to do the research needed to make sense of them. Gimpel et al. (2023: 37) point out that “[l]earners must possess adequate knowledge of the subject under scrutiny to achieve satisfactory outcomes”. In the context of translation, this means that students must have the skill to quickly search for

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resources (e.g. reference words, parallel texts, videos, images) to help them understand the ST. They also need to have efficient, purposeful reading skills to evaluate the usefulness of the respective resource for their own comprehension and its potential as a GenAI prompt.

Since GenAI can produce fluent, grammatically correct texts, the fostering of text production skills should focus on revising, editing and proofreading (see also [Koponen et al. 2021](#)). This represents a significant shift from the teaching of foreign languages and basic writing skills work that is still done in many entry-level degree programmes, especially for translation into the L2 ([Cerezo Herrero et al. 2021](#)) and consistent with recent proposals for curriculum development (e.g. [Sawyer et al. 2019](#)).

3.2.2 Technological and digital skills

The technology competence described in the [EMT \(2022: 9\)](#) framework potentially covers GenAI in stating that “students know how to use the most relevant IT applications [...] and adapt rapidly to new tools and IT resources”. The need for a basic understanding of MT and data literacy are also referred to. However, GenAI only became widely available for translation after the latest iteration of the EMT framework appeared, so even the most recent descriptions seriously underspecify what type of knowledge is needed and which skills should be fostered to achieve competence when working with this tool. It is only since then that more detailed frameworks specifically targeting professional MT literacy, translation-oriented data literacy and AI literacy have begun to appear (e.g. [Krüger 2024](#), [Krüger & Hackenbuchner 2024](#)), building on prior generic MT and data literacy frameworks (e.g. [Bowker & Buitrago Ciro 2019](#)). Especially relevant to the present context is the AI literacy framework for translation, interpreting, and specialised communication proposed by [Krüger \(2024\)](#). It contains various elements related to the functions and use of LLMs, some of which converge with the skills that we list in the final section.

Indeed, scholars outside Translation Studies have already started arguing that all students need to acquire specific GenAI literacy. For example, [Pretorius \(2023\)](#) points out risks and challenges such as inaccurate results, unequal access to the technology, bias propagation and ethical concerns about sensitive data. We would add that students need to understand the consequences of the data sources used for the LLMs and the time lags associated with the latter’s knowledge updates, which can be in the order of months if not years.

Perhaps the most noticeable change in technological and digital skills in an age of GenAI concerns the critical thinking required to formulate effective

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prompts, evaluate intermediate versions and adjust subsequent prompts accordingly. This also includes the ability to recognise the diminishing returns of continued prompting and the need to use one's own textual skills to polish the TT. Working with GenAI is different from simply using language technology in that the intermediate versions can be considered a type of priming or mutual prompting between the conscious student/translator and the non-conscious AI agent. Only continuous critical reflection can counteract potentially over-trusting GenAI output during the translation process. Technological and digital translation skills in an age of GenAI might therefore be more broadly conceptualised as key components of an HMI/HAI competence capable of accounting for the particular reciprocities of human and AI agency.

3.2.3 Interlingual competence

Prerequisites for professional translation have always included high reading proficiency in the SL as well as familiarity with the source culture and domain. No matter which aids are being used, the translator also needs high proficiency in the TL – both reading and revision skills – as well as in-depth knowledge of the domain and target culture. Interlingual transfer might be called for less often on the part of the professional translator deploying MT or GenAI, although they are always engaged in interlingual action.

As Lafeber (2023: 43) found in her survey of institutional translators, professionals are expected to “achieve high levels of accuracy in their translations, conveying not only nuances but also intended effect [...] draft well in their TL, compensating for poor wording in the original when appropriate while adhering to in-house conventions”. Just as actuaries attain high degrees of numeracy during their training but rarely do calculations by hand in their jobs, students should be given ample opportunity to translate with and without aids. The resulting interlingual competence in combination with HMI/HAI competence will contribute to their translation literacy (see also Massey 2021), effectively preparing them to work in the language industry in an age of GenAI.

3.3 Emerging roles in the language industry

Translation training institutions are assumed to prepare students for the language industry, yet up to two-thirds of graduates may not actually work as translators (Hao & Pym 2023: 223). However, according to the same authors, many of them work in language-related professions. This suggests that programmes would do well to also prepare their students for jobs without “translator” in the

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title. Even when translators are explicitly being recruited, though, a broad range of skills seems to be expected. In their analysis of job notices from 2005-2020, [Prieto Ramos & Guzmán \(2023: 53–55\)](#) found that the duties of translators at supranational and intergovernmental organisations also included (in order of overall average mention): assistance with other tasks, terminology work, revision, CAT-tool management and editing.

On the topic of how automation is changing the translation profession, [Pym & Torres-Simón \(2021\)](#) discuss various recommendations for translators to focus on what machines cannot (yet) do, such as language service advice, service provision, language consulting and high-stakes communication. This is consistent with comments from industry observers that text production with GenAI is less suitable for “content that has a regulatory or technical purpose” ([Slator 2024: 75](#)). The same report also predicts increased demand for multilingual experts to revise corporate-generated output to meet expected quality levels ([Slator 2024: 63](#)).

LSPs already offer a wide range of AI-related services ([Slator 2024: 101](#)), many of which require linguistic expertise that may still be difficult to find (cf. [Faes et al. 2024](#)). The latest European language industry survey ([ELIS 2025: 34](#)) shows the only increase in technology implementation over 2024 to be in GenAI. This represents an opportunity for both training institutions and their graduates and, encouragingly, 64% of the students participating in the same survey reported using GenAI in their training, 19% regularly ([ELIS 2025: 37](#)).

This raises one last important question about those who train translation students. The teachers themselves patently require the (Gen)AI literacy needed to understand AI techniques, critically assess AI productions and recommendations, and use AI creatively in their teaching ([OECD 2023: 401](#)).¹¹ And institutions need to have the staff development procedures in place that empower their teachers to do so.

4 Final take-aways

Whether employed as translators, linguists, transcreators, localisers, consultants or language professionals, our graduates have an important role to play in the language industry despite or because of the introduction of GenAI. But they will need appropriate skills to do so. And those skills will, above all, have to accommodate the dynamic interactions of human and AI agency. Though degrees of

¹¹Ideas for course design, materials and teaching scenarios that align well with the skills and competences outlined here can be found in [Pym & Hao \(2025\)](#).

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domain specialisation are open to debate and will need to be determined by local conditions, current TC models already adequately cover the old-school translation and textual skills that industry demands. But their transferable and technological and digital skills are underspecified, and the introduction of GenAI calls for greater precision. Krüger's (2024) AI literacy framework is a response, outlining the digital competences needed to harness AI efficiently, ethically and sustainably. It covers technical foundations (operating principles, training, etc.), assessing AI's usefulness for (domain-)specific tasks, interacting with (Gen)AI, implementing AI in workflows, and understanding ethical and societal aspects. Partially overlapping with elements of Krüger's framework, our own brief recommendations represent a set of concrete priorities for embracing GenAI in translator education.

We recommend that collaborative experiential learning should always integrate decisions about whether and how to deploy GenAI, together with critical evaluations of the resultant processes and products. Outcomes should focus on developing specific transferable skills:

- critical thinking
- adaptivity
- creative problem-solving
- cognitive and emotional self-regulation
- self-efficacy
- accountability and a sense of ethics
- collaborative ability in human and AI interactions.

The extension of the last point aims to develop what we have termed HMI/HAI competence, which includes:

- designing effective prompts (translation purpose, client specifications, target genre, audience, etc.), evaluating GenAI responses with supplementary research (where appropriate) and reprompting
- recognising priming effects and mitigating negative ones
- identifying and eliminating bias, additions, omissions, hallucinations

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- understanding LLM data sources and knowledge time-lags.

None of the above should be regarded as optional add-ons but must receive at least equal weight to so-called core skills. Only then will our graduates be properly equipped for the GenAI age, whatever roles they take on.

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Chapter 2

Generative AI as a facilitator of deliberate practice in translator training

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Advancement along an expertise trajectory, whether in training or professional contexts, stems from translators having ample opportunity to engage in deliberate practice, defined in the expertise studies literature as strategically designed activities to improve performance (Ericsson et al. 1993). In order for practice to be deliberate, a number of core conditions need to be met, including, among others, intrinsic learner motivation, focused and self-directed performance monitoring, informative (and relatively immediate) feedback, tasks being structured and undertaken at an appropriate difficulty level, and learners having opportunities to correct errors (Shreve 2006). In translator training contexts, complex, heterogeneous learner profiles, particularly when it comes to knowledge, skills, and competences (EMT 2022), pose inherent challenges in attempts to make sure these conditions are adequately met. In addressing these challenges, and in optimizing opportunities for deliberate practice in general, Generative AI offers pedagogical value as a vehicle to guide self-directed learning. This chapter will discuss how generative AI can be used to facilitate deliberate practice in translator training, where students engage in self-directed metacognitive activity to reflect on and assess facets of their own performance, and, in doing so, acquire and advance their expertise.

1 Introduction

Technological advancement, most recently in the realm of generative AI, is rapidly changing the roles and responsibilities of professional translators. It is also re-shaping the competences they need to possess to find success in the language industry, as reflected in the recently updated EMT Translation Competence Framework (EMT 2022). In times of seemingly perpetual change and a

Erik Angelone. 2025. Generative AI as a facilitator of deliberate practice in translator training. In JC Penet, Joss Moorkens & Masaru Yamada (eds.), *Teaching translation in the age of generative AI: New paradigm, new learning?*, 27–47. Berlin: Language Science Press. DOI: ??

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corresponding need to adapt (Angelone 2023), one aspect regarding competence has remained fairly constant, namely the need for translators to critically reflect on, assess, and optimize their own performance. This capacity is articulated in several of the sub-competences pertaining to overarching ‘translation competence’ in the EMT framework, such as translators being able to “analyse a source document, identify potential textual and cognitive difficulties and assess the strategies and resources needed to reformulate it”, and “analyse and justify their translation solutions and choices” (EMT 2022: 8). It is also manifest in the domain of personal and interpersonal competence as the ability to “manage workload, cognitive load, stress and critical professional situations”, and “continuously self-evaluate, update and develop competences and skills through personal strategies and collaborative learning” (EMT 2022: 10).

Competence, which can broadly be defined as the set of knowledge and skills needed to successfully translate, is closely tied to expertise, which, depending on research paradigm and theoretical foundation, can be described as ‘consistently superior performance’ in a given task domain (Ericsson & Charness 1994), or ‘optimal performance’ across varied, but interrelated task domains (Hatano & Inagaki 1986). Performance, when it comes to successful translation, involves not only quality (such as identifying and mitigating errors or successfully adhering to established project specifications), but also productivity (such as leveraging assistive technologies to expedite output or utilizing ergonomically sound approaches when facing time constraints). Expertise should be regarded as something developmental and incremental (Shreve 2018) rather than a desirable ‘end state’ in the sense of an ‘expert’ translator. With this in mind, training for translation expertise acquisition and advancement has an important place from the outset.

In order to advance along an expertise trajectory, for example from ‘novice’, to ‘advanced beginner’, to ‘competent’, to ‘proficient’, to ‘expert’ (Dreyfus & Dreyfus 1986), experience alone will likely not suffice. Instead, translators should intentionally seek opportunities to engage in ‘deliberate practice’, which can be defined as “individualized training activities especially designed by a coach or teacher to improve specific aspects of an individual’s performance through repetition and successive refinement” (Ericsson & Lehmann 1996:278-279). Deliberate practice necessitates moving beyond “plateaus where one is comfortable and confident” (Horn & Masunaga 2006:601). Unlike the somewhat predictable activity and approaches that translators might be inclined to encounter in their daily work, deliberate practice involves “exploring alternative methods with unknown reliability” (Ericsson et al. 1993:368). Indeed, common industry constraints on time, budget, infrastructure, and risk-taking in general make deliberate practice

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something distinct from ‘work’. It is more of an external, upfront investment that can ultimately optimize work performance.

2 Core conditions of deliberate practice

In order for practice to be deliberate, a number of important core conditions need to be met. Translation Studies scholars have outlined these as they pertain to the task of translation (Shreve 2006). Perhaps most important is the condition of translators receiving immediate, informative feedback on their performance. “In the absence of adequate feedback, efficient learning is impossible and improvement minimal” (Ericsson et al. 1993:367). Informative feedback ideally encompasses aspects of translation that are transferable and applicable across tasks. For example, feedback on the efficacy or inefficacy of a given external resource or assistive technology is likely more informative than feedback on a misspelled word that the translator might encounter in the context of one given translation but never again.

As a second condition, translators need to have opportunities to correct errors. In pedagogical contexts, this condition is often met through workshopping, where students submit and receive feedback on drafts, followed by the opportunity to submit revisions in which errors brought to their attention can be corrected. Learning is particularly enhanced when error correction stems from discovery-based learning, where errors are annotated, but deliberately not fully spelled out by trainers. The very presence of a trainer is yet another core condition of deliberate practice. The trainer, or ‘coach’, is responsible not only for providing informative feedback, but also for establishing individualized learning objectives (Miller et al. 2020) and designing tasks based on firm understanding of the learner’s pre-existing knowledge. Some proponents of deliberate practice-oriented training suggest that individualized supervision is ultimately preferred over group-based instruction (Ericsson et al. 1993:367).

The trainer also plays a pivotal role in fulfilling another core dimension of deliberate practice, namely making sure the translator is engaging in tasks of an appropriate difficulty level. Stagnation, due to strict adherence to the tried and true, can result in a plateau effect that stands in the way of expertise advancement. Translators may encounter this when working on tasks of a uniform difficulty level (same source text length, same readability level, same set of external resources, same project specifications, same assessment parameters, etc.). Trainers can push trainees to work outside of their comfort zones and help determine “when transitions to more complex and challenging tasks are appropriate” (Ericsson et al. 1993:367).

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Deliberate practice hinges on the learner's intrinsic motivation, another core condition. Translators looking to advance along an expertise trajectory need to embrace working outside of their comfort zones and have a firm belief that deliberate practice will ultimately improve their performance. Deliberate practice takes immense time and effort. The oft-debated 10,000-hour 'rule' to becoming an 'expert' (Gladwell 2011) is often cited in attempts to quantify just how much time and effort are required. This would amount to "two and a half years of sustained effortful practice every day for five hours" (Shreve 2019: 173-174), raising questions regarding feasibility of this 'rule' and its place in the deliberate practice model.

Deliberate practice calls for the learner's commitment to "conscious performance monitoring" (Horn & Masunaga 2006:601) and engaging in full concentration, as opposed to "mindless, routine performance" (Ericsson 2006: 692). This takes self-discipline and learner dedication to honing metacognitive capacities. Strategic scaffolding by the trainer, along with trainer/trainee dedication to the central ideas of cognitive constructivism (Piaget 1952), are instrumental in bolstering learner metacognition and performance monitoring processes.

Figure 1 provides an overview of the core conditions of deliberate practice outlined in this chapter. It does not represent an exhaustive list of all conditions mentioned in the Expertise Studies literature, but rather focuses on those conditions taken up in Translation Studies to date.

3 Challenges to the implementation of deliberate practice

It is worth noting here that empirical research on the benefits of deliberate practice on translation performance is still quite scant. This dearth can at least be partly explained by a series of inherent challenges associated with its implementation in formal training contexts. As mentioned in the previous section, several proponents of deliberate practice suggest that its benefits are best realized in contexts involving learners working one on one with individual trainers and in the absence of a pre-set curriculum (Ericsson et al. 1993:367). This type of design does not readily align with the fashion in which translators are usually trained for a number of different reasons, starting with the financial constraint of needing to hire a personal trainer.

Another constraint potentially standing in the way of meeting several of the core conditions of deliberate practice is the degree of learner heterogeneity commonly found in classroom-based training contexts. Students of translation often have widely varying levels of competence and experience, not to mention di-

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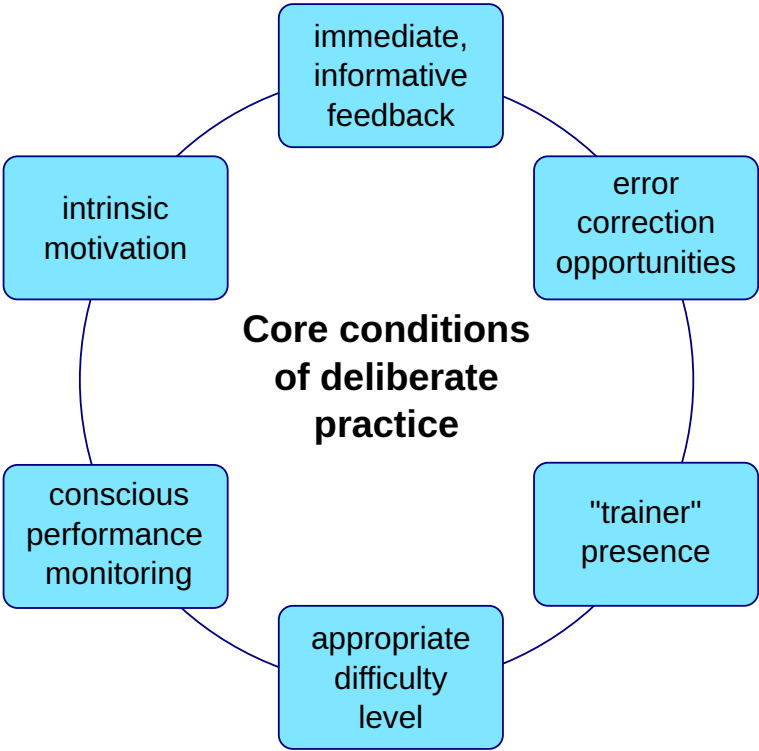


Figure 1: Core conditions of deliberate practice

verse learning needs and interests. This can make it difficult for trainers to establish truly individualized learning objectives and corresponding tasks to meet them. The problem is exacerbated when student enrollments are high. Feedback, arguably the most important dimension of deliberate practice, often becomes less detailed and, out of necessity, much less immediate. Peer feedback and self-feedback activities can help address this gap, but immediacy, as a criterion for practice to be deliberate, often remains very difficult to obtain.

Beyond time constraints, the deliberate practice condition of informative, immediate feedback might not be met if the kind of feedback being provided is not transferable in the sense of applying to future translation tasks. Oftentimes, assessment rubrics are used to mark up errors in accordance with various textual levels, such as grammar, word choice, and syntax. If the feedback given pertains to patterns along these lines that are applicable across tasks, it could be regarded as truly informative. If, on the other hand, feedback simply consists of marking up one-off errors pertaining to items that the translator may never encounter

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again in future translations, such as an isolated collocation error, it is relatively shallow and not particularly informative.

Learner heterogeneity in large cohort settings also presents challenges when it comes to meeting the deliberate practice condition of making sure tasks are being undertaken at an appropriate level of difficulty. Translation practice courses are often informed by trainer intuition and prediction of appropriate difficulty level, in turn based on such facets as level of study, in-take examination results, various cognitive process metrics (Sun & Shreve 2014), or previous student performance and corresponding ‘rich points’ (PACTE 2011). However, studies have shown a danger for misalignment between perceived or predicted problems and what actually proves to be problematic when translating (Angelone 2018). What is assumed to be difficult is actually not necessarily so, making attempts at predicting and setting an appropriate difficulty level challenging, particularly across a wide range of students with diverging needs. Furthermore, the idea of having students work outside of their comfort zone, or at the periphery of what they can realistically accomplish, for purposes of advancing expertise may contradict approaches to translator training that embrace the predictability of staying within the learner’s comfort zone.

According to deliberate practice guidelines, the learner’s intrinsic motivation needs to be constant. Having translation students work outside of their comfort zone in pedagogical contexts, particularly when grades are involved, runs the risk of hampering such motivation. The outcomes of deliberate practice would need to be gauged using metrics beyond formal grades, with a focus on helping learners become more self-reflective translators. Intrinsic motivation would come not so much from getting good grades or doing well in a given course, but rather from seeing the benefits of putting in hard work and embracing difficulty for purposes of becoming better translators.

4 Generative AI as a facilitator of deliberate practice

Against the backdrop of the conditions of deliberate practice presented in Section 2 and the constraints potentially standing in its way, as outlined in Section 3, the question remains: how can we best go about facilitating deliberate practice in translator training for purposes of expertise acquisition and advancement? In particular, how can we establish the requisite highly individualized, ‘coach’-oriented approach at the heart of deliberate practice? The pedagogical features of generative AI, in providing real-time feedback and prompt-driven interaction in a user-centered fashion, would seem to hold potential in this regard, as will be illustrated through a series of concrete scenarios in Section 6.

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At the time of writing, we are still witnessing the dawn of generative AI as a vehicle for optimizing translation, both in professional and pedagogical contexts. Its increasingly ubiquitous integration by LSP companies in project workflows has drawn attention to a need for artificial intelligence (AI) literacy (Krüger 2023), alongside MT literacy and data literacy in a broad sense. AI literacy can be defined as “a set of competencies that enables individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool online, at home, and in the workplace” (Long & Magerko 2020:2). In conjunction with discussions of translator performance, much of the discourse on AI intelligence to date has focused on how generative AI can be leveraged to facilitate human-in-the-loop translation, with an emphasis on the translation product. Outside of several of the chapters in this volume, relatively little discussion has been dedicated to the potential benefits of generative AI as a conversational agent, focusing less on generating translated content, and more so on enabling translators to reflect on their performance and engage in translation tasks to facilitate deliberate practice.

Extending on sociocultural learning theories (Vygotsky 1965), some have come to regard a generative AI tool like ChatGPT as a ‘more knowledgeable other’ (MKO), in essence taking on the role of a personalized trainer that can “lead the learner from the zone of current development to the zone of proximal development – the space where one cannot quite master a content/task of their own, but they can with the help of an expert” (Stojanov 2023: 2). As an MKO, ChatGPT can address the aforementioned conditions of deliberate practice, particularly the presence of a one-on-one personalized trainer. Through real-time responses to learner prompts, ChatGPT ensures the immediacy of feedback that is so difficult to obtain in a larger enrollment classroom-based translator training environment. The learner’s intrinsic motivation is likely to be heightened when training is personalized and self-driven, rooted in immediate, informative feedback, and interactive, driven by their own prompts in relation to aspects of their own performance.

5 *Generative AI as a scaffold for self-directed learning*

The utilization of generative AI for purposes of deliberate practice calls for the learner to partake in self-directed learning (SDL), where “individuals take the initiative, with or without the help of others, in diagnosing their learning needs, setting learning goals, identifying resources, choosing appropriate learning strategies, and evaluating their learning outcomes” (Knowles 1975: 18). In this case,

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generative AI helps scaffold learning in line with learner prompts. Translation trainers well-versed in the conditions of deliberate practice can provide learners with valuable information on the nature of prompts they should enter. However, self-directed learners need to be “able, ready, and willing to prepare, execute and complete learning independently” (Jossberger et al. 2010:419). The need for learner independence does not make the trainer superfluous, but rather shifts the focus of assignments undertaken and how they are assessed.

Models of self-directed learning bear very close resemblance to the deliberate practice model. One such model that is widely cited in the literature consists of three closely interrelated dimensions: 1) self-management, 2) self-monitoring, and 3) motivation (Garrison 1997). Self-management involves the learner establishing concrete learning goals and managing learning resources to achieve these goals. In other words, they take control, deciding on the tasks in which they will engage. From a deliberate practice perspective, through strategic prompts, the translator can leverage generative AI to annotate errors in their translations. Generated annotations could then serve as a framework for the translator to self-discover the nature of the errors. Generative AI could then be prompted to provide similar translation tasks, with the goal of engaging the learner in deliberate practice centered around a certain error pattern (such as avoiding false cognates, erroneous literal translation, or problematic translationese at a syntactic level). Beyond error detection and mitigation, the translator can also use generative AI prompting to self-manage the difficulty level of the tasks they are undertaking. Section 6 provides more concrete scenarios and descriptions along these lines.

Self-monitoring, the second component of Garrison’s model, pertains to the learner’s metacognitive processes. As an important dimension of self-directed learning, self-monitoring “requires learners to take responsibility to construct meanings” (Garrison 1997: 24). Through interactive feedback, generative AI can shed valuable light to help learners identify salient features of the translation task on which to focus their attention. For example, translators can enter a prompt asking generative AI to annotate source content that could be anticipated or predicted to present challenges in translation. Over the past decade, screen recording has found a place in process-oriented translator training for purposes of fostering self-monitoring and to enhance learner metacognition based on documentation of translation behaviors suggesting problems, including pausing, information retrieval, and revision (Angelone 2019). At present, generative AI tools like ChatGPT do not offer functionality where translators can upload screen recordings of their work for purposes of receiving analytic feedback at a granular level. Given recent advancements in this technology however, such as APIs that can provide

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automated video summarization, it is quite likely that utilization of generative AI tools for such purposes is not too far away.

The third component of Garrison's self-directed learning model, motivation, directly aligns with motivation as a core condition of deliberate practice. Whereas deliberate practice regards motivation as the learner embracing challenge through an inherent desire to become better, Garrison draws attention to the importance of motivation for purposes of staying on task. The conversational interface of generative AI tools such as ChatGPT requires active participation on behalf of the learner, and, thereby, a heightened need to stay on task. However, without the physical presence of actual trainers or peers, staying on task, and motivation in general, is not a given. ChatGPT is not inclined to openly praise the performance of the learner to pique interest and motivation. Indeed, at present, the ChatGPT interface itself is quite basic, lacking any structural or discourse elements that might lend themselves well to inherently facilitating learner motivation. It will be interesting to see if this changes over time, perhaps in line with empirical user experience studies.

6 Application scenarios

This chapter will now provide a series of concrete scenarios to illustrate how generative AI can be used to facilitate each of the core dimensions of deliberate practice put forward in Section 2 and to help translators advance along an expertise trajectory. The examples will be based on interaction with ChatGPT based on GPT-4o,¹ given the relative ubiquity and popularity of this particular generative AI tool at the time of writing. This focus on ChatGPT for illustrative purposes does not discredit the growing range of other generative AI tools available for use in a similar fashion. The translation scenario being used for purposes of contextualization is the German-English translation of web content from a German private health insurance company,² translated for an international, English-speaking audience for informative purposes. The English translation was generated using DeepL.³

Figure 2 provides side-by-side alignment of the source and target content.

As a point of departure, and to underscore the importance of strategic prompting when it comes to utilizing GenAI for facilitating deliberate practice, (1) de-

¹<https://openai.com/index/gpt-4/>

²<https://www.huk.de/gesundheitsvorsorge-vermoegen/krankenversicherung/krankenvollversicherung.html#huk>

³<https://www.deepl.com/en/translator>

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German (detected) ▾	↔ English (American) ▾	Glossary
Ihre private Krankenversicherung der HUK-COBURG Ihre Vorteile im Überblick Auf Sie zugeschnitten – Individuelle und bedarfsgerechte Krankenversicherung mit 3 leistungsstarken PKV-Tarifen. Faire Konditionen – Dauerhaftes, garantiertes Leistungsversprechen. Rückerstattung – Aktuell bis zu 2 Monatsbeiträge bei Leistungsfreiheit*. Unser Schutz ist ausgezeichnet – Focus Money verleiht uns zum wiederholten Mal das Prädikat „Fairster privater Krankenversicherer“ – Note „sehr gut“ (Ausgabe 04/2024).	×	Your private health insurance from HUK-COBURG Your advantages at a glance Tailored to you – Individual and needs-based health insurance with 3 high-performance private health insurance tariffs. Fair conditions – Permanent, guaranteed benefit promise. Reimbursement – Currently up to 2 months' premiums in the event of no benefits*. Our protection is excellent – Focus Money has once again awarded us the title of "Fairest private health insurer" – rating "very good" (issue 04/2024).

Figure 2: German-English translation of private health insurance content

picts the relatively shallow output proposed by ChatGPT based on the following generic prompt:

- (1) “Set up translation training tasks with the goal of facilitating deliberate practice”.
- (2) ChatGPT output for setting up translation tasks to facilitate deliberate practice

Setting up translation training tasks for deliberate practice involves focusing on specific skills, receiving immediate feedback, and gradually increasing difficulty.

 - 1. Identify key skills and goals**

Skills: Vocabulary, grammar, idiomatic expressions, cultural context, speed, accuracy, style, and tone.

Goals: Improve comprehension, enhance translation quality, increase speed without losing accuracy, and develop a nuanced understanding of both source and target languages.
 - 2. Feedback and reflection**

Immediate Feedback: Provide prompt feedback on translations, highlighting errors and explaining corrections.

Self-Assessment: Encourage learners to reflect on their work and identify areas for improvement.
 - 3. Deliberate practice cycle**

Plan: Identify specific skills to focus on in each session.

Practice: Perform targeted translation tasks.

Feedback: Receive and review feedback from instructors or peers.

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Reflect: Reflect on the feedback and identify areas for improvement.

Repeat: Repeat the cycle with new tasks, gradually increasing difficulty.

(3) depicts corresponding training tasks proposed by ChatGPT in parallel to the content in (2).

(3) **Week 1: Basic sentences**

Translate 10 simple sentences from the source language to the target language, with a focus on vocabulary and basic grammar.

Week 2: Themed paragraphs

Translate a paragraph related to a specific theme (e.g., technology), with a focus on thematic vocabulary and coherence.

Week 3: Idiomatic expressions

Translate sentences rich in idiomatic expressions, with a focus on cultural nuances and idiomatic accuracy.

Week 4: Time translations

Translate a 200-word article within 20 minutes, with a focus on speed and maintaining accuracy under time constraints.

Week 5: Reverse translation

Perform a back translation exercise, with a focus on understanding equivalence and identifying subtle changes in meaning.

While tapping into the core conditions of deliberate practice put forward in the literature, this generic framework proposed by ChatGPT seems to focus first and foremost on group-based (rather than truly personalized) training and also seems to make the assumption that learners are starting ‘from scratch’ (moving from basic vocabulary to field-specific terminology, to cultural and idiomatic expressions, for example). Given the inherent learner heterogeneity described in Section 3, many translators would not directly benefit from such a lockstep, linear approach. As far as difficulty is concerned, it seems as if there is less focus on working at an appropriate difficulty level per se and more of an emphasis on incorporating a variety of potential sources of difficulty in a more fragmented sense (thematic vocabulary, idiomatic accuracy, time pressure, etc.). This is not to suggest that ChatGPT cannot personalize deliberate practice and cater to the needs of the individual user, but rather highlights the important, active role that the individual learner needs to play in order to benefit from generative AI. This

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starts with utilizing more granular prompts in accordance with each specific condition of deliberate practice rather than taking a broad-brush approach such as that associated with the generic prompt that generated the output here.

7 Immediate, informative feedback

At a very basic level, translators can utilize ChatGPT to get immediate, informative feedback on their performance by entering source content and their corresponding translations, along with correlating project specifications for the task (such as audience and purpose), in conjunction with the following prompt:

- (4) “Provide me with feedback on the following translation, German to English, of content from a private health insurance company in Germany for an international English-speaking audience, with the purpose of providing information”.

(5) depicts the feedback obtained from ChatGPT in relation to this prompt.

- (5) ChatGPT feedback on a translation

Your translation is clear and accurate, but I have a few suggestions to make it sound more natural for an international English-speaking audience. Here are the changes made:

- 1.Changed "advantages" to "benefits" for a more common term in health insurance.
2. Changed "tariffs" to "plans" to better fit the context.
3. Added a colon after "Your benefits at a glance" for better readability.
4. Capitalized "Private Health Insurer" for consistency with the title awarded.

This feedback was generated in a matter of a few seconds, overcoming the time constraint often standing in the way of feedback being immediate in a typical translator training environment. As mentioned in Section 3, feedback becomes truly informative when it is applicable across translation tasks under similar conditions. With this in mind, the translator could enter the following prompt in response to the obtained feedback:

- (6) “Which of this feedback would be applicable when translating other private health insurance content from German into English in the future”?

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(7) provides an overview of the response provided by ChatGPT, illustrating how feedback can become truly informative and transferable, as opposed to isolated to the one translation task at hand.

(7) Informative, transferable feedback

When translating content for other private health insurance companies in the future, you can apply the following feedback to ensure clarity, accuracy, and appropriateness for an international English-speaking audience:

1. Use Clear and Common Terms:

Replace terms like "advantages" with "benefits" for more common understanding.

2. Ensure Readability and Natural Flow:

Add colons or other punctuation to improve readability (e.g., "Your benefits at a glance:").

3. Maintain Consistent Formatting:

Use bullet points to list benefits for easy reading and better presentation.

Interestingly, ChatGPT seems to attribute transferability to each piece of general feedback provided in response to the initial prompt, as depicted in (1). This suggests a potential tendency for ChatGPT to conflate task-specific and transferable feedback. However, this did not hold true in the context of a different simulated translation task, where some of the task-specific feedback was deliberately filtered out and not inherently regarded by ChatGPT as also being transferable. Nevertheless, from a prompting standpoint, for purposes of obtaining truly informative feedback, the translator may want to ask: "Which of this feedback is likely specific only to this one translation task and which of the feedback is transferable across translations in the same domain?", with the latter being more beneficial from a deliberate practice perspective.

7.1 Error correction opportunities

Immediate, informative feedback, as a core condition of deliberate practice, provides the translator with granular insight into the nature of errors. In hopes of advancing along an expertise trajectory, it is just as important for the translator to have opportunities to correct errors they have made by virtue of having them annotated but not spelled out, and to translate additional source content in which

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they would be prone to making similar errors. ChatGPT can be prompted to annotate translation errors without correcting or describing them, using the same approach outlined in the preceding subsection. For this purpose, the following prompt could be used:

- (8) “Annotate errors in the following English translation of German source content for a private health insurance company using a bold font. The translation has an informative function and targets an international audience. Don’t provide any descriptions or corrections for the errors”.

(9) provides such annotation, with bold content reflecting errors in need of change, but not described in any capacity.

- (9) Annotated errors, with no descriptions or corrections
- Your private health insurance from HUK-COBURG
- Your advantages** at a glance
- Tailored to you - **Individual** and needs-based health insurance with **3** high-performance private health insurance **tariffs**.
- Fair conditions** - Permanent, guaranteed benefit promise.
- Reimbursement - Currently up to 2 months' premiums in the event of no benefits*.
- Our **protection** is excellent - Focus Money has once again awarded us the title of "Fairest private health insurer" - rating "very good" (issue 04/2024).

In the event that the translator needs hints as to the nature of the annotated errors, or, ultimately, more detailed descriptions, ChatGPT can readily provide both when prompted to do so, thereby linking the deliberate practice conditions of having opportunities to correct errors and receiving immediate, informative feedback. The translator can make use of a prompt such as: “Provide me with additional content to translate from German into English where similar errors would potentially be made” to obtain more goal-directed, pattern-specific deliberate practice. (10) depicts what ChatGPT generated in response to this follow-up prompt.

- (10) Generation of similar content in line with previous error patterns
- Ihre private Rentenversicherung der HUK-COBURG
- Ihre Vorteile auf einen Blick
- Individuell angepasst – Maßgeschneiderte Rentenversicherung mit flexiblen Beitragsoptionen.

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- Sichere Altersvorsorge – Garantierte Rentenzahlungen ab Rentenbeginn.
- Zusätzliche Leistungen – Attraktive Zusatzleistungen wie Hinterbliebenenschutz und Berufsunfähigkeitsabsicherung.
- Ausgezeichneter Service – Kundenzufriedenheit auf höchstem Niveau, mehrfach ausgezeichnet von unabhängigen Testern.

The content generated by ChatGPT would seem to potentially trigger similar terminology and collocation errors based on the errors produced in the previous task. Interestingly, and perhaps concerningly, the proper noun name of the same private insurance company, HUK-COBURG, is used, yet, the company itself did not author this content. ChatGPT, in generating similar content, went with the same company name, potentially crossing an ethical line in doing so. That said, and while not to downplay this ethical issue, ChatGPT is successful in providing the translator with similar tasks for purposes of error mitigation and correction practice in relation to previous performance.

7.2 **Trainer presence**

Somewhat inherently, ChatGPT, as an MKO, takes on the role of coach or trainer in facilitating deliberate practice. This is not to suggest that generative AI can think or train in the same capacity as a human trainer, or that human trainers become superfluous through automation. Instead, as a mechanism for self-directed learning, ChatGPT serves as a conversation partner, dependent on optimal prompting and interaction from the learner. It is also a trainer that comes with flaws, reliant on large language models and transformer input rather than years of experience in training translators or working in the language industry. As a trainer, ChatGPT may very well make mistakes, and the success of its implementation in deliberate practice contexts hinges on the learner having sufficient knowledge to detect erroneous information and question the tool accordingly.

7.3 **Appropriate difficulty level**

Building on the deliberate practice conditions of informative, immediate feedback and opportunities for error correction, the translator can prompt ChatGPT to provide content to be translated that is slightly more difficult in relation to previously completed tasks. This approach can address the condition of making sure work is undertaken at an appropriate difficulty level, or, ideally, at a difficulty level that is at the periphery of the translator's comfort level. The content depicted in (12) was generated by ChatGPT in response to the following prompt:

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- (11) "Provide me with a German source text to translate in this same domain that is a little more difficult than my previous translation".
- (12) Generation of a slightly more difficult translation task
- Ihre private Krankenversicherung der HUK-COBURG
- Warum sich eine private Krankenversicherung lohnt
- Umfangreicher Schutz – Unsere Tarife bieten umfassende Leistungen, die weit über das hinausgehen, was gesetzliche Krankenkassen abdecken.
 - Schnelle Terminvergabe – Bevorzugte Behandlung bei Fachärzten und kürzere Wartezeiten.
 - Hohe Flexibilität – Wählen Sie aus verschiedenen Selbstbeteiligungsstufen und Tarifoptionen, die sich Ihren individuellen Bedürfnissen anpassen.
 - Präventionsangebote – Zugang zu exklusiven Gesundheitsprogrammen und Vorsorgeuntersuchungen.
 - Internationaler Schutz – Weltweite Absicherung für Reisen und Aufenthalte im Ausland.
 - Transparente Beiträge – Keine versteckten Kosten, klare und nachvollziehbare Prämienberechnung.

Establishing an appropriate difficulty level often proves to be a daunting task for trainers in formal translation classroom settings. It is interesting to see how ChatGPT goes about describing what exactly it is that makes this task more difficult when prompted to do so, as reflected in (13).

- (13) ChatGPT's description of what makes the proposed task more difficult
- This translation task is more difficult because it includes:
- Complex Vocabulary and Phrasing: The German text uses more specific and technical terms related to insurance, such as "umfangreicher Schutz," "Selbstbeteiligungsstufen," and "Prämienberechnung." Translating these terms accurately requires a good understanding of insurance terminology in both languages.
 - Longer Sentences and More Detailed Content: The sentences are longer and contain more detailed information, which requires careful attention to maintain clarity and coherence in English.

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- **Varied Benefits and Features:** The text lists multiple benefits and features of the insurance, each with its own nuances. Ensuring that each point is clearly and accurately translated without losing meaning or context adds to the complexity.
- **Idiomatic Expressions:** Certain phrases in German, such as "weit über das hinausgehen" and "bevorzugte Behandlung," might not have direct equivalents in English, requiring the translator to find idiomatic expressions that convey the same meaning.

At a quick glance, much of what is deemed to be “more difficult” is somewhat generic (more complex field-specific terminology, a longer text, potential lexical gaps, etc.), as opposed to personalized for the individual translator based on previous performance. This condition of deliberate practice may very well prove to be the most challenging for not only human trainers, but also ChatGPT to meet. Perhaps ChatGPT’s efficacy in doing so hinges on the translator entering multiple translations as part of the prompt. These translations could then serve as a corpus of ‘big data’ for training the system to base what is being proposed as more difficult on concrete patterns (errors, text attributes, etc.).

7.4 **Intrinsic motivation**

The extent to which learners are intrinsically motivated to partake in the scenarios outlined in this chapter is intertwined with the extent to which they value and trust ChatGPT as a ‘trainer’ and the extent to which they embrace self-directed learning. Both can take some time and necessitate adjustment, particularly in the case of translation students who might be used to interacting with peers and instructors. As previously mentioned, intrinsic motivation in the context of self-directed, generative AI-based deliberate practice will likely stem first and foremost from the level of personalization and one-on-one interaction the technology enables. Learners are very much in the driver’s seat. It is their prompting and content input that serves as a catalyst for the entire deliberate practice process. Nevertheless, intrinsic motivation could quickly falter in instances where generative AI makes errors or provides unhelpful feedback. To maintain motivation, translators would need to look beyond such isolated instances and see the bigger picture of how the technology can foster their expertise acquisition on the whole. Depending on learning styles and preferences, some learners may even find motivation at the idea of being able to openly question and challenge the

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non-human ‘trainer’ in such contexts, and would perhaps be less inclined to do so in contexts involving face-to-face instruction involving human trainees and trainers.

7.5 Conscious performance monitoring

Successful deliberate practice calls for the learner to be consciously engaged in reflection on their performance. In the absence of an actual human trainer and face-to-face learning, it is basically up to the learner to make sure they are on task and actively engaged in such monitoring. This important condition of deliberate practice may prove to be difficult for ChatGPT to facilitate in a fashion that is inherently advantageous in relation to what one might find in a face-to-face learning environment. When prompted to describe ways in which it can facilitate learners’ conscious performance monitoring when utilized as a sole ‘trainer’, ChatGPT proposed such strategies as programming the tool to periodically ask users for feedback on their performance, yet also points out that it could not track detailed usage analytics or goal progression. Instead, it can guide users on how to track their usage manually. In other words, the onus is truly on the learner when it comes to meeting this condition of deliberate practice.

8 Conclusion

Whereas many of the current discussions surrounding generative AI involve its potential place in translation project workflows and in the generation of translated content, this chapter advocates for leveraging its pedagogical features as an MKO and conversational agent in facilitating deliberate practice and expertise acquisition for human translators. Its primary advantages in this capacity lie in its ability to generate immediate, informative feedback in a highly personalized fashion. As far as the nature of feedback is concerned, a tool like ChatGPT is equally efficacious at annotation for purposes of error correction and description for purposes of facilitating nuanced understanding and error mitigation. It can readily provide translators with deliberate practice in relation to a given text attribute or error type, with an eye towards transferability across tasks. Personalized feedback, one-on-one training, and ongoing interaction through learner prompt entry and tool response can foster intrinsic motivation. The core deliberate practice conditions of making sure tasks are undertaken at an appropriate difficulty level and that learners engage in conscious performance monitoring, at least at the time of writing, are still somewhat of a challenge for ChatGPT, yet both can be addressed through strategic prompting.

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Truly experimental studies on the benefits of self-directed, deliberate practice in relation to more traditional learning approaches are still limited in scope, for the main reason that the core conditions of deliberate practice have been quite hard to address to date (Miller et al. 2020:8). This is not something specific to translator training, but training in general, in all situations where learner heterogeneity, time constraints, and the absence of one-on-one training opportunities emerge as roadblocks. ChatGPT offers untapped potential to be a game changer in this regard and will hopefully draw renewed interest in both pedagogical and empirical research on deliberate practice and its place in translation expertise acquisition.

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Chapter 3

AI Literacy: The concept of suitability and core translation skills

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The release of ChatGPT in November 2022 reignited the debate on the future of the translation profession and, consequently, on how translator training programmes should change and adapt. Similar to discussions that occurred with the introduction of neural machine translation a few years earlier, some argued that core translation skills, such as language knowledge, translation ability and cultural expertise that have long been essential components of many translator training programmes, had now been rendered obsolete by the arrival of generative AI (GenAI).

Inspired by the concept of machine translation literacy, a case will be made that these core skills (together with some other complementary skills, such as selection and assessment) are absolutely essential in order to ascertain whether the content produced by GenAI tools is not simply accurate or inaccurate, but, more importantly, suitable for the requirements of any given translation project, as specified in the relevant translation brief.

Examples will be given of specific AI-based tasks (such as translation of content, terminology extraction, multilingual glossary creation and machine translation post-editing) in which the suitability of the results cannot be determined without recourse to so-called traditional core translation skills.

1 Machine translation literacy and artificial intelligence literacy

The concept of machine translation (MT) literacy, introduced by [Bowker & Buitrago-Ciro \(2019\)](#), has attracted a lot of interest, both in academia and in

Ramon Inglada. 2025. AI Literacy: The concept of suitability and core translation skills. In JC Penet, Joss Moorkens & Masaru Yamada (eds.), *Teaching translation in the age of generative AI: New paradigm, new learning?*, 49–64. Berlin: Language Science Press. DOI: ??

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the language industry, as well as in other sectors. Furthermore, Bowker's MT Literacy project¹ aims to educate users on the dos and don'ts of using MT output and its infographics have been very popular on social media. It also highlights the importance of confidentiality and privacy when using programs such as DeepL and Google Translate, with an emphasis on different use cases of free online MT systems. The project's focus on enhancing digital literacy skills and the responsible, ethical, and sustainable use of MT systems can be used as the basis to further develop and adapt the concept of Artificial Intelligence (AI) literacy. Long & Magerko (2020: 2) defined AI literacy as "a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace". In this chapter, this definition will be used as the basis for a different definition specifically created with the translator training context in mind. This approach aligns with (Krüger 2024) framework for AI literacy in translation, which emphasises the need for translators to develop competencies in understanding and effectively using AI technologies in their work.

It is undeniable that the release of ChatGPT in November 2022 generated a lot of hype. It also reignited the debate on the future of the translation profession and, consequently, on how translator training programmes should change in order to adapt to this latest technological advancement. Similar to discussions that occurred with the introduction of neural machine translation (NMT) systems a few years earlier, some argued that core translation skills that have long been essential components of translator training programmes had now been rendered obsolete by the arrival of GenAI. However, it could be argued that this is an oversimplistic approach.

This chapter advocates for technological advancement, as it can bring many benefits to society and offer solutions to some of the greatest challenges humanity currently faces. In the field of translation, when used sensibly as part of well-designed workflows, computer-assisted translation (CAT) tools and MT can be a great asset for professional translators. The same applies to GenAI chatbots based on large language models (LLMs), such as OpenAI's ChatGPT, Microsoft Copilot and Google Gemini. However, given how recent this technology is and the pace at which it evolves, it could be argued that the translation industry might struggle to agree on what a 'sensible' and 'well-designed' AI-based workflow constitutes in the field of professional translation. In any case, it is always worth keeping in mind that no technology-based solution is infallible. This obviously applies to ChatGPT (and to the other LLMs) and it could be said that, to

¹<https://sites.google.com/view/machinetranslationliteracy/>

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a certain extent, even ChatGPT itself reminds us to be vigilant and use our own judgment and common sense. At the time of writing, this is the message that appears underneath ChatGPT's chat interface:

ChatGPT can make mistakes. Consider checking important information.

2 Core translation skills

The purpose of this chapter is not to provide an exhaustive list of core translation skills, describing what they are and why they can be considered as key skills all translators should have, as many others have already done this very effectively in the past. Instead, it will focus on three very general skills which, arguably, most translation scholars would generally agree are essential for all professional translators (and which, therefore, should feature prominently in all translator training programmes). This is also done for the sake of simplicity as, ultimately, these skills are mere examples. This selection of core skills has been based on the reference standards for translator training set out in the European Master's in Translation (EMT) Competence Framework (2022) and on the translation competence model created by PACTE (2003). Different translator training programmes, institutions and settings could naturally select other skills that they would consider as essential.

These chosen three core skills are:

- Linguistic knowledge
- Translation ability
- Cultural expertise

Core translation skills such as the ones mentioned above should not be allowed to disappear from translator training programmes, but rather should now be considered more important than ever, precisely because of the advent of GenAI. This is not to say that translator training curricula should not be adapted: they should, particularly as concepts such as augmented translation and human in the loop are becoming increasingly important, and discussions around translators now becoming 'language specialists' or 'language experts' are happening with increasing frequency. However, the debate should not be about these core skills becoming obsolete, but about how essential they are when GenAI is used in translation.

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3 The concept of suitability over correctness

When translators or translation students use ChatGPT or other similar tools to assist them in their translation tasks, the focus should not be on whether the AI-generated output is ‘correct’ (something that could also entail a degree of subjectivity), but rather on whether such output is suitable (or fit-for-purpose) for any given set of translation requirements, as described in the translation brief, for instance in terms of tone, register, target audience, vocabulary and text type. The priority should be placed on the concept of ‘suitability’ of AI-generated content, rather than on its ‘correctness’. This cannot be successfully achieved if the aforementioned core skills (as a minimum, linguistic knowledge, translation ability and cultural expertise) are not present and well honed. Translators would struggle to decide whether AI-generated content is suitable for any given translation project (and its set of unique needs) if they do not know how to translate (and they have not developed their core translation skills). Using GenAI chatbots is easy; using their output critically requires thought.

4 New and complementary skills

It could also be argued that not only are core translation skills are still essential, but that they should also be complemented with the ‘new’ skills of selection and assessment. These are obviously not new skills — after all, many translators have been already selecting and assessing translation memory matches for decades — but they have now become fundamental when using GenAI in translation. Professional and trainee translators need to be able to select among the alternatives offered by ChatGPT and other similar tools. They also need to be able to critically assess the suitability of the output being presented to them. Therefore, a renewed emphasis should be given to selection and assessment as core skills to be integrated and developed in translator training programs.

The reasons why ‘traditional’ core language, translation and cultural expertise skills should still be a pivotal part of translator training programmes have already been discussed, in combination with the reasons why they should also be complemented by the not-so-new skills of selection and assessment. However, is there any additional new skill that should be added to this list of core skills for the training of translators in the new era of GenAI? The answer is a resounding yes. This skill is prompting (also known as prompt engineering). Prompt engineering can be defined as the process of designing, crafting, and refining inputs to elicit specific responses from a GenAI model, aiming to optimize interaction outcomes through careful consideration of the prompts (Bozkurt 2024). The model

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then generates a response based on the input it receives. Effective prompting is crucial for obtaining relevant, coherent and suitable answers. Translator training programmes should seek to integrate prompting skills into their curricula, so that future translators can ensure a greater degree of suitability in the output generated by GenAI. When adapting existing translator training programmes (or when designing new offerings), an emphasis should be placed on processes aimed at the careful creation of effective prompts (or sequences of prompts) with different roles/personalities, different levels of complexity or even different degrees of creativity.

5 AI Literacy in translation: a simple definition

Based on all the principles discussed before, AI Literacy for Translation could be defined as the combined application of a definite set of basic core skills in order to maximise the usefulness and relevance of GenAI output and ensure its suitability in relation to the requirements set out by the translation brief in any given translation task. The basic core skills are linguistic knowledge, translation ability and cultural expertise, complemented by selection and assessment. All these skills should be applied to GenAI output produced as a result of effective prompting. The overarching notion that underpins the application of AI Literacy in translation is that of suitability, rather than ‘correctness’, of the generated output.

6 AI Literacy applied to practical language and translation tasks

Five examples will now be provided of translation or translation-related tasks, chosen to represent some of the tasks that professional translators carry out on a regular basis (and which, as such, are also commonly observed within the confines of the translation classroom). The concept of AI Literacy will be applied to these examples. This means that, in all cases, an argument will be presented to emphasise the importance of relying on the three generic core skills of linguistic knowledge, translation ability and cultural expertise (presented earlier in this chapter) in order to select the most appropriate GenAI output and assess the suitability of this output. An example of the prompt (or prompts) used to request the completion of the task in hand by the GenAI chatbot (ChatGPT based on GPT 3.5) will also be provided for each example.

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In the first example, ChatGPT itself was used to request the creation of a sentence related to renewable energy containing some grammar issues and awkward word groupings. In the other four examples, the three initial paragraphs from the English version of the Wikipedia article on the topic of renewable energy (from July 2024) were used as the reference source text.

These five examples are: sentence reformulation, translation of content, terminology extraction, multilingual glossary creation and MT post-editing (MTPE).

6.1 Example 1: Sentence reformulation

For our first example, I will start with one of the simplest tasks LLMs can be used for. This task has also been chosen as an introductory example as anecdotal evidence seems to suggest that this represents one of the primary ways in which GenAI chatbots are commonly utilised, especially by non-native speakers of the language they are engaging with.

ChatGPT can be a valuable tool for reformulating badly written sentences. By providing a poorly constructed sentence to ChatGPT, users can receive suggested revisions that address spelling mistakes, grammar errors, improve clarity and enhance overall readability. ChatGPT leverages its model of language patterns and grammar rules to generate alternative phrasing that is more coherent and idiomatic. Users can input sentences from various fields, such as renewable energy in our example, to receive tailored suggestions for improvement. With ChatGPT's assistance, individuals can refine their writing skills, produce clearer communication and convey their messages effectively across diverse contexts.

In this example, the following prompt was used to ask ChatGPT to create the sentence I would use as a basis for reformulation (see (1)):

- (1) Prompt used to elicit a poorly constructed sentence from ChatGPT 3.5.
(Created by Ramon Inglada)

Can you please give me an example of a poorly worded phrase in English? It has to be related to the field of renewable energy and include grammar errors and non-idiomatic collocations.



This was the sentence resulting from the prompt above:

- (2) 'The sun is giving us muchly energy, so we should put many solar panels to catch it all'.

The following prompt in (3) was then used to request an improved version of the sentence, correcting any grammar issues and changing any unnatural word collocations:

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(3) Sentence reformulation prompt in ChatGPT. (Created by Ramon Inglada)

You are an expert in English language. Please reformulate the following sentence to correct any grammar issues and non-idiomatic collocations: The sun is giving us muchly energy, so we should put many solar panels to catch it all.



This was ChatGPT's reply to our request:

(4) 'The sun provides us with abundant energy, so we should install numerous solar panels to capture it efficiently'.

It could certainly be argued that this version corrects existing grammar issues and uses more idiomatic collocations to convey the intended meaning more clearly. However, it would only be possible to do that if our knowledge of the English language is advanced enough. If I now try to apply the concept of AI Literacy in this first example, it is immediately evident that I would need to rely on one of the aforementioned core skills (in this case, knowledge of the English language) firstly to realise that the original sentence contains some issues, and secondly to assess the suitability of the suggested 'improved' version provided by the LLM. Furthermore, in this specific example, ChatGPT decided that three elements in the original sentence needed to be changed. These are 'muchly energy', 'many solar panels' and 'catch it all'. If my English language skills were insufficient, I could simply accept all three suggested improvements, assuming (or even hoping) that the resulting sentence is now much more suitable for my needs. However, I could also decide to select only some of these changes, so I would only keep those that I (and not ChatGPT) consider as suitable. I could even accept them all and then go on to further modify them to end up with a final sentence which would be the collaborative result between me and the LLM. This process exemplifies the concept of "centaur tasks" as described by [Mollick \(2023\)](#), where there is a clear division of labour between human and AI, leveraging the strengths of each. Once more, the decision on whether this final sentence is the most suitable option for our needs is something that can only be achieved if our core skills have been developed enough.

6.2 Example 2: Translation of content

This second example is intended to cover the use of LLMs as MT tools. While LLMs were not in principle designed to be used as MT providers, this is certainly one of the many tasks these tools can perform. It is also probably one of the first uses that comes to mind when one thinks about the potential uses of GenAI in

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the field translation. There is, indeed, ample anecdotal evidence of the use of LLMs as MT tools.

(5) below is an example of a prompt that could be used to ask ChatGPT to translate a piece of text. Yamada (2023) explored how incorporating the translation's purpose and the target audience into prompts affects the quality of translations generated by ChatGPT. Therefore, in this specific example, the chatbot will be given a concrete personality, the language combination to be used will be English into French, the content to be translated is related to renewable energy, and the translation is to be published in a government report.

(5) Content translation prompt in ChatGPT. (Created by Ramon Inglada)

You are a professional English into French translator specialised in renewable energy. Please translate the following text into French for publication on a government report.



If I were to go ahead and enter this prompt in ChatGPT followed by a text for translation, the LLM would promptly generate a translated version of the content in question (in French in this case). As mentioned earlier, the three initial paragraphs from the English version of the Wikipedia article on the topic of renewable energy have been used as the source text. At this point, I would have two options. Option 1 would simply entail accepting the generated translated content at face value, without conducting any checks and without verifying the suitability of the said content (depending on a series of factors that could include overall quality, register, tone, intended audience, publication media and so on). Option 2 would involve operationalising the concept of AI Literacy. Using my core skills, I could perform all the checks mentioned beforehand. I could select other potential translation alternatives requested to the LLM for any passages that I might not be satisfied with (for instance, asking ChatGPT to rephrase any given part of the translated text). Finally, I could also assess the overall suitability of the generated output. The relevance and usefulness of both the initial translation-generating prompt and any additional prompts entered in the chatbot (for instance to check for alternative formulations or to change the level of formality in the translated text) would be greatly dependent on the use of effective prompting techniques.

6.3 Example 3: Terminology extraction

One of the potential uses of GenAI tools that was quickly identified as potentially highly interesting for translators, interpreters, terminologists, researchers and other professionals was that of a terminology extraction tool. Users can request ChatGPT to create a list of key terms from a given passage of text. Users can ask

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for a specific number of terms to be identified, and they can even request the tool to present the results in a table format, to facilitate further work. An example of a prompt that could be used for this purpose (using a text in the field of renewable energy, including all the requirements mentioned above and also a personality) is shown below:

- (6) Prompt: You are a professional terminologist. Please extract a list of 15 key terms from the following text in the field of renewable energy and present them in table format. Please also include a definition.
- (7) The first four terms related to renewable energy and their definitions, as provided by ChatGPT (Created by Ramon Inglada)

Term	Definition
Renewable energy	Energy derived from naturally replenished resources on a human timescale, including sunlight, wind, water movement, and geothermal heat.
Sustainable	Practices or sources of energy that can be maintained or continued over the long term without causing harm to the environment or depleting resources.
Biomass	Organic materials, such as wood or agricultural crops, used as fuel or for industrial production, considered a renewable resource but can be unsustainable at high rates of use.
Electrification	The process of expanding the use of electricity to replace traditional methods of energy consumption, often associated with renewable energy deployment.

As can be seen in (7), when using this prompt and providing ChatGPT 3.5 with a text to perform the terminology extraction on, the LLM will very quickly produce a convenient table containing the specified number of terms in the field in question and also their definitions. Once again, I could simply accept the automatically created selection of terms and their corresponding definitions as relevant and suitable for our needs (whatever they might be). However, I could also use critical thinking and in this instance, again, apply the concept of AI Literacy. The speed and convenience with which ChatGPT has created this list of terms and corresponding definitions is astounding (and several orders of magnitude faster than performing a similar process manually), but how can any user ensure the suitability of the generated terms and definitions if the core skills are lacking? How can I select the candidate terms that might indeed be useful for our needs in any given task? How can the suitability (and even accuracy) of the definitions provided be assessed? And how can I specify a concrete number of terms to be

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provided in a very convenient table format if I do not have even the most basic notions in the craft of effective prompting?

6.4 Example 4: Multilingual glossary creation

Following the previous example, Example 4 also focuses on a terminology-related task that can be performed using ChatGPT or any other of the now widely available GenAI tools. As illustrated in Example 3, when using the right prompt, it is very quick and simple to create a monolingual list of key terms related to virtually any topic, with their corresponding definition and in a convenient table-based format. By using a slightly different prompt, it is an equally simple and quick task to create a multilingual glossary on virtually any given domain. Once more, I can specify the exact number of terms I would like the tool to generate, and I can also request the output to be presented in an orderly table format. An example of the prompt that could be used to achieve this task is presented below:

- (8) Prompt: You are a professional terminologist. You work in the field of renewable energy. Please extract a list of 15 key terms from the following text in the field of renewable energy and present them in table format. Please also include the translation of these terms in both French and Spanish.
- (9) Multilingual glossary (English, French and Spanish) in the field of renewable energy. (Created by Ramon Inglada)

English Term	French Translation	Spanish Translation
Renewable energy	Énergie renouvelable	Energía renovable
Renewable resources	Ressources renouvelables	Recursos renovables
Sunlight	Lumière du soleil	Luz solar
Wind	Vent	Viento
Movement of water	Mouvement de l'eau	Movimiento del agua
Geothermal heat	Chaleur géothermique	Energía geotérmica
Sustainable	Durable	Sostenible
Biomass	Biomasse	Biomasa
Electricity generation	Production d'électricité	Generación de electricidad
Heating and cooling	Chauffage et refroidissement	Calefacción y refrigeración
Large-scale	Grande échelle 	Gran escala

As can be seen in the example prompt above, ChatGPT has been provided with a role (that of a professional terminologist), as it was already the case in

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Example 2. However, in this example, this role (or personality) has been further fine-tuned by also including the field (renewable energy) in which this imaginary professional terminologist would work (as this is the topic of the text being used in previous examples). It is equally worth noting that, by using the prompt above, a request has been sent for ChatGPT to create a multilingual glossary containing the source terms in English and their corresponding translations in both French and Spanish (9). However, by slightly modifying the instructions in the prompt and simply reducing the number of languages to two (e.g. English as source language and French as target language), the user would obtain a bilingual (rather than a multilingual glossary), a resource that could potentially be more useful to assist the work of a professional translator (or translation student) rather than that of a professional terminologist.

This chapter does not intend to praise the usefulness of glossaries (both bilingual and multilingual) in assisting translators with their day-to-day work (particularly when working with technical content). However, in the case of an AI-generated glossary, it could be argued that its usefulness could be questioned if the aforementioned principles of AI Literacy were not applied. Once more, how could a trainee (or professional) translator be confident about the suitability of the terms offered by the GenAI tool? In case of doubt about such suitability, how could a translator check, select and assess if these important skills had not been honed during their formative years? How would a translator (or terminologist) be able to determine that the selected terms are, indeed, the most representative key terms to be found in the chosen text? How would a terminologist (or other language professionals) be able to conduct extensive good quality research in order to compare and contrast the AI-generated terms if they have not developed adequate research practices and techniques (and reflected on how undoubtedly essential these are) as part of their training?

6.5 Example 5: Machine translation post-editing

In this fifth and final example, I will shift the focus from terminology-related tasks to examine a different type of task, one that ChatGPT (and other similar tools) is also capable of performing but which remains relatively unknown. Many users very quickly discovered that GenAI chatbots based on LLMs could be used to correct sentences (Example 1), to translate text (Example 2), and to conduct many terminology-related tasks (Examples 3 and 4). This is of course not an exhaustive list of translation- or language-related tasks that can be performed with these tools and naturally many others are possible (such as summarising or simplifying a text, looking for synonyms or antonyms, checking for alternative

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translations of the same text, changing verb tenses, rephrasing fragments of text, and requesting singular/plural/more inclusive versions of the same sentence, just to name a few).

It is equally possible (yet again by using effective prompting) to develop a full AI-based MTPE workflow. In this case, given that this is a more complex task (it consists of creating a workflow rather than one specific action), three prompts need to be used. It is also recommended to use this workflow on a paragraph-per-paragraph basis, for better results. The suggested prompts are as follows:

- (10)
 - a. Prompt 1: You are a professional English into French translator specialised in renewable energy. Please translate the following text into French for publication on a government report.
 - b. Prompt 2: Please reformulate the translation you have just produced to ensure it's accurate and idiomatic in the target language.
 - c. Prompt 3: Please show a list of the changes you have introduced in the translated text.
- (11) ChatGPT showing some of the changes introduced in the reformulated translation after entering prompt 3 (complemented with an explanation and justification). (Created by Ramon Inglada)



ChatGPT

Certainly, here's a list of changes made in the reformulated translation:

1. **"provient de" instead of "est issue de"**: This change offers a more idiomatic and concise expression for "comes from" in the context of renewable energy.
2. **"régénérées" instead of "se reconstituent"**: The term "régénérées" is a more direct translation for "replenished" and is commonly used in the context of renewable resources.
3. **"Ces ressources comprennent" instead of "Les ressources renouvelables comprennent"**: This adjustment provides a smoother transition to list the renewable resources.
4. **"On utilise fréquemment" instead of "est souvent utilisée"**: This change maintains accuracy while introducing a more fluid and natural expression for "is often used."

By using this sequence of three simple (but effective) prompts, I can create what is, in essence, a basic MTPE workflow. However, it is Example 5 the one that serves as the most evident illustration of the need to apply the principles that underpin the concept of AI Literacy. GenAI chatbots can translate text (as demonstrated in Example 2), they can also reformulate (i.e., post-edit) the translated text in order to refine it (as demonstrated in Example 5), and they can even provide a

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list of the edits and reformulations that have been introduced (with explanations and justifications for those changes, also as demonstrated in Example 5). However, how can trainee and professional translators determine the suitability of not only the produced text, but also of the suggested ‘refinements’ and ‘reformulations’ without recourse to their core skills of linguistic knowledge, translation ability and cultural expertise? How can they assess the reformulations suggested by the tool and select the changes that fit their requirements, as dictated by the translation brief, if such practices have not been embedded in their training? And how can they further modify, combine and test the prompts they need to enhance this AI-based workflow (and potentially create others) without effective prompting skills?

7 Concluding remarks

The concept of AI Literacy has been developed, first and foremost, with simplicity in mind. At its most basic level, it can be considered as a simple framework based on three pillars. These are:

- Core skills (such as linguistic knowledge, translation ability and cultural expertise).
- Complementary skills (selection and assessment).
- A new skill (effective prompting).

At its heart, AI Literacy seeks to reinforce the key message that GenAI has many potentially useful applications in the field of translation. However, certain skills are essential to ensure the suitability of AI-generated content (as instructed by the requirements of the translation brief). This is a hopeless task if those skills are lacking and AI-generated content is used without any scrutiny. Translator training programmes should evolve and adapt to reflect advancements in the field of AI, while focusing on this set of essential skills. The overarching aim would be to foster a collaborative relationship between human expertise and machine efficiency.

Why does it seem to be necessary to disseminate this key message via the concept of AI Literacy? The main reason is the evident fact that the overall level of hype surrounding AI in the field of translation has been very high, particularly since the release of ChatGPT in November 2022. One of the main objectives linked to the application of the fundamental principles of AI Literacy is an

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attempt to overcome these high levels of hype. This objective is critical because it helps to ensure that the expectations of AI are realistic. It could be said that, in some cases, the hype surrounding AI has led to unrealistic expectations. Yes, GenAI can be very useful in the translation field and it has many potentially useful applications (as we have seen in this chapter and as many others have explained elsewhere). However, it is still essential for translation students to acquire core translation skills and for professional translators to continue developing and enhancing them throughout their careers. By applying the principles of AI Literacy, it is hoped that we can contribute to ensuring that expectations linked to the use of GenAI in translation are realistic and that the technology is used in a way that is beneficial to all stakeholders (including translators). It is equally important to note that AI is not a panacea for all translation ‘problems’, and it is essential to use it in conjunction with other technologies and approaches. GenAI should become another tool in the extensive translator’s toolbox, but it should not aim at replacing translators altogether.

It is also important to mention that the concept of AI Literacy and its key principles have been adapted with flexibility firmly in mind. They are not necessarily meant to be taken literally as they have been presented in this chapter (although it is hoped that this would indeed be possible), but rather as guiding principles open to interpretation and adaptation based on specific contexts and evolving requirements and perspectives. As [Long & Magerko \(2020\)](#) discuss, AI Literacy encompasses a range of competencies that can be tailored to different needs and contexts. Similarly, [Krüger \(2024\)](#) outlines an AI Literacy framework that emphasises the adaptability of several skills. The core skills (linguistic knowledge, translation ability, and cultural expertise), complementary skills (selection and assessment), and the new skill (prompting) that have been presented in this chapter as underpinning the concept of AI Literacy, have been included as examples. The number of skills is flexible and can be changed based on requirements. In addition, these skills can be tailored or modified to align with diverse needs. AI Literacy is not meant to be an unmovable or fixed entity, but rather a dynamic and evolving concept that adapts to changing circumstances, contexts, and priorities.

A point that warrants reiterated emphasis is that GenAI should not be viewed as a replacement for human translators. Instead, it should be considered as a tool that could be used to augment the work of translators. The concept of AI Literacy and its underlying principles emphasise the need to use core translation skills when using GenAI for translation tasks. These skills include not only linguistic knowledge but also the ability to understand the context, culture, and nuances of the language. Translators can add value by leveraging these skills to improve the

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quality and suitability of the output produced by GenAI. Moreover, the potential for AI to hallucinate should never be underestimated. [Waldo & Boussard \(2024\)](#) analysed the reasons why LLMs hallucinate and concluded that these models encountered challenges with topics for which limited data was available online, often generating inaccurate responses presented in a realistic format without acknowledging the inaccuracies. The responsibility to find issues and fix them, and to make the output produced by GenAI suitable for any set of specific needs at any given time, is routinely placed on translators. Without the skills to conform to the basis of AI Literacy, this would be an impossible endeavour.

Debates around the emergence, evolution and implementation of AI in our modern societies should be based on the overarching principles of ethics, sustainability and data privacy. Discussions around AI Literacy should not be an exception and, as the application of GenAI in translation continues to evolve, attention should be devoted to concerns about its ethical implications, environmental impact, and potential impact on the translation profession. The development and deployment of AI in the field of translation raise a wide range of ethical concerns, including those related to accuracy, bias, transparency, authorship, privacy and personal data. GenAI in translation has many potentially useful applications and this has been widely discussed in this chapter. However, its ethical, sustainability, and professional implications must be carefully considered and addressed to ensure that AI-based translation-related tasks and workflows are developed and used responsibly and sustainably. It is hoped that promoting the notion of AI Literacy (and integrating this concept in translator training curricula) will also contribute to better decision-making and to more mature debate (and scrutiny) among all stakeholders in the translation industry.

As mentioned earlier in this chapter, using GenAI chatbots is easy, but using their output critically requires thought. Ultimately, AI Literacy in the field of translation (and, more specifically, in the field of translation training) aims at equipping future translators –and other language professionals– with this fundamental ability for critical thinking.

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Part II

New paradigm: New knowledge

Chapter 4

Teaching translation students about data in the age of generative AI

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AI translation tools are now a key part of translation education, but many educators are searching for effective ways to teach the essentials of these tools to students with no background in computer science. This chapter explains why corpora make a good entry point to learning about AI translation tools, and it explores how science communication techniques such as framing, analogies and visualization can be used to help translation educators and students come to grips with data and machine learning.

1 Introduction

Though translator education often takes place in an arts or humanities faculty, technology has been a vital part of the translation profession, and hence of translator education programmes, for at least thirty years. Over this period, translators and translation students have shown remarkable resilience as they adapt to new tools and new technology-based ways of working. However, this does not mean that teaching technologies to translation students is easy, and educating the educators can be a particular challenge (Bowker 2023, Kenny 2020). As the pace of new tool releases gets faster, it can be difficult for translator educators to know where to begin. This chapter proposes that teaching translation students about data is one of the key building blocks in preparing them to use translation technology effectively. In the case of translation, data frequently takes the form of texts organised into corpora. This chapter therefore begins with a brief review of corpora and corpus-based tools, noting how AI translation tools have influenced

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the nature of corpora. Next, the chapter outlines why a science communication approach, rather than traditional scientific communication, could be useful in the context of translation technology education. Three science communication techniques – framing, analogies and visualisation – are combined with examples to demonstrate how science communication could be adapted to teach translation students about data-related topics in the age of AI.

2 Corpora and corpus-based translation tools

2.1 The changing characteristics of corpora in the age of AI

Over the past few decades, translators have seen the introduction and integration of a range of different tools, including concordancers (Zanettin 2023), term extractors (Korkontzelos & Ananiadou 2022), translation memory systems (Melby & Wright 2023), machine translation systems (Way 2020), and most recently, generative AI (GenAI) systems (Siu 2024). The various tools have become increasingly sophisticated with regard to their capabilities and their underlying architecture, but a common feature for all of the tools named above is that their core functionality revolves around processing data in the form of texts. Collections of texts are usually referred to as corpora, and these can take different forms depending on the nature of the texts and the way that they are organised (McEnergy 2022). Translation tools that process corpora are frequently described as corpus-based or data-driven tools (Wang et al. 2022).

As pointed out by Isabelle et al. (1993: 205), “existing translations contain more solutions to more translation problems than any other available resource”. Therefore, one type of corpus that has been used very often by translation tools is the bilingual parallel corpus (Simard 2020). In this type of corpus, a collection of source texts are aligned – usually at sentence level – with their counterpart target texts. In other words, each sentence in the source text is linked to its corresponding translation in the target text. Some translation tools might also use monolingual corpora of original texts in the source and/or target language to act as a linguistic model for that language.

For students who have already learned about concordancers, term extractors or translation memory systems, the notion of a corpus is already familiar because the corpus is a very visible resource in such tools. When using these tools, the translator or translation student often has a hand in creating the corpus or may need to upload the corpus that has been provided by the client or educator. Owing to their familiarity, corpora make a good starting point for learning about neural machine translation or GenAI tools.

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According to Bowker & Pearson (2002: 9),

A corpus can be described as a large collection of authentic texts that have been gathered in electronic form according to a specific set of criteria. There are four important characteristics to note here: ‘authentic’, ‘electronic’, ‘large’ and ‘specific criteria’.

While the general notion of a corpus remained relatively stable in the period before AI translation tools appeared, the introduction of these tools has ushered in some changes with regard to the features of corpora. Therefore, it is important for translation students to understand how AI tools have influenced and altered the nature of corpora.

2.1.1 Machine-readable form

The need for the corpus to be in electronic or machine-readable form has not changed. Indeed, we could say it is more important than ever since AI translation tools take on an even greater degree of corpus processing than do tools such as concordancers or translation memory systems. While concordancers and translation memory systems conduct pattern matching and then sort and display information for the tool users to interpret, AI translation tools go further by attempting to interpret the results and present fully formed translation solutions.

2.1.2 Size

Corpora are used to reveal linguistic patterns, which only become apparent when there are multiple examples of a given linguistic phenomenon. Therefore, another feature of corpora is that they are usually very large collections of text. However, our understanding of what constitutes “large” has evolved over time. The first generation of corpora created in the 1960s contained hundreds of thousands of words and were mainly consulted by linguists (McEnery 2022). These linguists used corpus analysis tools (e.g. concordancers) to help them sort and display the text data, but the linguists were still responsible for interpreting it. Today, the corpora used to power AI tools contain hundreds of billions of words (Hughes 2023). This is in large part because these AI tools do not understand text in the way that people do, and they need a much larger number of examples in order to predict patterns with confidence. However, this need for extremely large corpora is influencing other characteristics of corpora, such as the texts selected for inclusion (see 2.1.3) and sometimes even the authenticity of the texts (see 2.1.4).

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The availability of machine-readable texts can differ dramatically from one language to another, creating disparities with regard to an AI tool's performance in different languages. A high-resource situation occurs when it is relatively straightforward to gather a large number of high-quality resources for a given language or language pair. For instance, both English and French are widely used languages, and there is a lot of translation activity between these two languages. As a result, it is not too difficult to compile monolingual and bilingual parallel corpora for these languages, and they are thus referred to as high-resource languages. In contrast, a low-resource situation can occur when languages (or language varieties) are less widely used, such as some of the Indigenous languages of the Americas or the variety of French used in Canada. For languages or language varieties of limited diffusion, it can be more challenging to build a large corpus. Moreover, even if two languages have a large number of speakers (e.g. Russian and Hindi), there may not be a lot of translation activity between them, making it hard to create bilingual parallel corpora for this language pair. Therefore, languages, language varieties or language pairs for which there are few corpora available are described as being low resource.

2.1.3 Specific criteria

As emphasised by [McEnery \(2022\)](#), in order to be most useful, a corpus cannot consist of texts that have been gathered at random or in a purely opportunistic way. Rather, the texts in a corpus are selected because they correspond to specific criteria and are representative of a larger set of texts with those characteristics. One clear criterion in the context of translation is that the texts should be of high quality. Beyond this, there are many different options for designing a corpus depending on its intended purpose, but the key point here is that the choice of which texts to include is motivated. For example, in the context of a corpus to be used for translation, it could be important to select texts that are on a given topic, of a certain text type or register, or from a particular time period.

Evidence of the importance of corpus design can be seen in the way that translators construct translation memory databases, such as by creating different databases (or adding relevant metadata) for different domains or for different clients (e.g. to respect their preferred terminology or house style). In this way, they can restrict a search to texts that have specific features ([Melby & Wright 2023](#)). Likewise, neural machine translation tools are known to achieve better quality when the corpus is adapted for a specific domain ([Chu & Wang 2018](#)). However, as noted in section 2.1.2, AI translation tools need to have an enormous number of texts in the corpus. As a result, it can be challenging to achieve the

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necessary size while trying to be selective about the content. The low-resource situation described in section 2.1.2 can be further complicated when it comes to finding certain text types or texts on very specialised topics in less widely used or translated languages and language varieties. A consequence of not having a large enough high-quality corpus is that the AI tool does not have enough good examples to draw on, and so the tool's performance may be poorer in low-resource situations (Way 2025). If lower quality texts are included in the corpus in order to increase the size, then the tool may generate low-quality translations. The implications of insufficient quantity and quality of texts in training corpora used by AI translation tools is discussed in more detail in sections 4.2. and 4.3.

2.1.4 Authenticity

Finally, the need to have authentic texts in the corpus used to be sacrosanct. For instance, McEnery (2022) describes a corpus as “a large body of linguistic evidence composed of attested language use” (494) and “a collection of naturally occurring language data” (495). In the case of translation, there is a desire to have high-quality data, which means using texts that have been translated by language professionals. As observed by Kenny (2011: 2), the reason that the developers of translation tools use corpora of human translations to train their systems is because such corpora are assumed to contain good answers to translation problems; and they are assumed to contain good answers precisely because they contain translations performed by human beings.

One consequence of the need for bigger and bigger bilingual parallel corpora for training AI translation tools has been that such authentic high-quality human translated texts have become increasingly valuable commodities. This in turn has raised many ethical questions about ownership of translation data and permission to use it, prompting an explicit need to discuss these issues with translation students. Moorkens (2022) contains a detailed examination of such ethical issues, along with suggestions for how these can be integrated into translator education (see 4.1).

Another response to the potential shortage of text needed for constructing very large corpora for use with AI translation tools has been to set aside the long-established tradition of using authentic data and to explore the use of synthetic data. In the context of translation, synthetic data is created by using a machine translation tool to translate additional texts, and then adding these machine-translated texts to the corpus (Sennrich et al. 2016). The quality of synthetic data can vary, and while the need for additional data tends to be for low-resource

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languages, these are the languages for which machine translation already underperforms. Therefore, using a low-performance tool to generate more translations in that same language is likely to result in poor quality text. If this poor quality text is then used to further train the tool, there is a risk that using synthetic data could perpetuate a cycle of mediocrity.

This section has elaborated why understanding corpora is important for translation students, and has outlined some of the ways that corpora are changing in the age of AI. The following section moves beyond what students need to know about translation data (i.e., corpora) to consider how educators can communicate this information effectively.

3 From scientific communication to science communication

3.1 Scientific communication

Scientific communication (sometimes called internal science communication) is a type of expert-to-expert communication (Hanauska 2019). It occurs when one subject field specialist addresses another, with both parties having a deep knowledge of the complex matter in question. Academics are typically very comfortable with this type of communication since they spend much of their time discussing their research with their peers. Even as part of their teaching, they are normally communicating information directly from their own field, helping their students to acquire the necessary expertise to become specialists in their own right. However, the arrival of AI tools – more than any other technology – has changed that situation for many translation educators.

As noted in the introduction, technology is not new either to the translation profession or to translator education programmes. Yet as pointed out by Kenny (2020), when it comes to translation technology, identifying what to teach has often been easier than working out how to teach it. As the field of translation became more technologised, some educators admitted to feeling challenged by the demands of keeping up with technology (e.g. Kenny 2007, Marshman & Bowker 2012), but this responsibility lay primarily with those educators who had chosen to specialise in technology. Technology-related teaching was often restricted to a core course on translation technologies, where a large focus was on learning how to use the tools, which offered an array of sophisticated features (Bowker 2023).

AI translation tools are different in several respects. Firstly, the user interface is comparatively simple – often requiring the user to do little more than select

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the source and target languages and paste or type a text. In some cases, the AI tool may even be called automatically by another tool. Therefore, the “how-to” aspect of translation technology teaching is no longer a principal focus. A more important difference, however, is that the technology behind the interface is far from simple. As [Kenny \(2018\)](#) points out, older generations of technology were relatively transparent and comprehensible in their inner workings, while AI translation tools are more opaque in that the people using them – and even the people developing them – cannot always understand how the tool arrives at its proposals or output. In the field of AI more broadly, this opacity has led to a push for more explainable AI (XAI) ([Ridley 2025](#)), while machine translation researchers are also beginning to work on explainable aspects of this technology (e.g. [Lankford et al. 2023](#)). In the meantime, AI translation tools are now starting to appear in courses across the translation education curriculum rather than solely in core technology courses, and translator educators are seeking ways to help themselves and their students understand how these tools can affect translation processes and products, and how to use the tools responsibly.

In these circumstances, scientific communication is not feasible since translation educators and students are not computer scientists. The technology behind neural machine translation and GenAI tools is highly sophisticated. The tools may appear to be simple because the user interface is simple, but behind the scenes, there are very complex artificial neural networks, word embeddings, vectors, transformers, and more. It is not necessary for most translation educators or students to understand all the details of how an artificial neural network operates in order to use AI-based translation tools.¹ However, understanding the role of data (i.e., corpora) in a data-driven AI translation tool is valuable and can help translation students to be more informed and responsible users of this type of technology. At the very least, understanding the role of data is a useful first step, and the approach proposed in section 4 is intended as way to introduce data on programmes aimed at educating students who will work primarily as language

¹This is currently true for many translation-oriented jobs and translation education programmes. However, as [Briva-Iglesias & O’Brien \(2022\)](#) point out, new jobs are emerging that straddle the boundary between translator and technologist and that require a deeper grounding in technologies. To address this changing market, new education programmes are also emerging that place more emphasis on technological skills, such as the MSc in Translation Technology at Dublin City University in Ireland, and the MSc in Multilingual Digital Communication at McGill University in Canada. In these more technology-oriented programmes, a deeper understanding of the underlying architectures is both required of the educators and offered to the students, and data literacy training may also be deeper and more technical, such as the approach described by [Hackenbuchner & Krüger \(2023\)](#) as part of the DataLit^{MT} project.

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professionals rather than as technology specialists. But if scientific communication is not a feasible approach, can science communication help?

3.2 Science communication

When translation students hear people talking about AI-based translation tools using terms such as transformers or word embeddings, they may feel intimidated and lack confidence in their ability to use these tools effectively. Therefore, when introducing these tools, translation educators may find it helpful to draw on science communication techniques. As pointed out by Burns et al. (2003), science communication is tricky to define precisely, and there are myriad competing and overlapping descriptions of this concept in the literature. However, there is general agreement that the essential goal is to make expert knowledge accessible to non-experts. This is why this activity is sometimes described more specifically as external science communication (Hanauska 2019). In the case of translation students, while they are developing expertise in translation, they are not experts in AI or machine learning.

Academics in all disciplines are increasingly encouraged to engage in science communication to share the results of their research with a wider audience, which could include policy makers, participants in research studies, funding agencies and even the general public (e.g. National Academies of Sciences, Engineering, and Medicine 2017). Yet while science communication is strongly encouraged, many academics receive little training in this area, even though it is a complex and skilled task (Borowiec 2023). Fortunately, tips, guides and recommendations are emerging to help fill this gap (e.g. Borowiec 2023, Cooke et al. 2017, Henville 2020). From these, we can glean several suggestions that can be integrated into teaching to help explain data-related concepts to translation students as a foundation for learning about AI translation tools.

Frame your message: Frame your message in terms that are accessible, relatable and meaningful for your audience. Framing is not about marketing your point of view but about finding a way to actively engage the audience with an issue by showing them why they should care about it.

Make the abstract concrete and connect unfamiliar concepts to familiar ones: Use metaphors or analogies that connect at the level of underlying commonalities to help the audience understand an unfamiliar or abstract concept in relation to a familiar and concrete one.

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Visualise the content: Use visual aids (e.g. images, concept maps, graphs) to support your message. Making the information available in different modes or formats can help to crystallise the ideas in the mind of your audience.

The following section considers how these three science communication techniques could be implemented to help translation educators teach concepts related to data.

4 Applying science communication techniques to teach translation students about data

4.1 Framing the message about data

It was noted in section 2 that corpora can be a good entry point for introducing data-driven AI tools to translation students because they have likely been exposed to corpora when using other tools (e.g. concordancers, term extractors, translation memory systems). Corpora are less immediately visible in the more automated AI translation tools, but this makes it all the more important to actively discuss them. As [Kenny \(2011\)](#) emphasises, corpus-based tools cannot function without corpora. If no corpus is present, the tool itself has nothing to offer as a translation aid. This is a very important point to share with translation students because it demonstrates the essential contribution of translators. Of course the tool developers play a key role in both programming the tools and determining how corpus data can be processed effectively, but the data itself – the translated text – is indispensable, and this fact is often glossed over when the tools are described or promoted. Yet as part of translator education, it is important to make students aware of the value of corpora because this helps them to understand their own worth as language professionals in addition to understanding how corpus-based or data-driven tools work, and where the potential pitfalls might lie.

As mentioned in section 2.1.4, translation corpora have become valuable resources, leading to some questionable practices. Framing the discussion of AI translation tools as part of a broader issue of professional practice, ethics, and digital citizenship, rather than simply as an instrumental means of making their job easier, can help translation students to recognise not only the value of their work, but also the fact that technology does not exist in a vacuum. Discussions about the role of data in data-driven tools can encourage translation students to reflect on how AI translation technology can impact tool users and end users of translated products in various ways. It also empowers language professionals to

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be able to participate in broader conversations about AI technology development and use in other areas of life.

4.2 Connecting unfamiliar concepts to familiar ones

The concept of data is relatively abstract. For instance, some popular dictionaries define data in the following ways:

- **information** in an **electronic form** that can be **stored** and **processed** by a **computer** (Cambridge Dictionary)
- information in digital form that can be transmitted or processed (Merriam-Webster Dictionary)

Of course we can make this abstract concept more concrete in the context of translation by explaining that the data in question is text, and usually a series of source texts aligned with their corresponding translations (i.e., a bilingual parallel corpus) (see 2.1). But can we find an even clearer analogy to help students understand the nature of the relationship between the data (corpus) and AI-based translation tools?

In discussions about AI more broadly, data is frequently compared to oil or water. However, as pointed out by [Kenny \(2011\)](#), this is problematic because it suggests that data is a naturally occurring resource that can be exploited (e.g. through harvesting or mining), whereas in reality, translation data has been created by skilled translators who deserve to be recognised for their contribution. As part of a discussion about why the metaphor of natural resources is not appropriate for describing data, translation educators and students can explore where translation data comes from, and what might happen to it after a translator delivers it to a client. For such conversations, translation educators can be inspired by [Moorkens \(2022\)](#), who undertakes an in-depth examination of the concepts of translation data ownership, permission to use translation data, and related questions concerning the ethical use of AI translation tools.

While the metaphor of translation data as a naturally occurring resource is clearly problematic, an analogy such as fuel for a vehicle could be useful for helping translation students to understand the relationship between the translated texts and the AI translation tool. Most car drivers do not have a detailed understanding of how their car's engine works. However, they do understand that it is necessary to keep fuel in the tank, and they know that it is important to select the right type of fuel when they go to the filling station. Just as a vehicle needs fuel in order to function, so too does a data-driven AI tool. Moreover,

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the concept of fuel can be used to foster discussions about both the quantity and the quality of data needed for the AI tool to perform well. For instance, a car's performance could go down when the fuel tank is very low (i.e., a low-resource situation), but it can go up when more fuel is added (i.e., a high-resource situation) (see 2.1.2). And in addition to needing enough fuel, a car will perform best when it has the right type and quality of fuel. Using the wrong type of fuel, or lower quality of fuel, will negatively affect the car's performance.

4.3 Starting with image data

4.3.1 Machine learning

The science communication tips also recommend incorporating visuals as a means of supporting explanations. In the case of AI tools, it might be helpful for educators to use image-based data as a way to explain machine learning before moving on to translation data.

Machine learning is an AI technique that is used by translation tools, chatbots, predictive text, image generators, and more (Flach 2012). It is a subfield of AI that enables computers to learn things that have not been explicitly programmed. Machine learning begins with data. The type of data needed depends on the task that the AI tool is being asked to do. Machine learning has four main steps:

- Identify a task for the AI tool.
- Show the tool examples of what you want it to learn (i.e., the training corpus).
- Give the tool some feedback (e.g. confirm the correct answers).
- Test the tool on new data that it has not seen before.

The following sections contain some examples that can be shared with translation students to help them understand how AI tools interact with data as part of machine learning.

4.3.2 Example 1: The lone banana problem

In this example, the task for the AI tool is to generate images – a concept that is already likely to be familiar to students since some of the earliest best-known GenAI tools were for image generation (e.g. DALL-E, Midjourney). For the purpose of this example, the task can be narrowed down even more to generating images of bananas.

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To see examples of images of bananas, translation educators and students can navigate to open access image sites such as Pixabay.com or Unsplash.com and use the search term “bananas”. To perform well, an AI tool would need to be provided with many, many examples of pictures of bananas in order to be able to generate a new image based on these examples. People typically need fewer examples from which to learn.

A fun exercise to do in class is to ask the students to look at a dozen pictures of bananas on an open access site, and then to imagine that they are an AI image generation tool that has been asked to generate (i.e., draw) their own image in response to the following prompt: “Draw a banana.”

This will not likely pose too much of a challenge for the students – even those who are not artistically inclined! As a follow-up, share with students the results of the Daniel Hook (2023) experiments in which he asked the GenAI tool Midjourney to generate images of a banana. What Hook found was that the AI tool only generated images with two or more bananas, rather than a single banana. Hook refined the prompt several times asking for “a banana”, “a single banana”, “one banana”, and so on, but the tool continued to generate images containing multiple bananas.

This example, which Hook (2023) dubs “The lone banana problem”, can be used as a simple yet effective way of explaining the issue of data bias. If the training data does not contain a sufficient number of images of individual bananas, then the AI tool will not be able to generate an image of a single banana. Of course, the problem can be fixed by adding more data (i.e., images of lone bananas), but as Hook warns, we do not always realise what gaps are there. Hook (2023) also stresses that it is important to recognise that AI tools do not understand the world the way that people do: they only understand commonly occurring patterns.

While humans are amazing pattern matchers, that skill is augmented by common sense (in many but not all cases), context and an evolved and subtle understanding of the physical world around us. AIs don’t yet have those augmentations – they are pure pattern matching power. And hence, they are only as good as the data that we input into the training set and hence can be no more than the statistical average of those inputs. In the lone banana problem, the statistics suggested that bananas only appear in twos (or more) and so the AI could not imagine a single banana. (Hook 2023, n.p.)

4.3.3 Example 2: The snow detector

As a follow-up to the lone banana problem, it may be useful to share the example of an AI tool whose task is image classification. This example comes from Ribeiro

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et al. (2016), who trained an AI tool using two different sets of images – ten images of husky dogs and ten images of wolves. After the training session, the researchers tested the AI tool's ability to classify images correctly by showing it ten new images (five huskies and five wolves) that had not been part of the initial training data. The AI tool classified most of the images correctly, but it made two errors. In investigating the cause of the errors, the researchers found that the AI tool was making decisions not based on the animals in the pictures but based on the background. In the training data, all the pictures of the wolves contained snow, but none of the pictures of huskies contained snow. Therefore, in the test phase, if the image showed an animal against a snowy background, the AI tool classified the image as a wolf, but if there was no snow, then the AI tool classified the image as a husky. In other words, the AI tool was looking for patterns, but the pattern that it used as a basis for its decision-making was not a sensible one.

The snow detector is a funny example, but the medical literature contains more serious examples of similar phenomena. In Narla et al. (2018), researchers tried to train an AI tool to distinguish between images of cancerous and non-cancerous skin lesions. However, in the training corpus, the majority of images of cancerous lesions also contained a ruler to measure the size of the lesion. During the testing phase, it became clear that the AI tool was equating the absence of a ruler with the absence of cancer. In another example, an AI system was trained to distinguish lung X-rays of people with pneumonia from lung X-rays of people with COVID-19 (Roberts et al. 2021). Unfortunately, the training data containing examples of pneumonia consisted mainly of pediatric patients, while the training data with examples of COVID-19 was from mainly adult patients. As a result, the AI tool learned to distinguish between children and adults rather than between pneumonia and COVID-19.

4.3.4 Connecting the dots: From images to translation

Though humorous, the lone banana and snow detector examples are intended to drive home the very serious point that AI tools do not think, understand or even process information the way that people do. By focusing on relatively straightforward, concrete and visual tasks such as image generation and image classification, these examples are easy for students to relate to, and they can then be connected to translation-related issues. For example, the lone banana problem can lead into a discussion of data bias and of what happens when a translation corpus has gaps in the training data. For instance, Vanmassenhove et al. (2018) have demonstrated how data bias can lead to issues of gender bias in translation,

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while Bowker & Blain (2022) point out how a lack of data in one language variety can lead to content from another variety being used erroneously in a translation.

In addition to reinforcing the lesson about data bias, the snow detector example, along with the more serious examples of the ruler detector and the child detector, can be used to promote discussions on opacity and what can go wrong when it is not clear what patterns the AI tool is using as the basis for decisions. These examples can also feed discussions about risk assessment – an important element of any decision about whether or not to use an AI translation tool (Koponen & Nurminen 2024). In some cases, the translation task at hand might be relatively low-stakes (e.g. translating a text for the purposes of entertainment), and so using an AI tool could be a good option. However, in other circumstances, the translation task might be one with higher stakes, where the consequences of a poor translation could negatively affect someone’s health or life (Way 2013). Another type of information that can be important as part of risk assessment is understanding whether the situation is more likely to be a high-resource or a low-resource situation, given that the performance of the AI tool is likely to be better for high-resource languages, language varieties, language pairs and domains, and poorer in low-resource situations (see 2.1.2).

5 Conclusion

AI translation tools are rapidly inserting themselves into the translation industry and must also be addressed in translator education. The technology behind these tools is very sophisticated and may be intimidating for translation educators and students alike, given that most do not have a background in computer science. However, most language professionals do not need to understand all the details of how AI translation tools work in order to use them in an informed and responsible way. In contrast, it is extremely useful for language professionals to understand how these tools interact with data, and how the data used for machine learning can affect tool performance and translation quality. In the context of AI translation tools, the data is typically a corpus. Translation students are already familiar with corpora, which are also used with other types of translation tools (e.g. concordancers, term extractors, translation memory systems). However, students need to learn how AI tools are impacting the characteristics of corpora.

Rather than approaching the teaching of AI translation tools from the more traditional perspective of scientific or expert-to-expert communication, translation educators may find that science communication techniques (e.g. framing,

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analogies, visualisation) are more useful for teaching students about data and machine learning.

In the spirit of transversal learning, translation educators may even find that developing effective techniques to teach their translation students about data and its role in AI translation tools can pay off in other ways. As Borowiec (2023: 1) laments,

Despite there being many excellent reasons for scientists to engage in science communication, they often lack the tools to do so. [...] Scientists frequently cite lack of training and/or confidence in their science communication skills as a barrier to their participation in public-facing activities.

Since many translator educators are also researchers, they may be able to apply these science communication techniques to their research-oriented activities. Moreover, Ehrensberger-Dow et al. (2023) suggest that translator trainers are well placed to take on a role such as machine translation literacy consultant, which could involve helping clients (who are unlikely to be computer scientists) to understand how AI tools process data or how data resources can be curated to improve the tools' performance.

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Chapter 5

Teaching translation with AI: Bridging theory and practice through prompt engineering

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This chapter explores the innovative application of large language models (LLMs) in translator training, focusing on the use of few-shot prompts and chain-of-thought prompting. It proposes a novel approach that integrates metalanguages and concepts of Translation Studies into prompt engineering, moving beyond traditional natural language processing goals of improving machine translation quality. The chapter demonstrates how this method can create interactive and engaging learning experiences for translation students, allowing them to explore various translation strategies and develop critical thinking skills. Through concrete examples, the chapter illustrates the potential of LLMs to generate diverse translation variations and provide insightful analyses of translation processes. While acknowledging limitations and the need for critical evaluation, the research emphasises the positive and proactive possibilities of LLMs in translator training. This approach not only bridges the gap between translation theory and practice but also opens new avenues for autonomous learning and the development of essential skills for future translators in the AI era.

1 Introduction

Recent advancements in artificial intelligence (AI), particularly the emergence of Large Language Models (LLMs), have generated extensive debate regarding their benefits and drawbacks. In response, the UK's Institute of Translation and Interpreting (ITI) has articulated the Slow Translation Manifesto (ITI 2024). Drawing

Masaru Yamada. 2025. Teaching translation with AI: Bridging theory and practice through prompt engineering. In JC Penet, Joss Moorkens & Masaru Yamada (eds.), *Teaching translation in the age of generative AI: New paradigm, new learning?*, 87–104. Berlin: Language Science Press. DOI: ??

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on an analogy between fast food and AI-generated translations, this manifesto highlights the potential societal and professional harms of prioritising speed over quality in translation practices. It advocates for a renewed appreciation of slow, human-led translation, emphasising its artistry, ethical rigour, and cross-cultural competence—qualities that are often sacrificed in rapid, machine-driven translation processes.

The Société française des traducteurs (SFT), the French translators' association, has also raised concerns about the inadequate quality of AI-based translations and the transparency of the resources used for machine learning in LLMs. The SFT warns that unfettered access to AI tools, such as ChatGPT, could lead to a decline in respect for language professionals or result in an “unchecked rush of translations that we are asked to enhance” (Slator 2024). Similar statements regarding the potential crisis posed by AI technology in translation have been issued by other organisations, including CEATL (European Council of Literary Translators' Associations 2024), ATA (American Translators Association 2024), and JAT (Japan Association of Translators 2024).

The EU's Language in the Human-Machine Era (LITHME) project, focusing on language in the technology era, aims to “prepare language researchers for what is coming” and “facilitate longer-term dialogue between linguists and technology developers” (LITHME Project 2021; see also Sayers et al. 2021). This comprehensive project compiles expert opinions and views, aiming to counteract the potential overreliance on AI-based technology and the market-driven overselling of AI (Sayers et al. 2021, Way 2025). By promoting such perspectives, the project plays a crucial role in encouraging a more cautious and thoughtful approach towards AI advancements. These investigations and statements from this project serve as a reminder to pause and critically reflect on the rapidly accelerating trends in our society.

However, upon closer examination of these statements and surveys, it becomes apparent that the understanding and predictions about technology are not always accurate or up-to-date, and may not be entirely evidence-based. For instance, while LITHME's 2021 report (Sayers et al. 2021) does touch on concepts relevant to LLMs and Generative AI (GenAI), these references are minimal and do not encompass the transformative developments that have since occurred, such as chat-based LLMs. This highlights the challenge of keeping pace with the rapid evolution of technology, which even at that time was difficult to fully anticipate.

Similarly, the SFT states that the output of machine translation (MT) remains unreadable in its raw state and requires human correction through post-editing. However, it also adds that “70 percent of our member translators who responded to our survey considered PE (and by extension AI) a threat to their profession”

(Slator 2024). The conflicting statement leaves unclear which era or type of MT it refers to, and the apparent emotional reaction reflecting fears of job loss suggests a lack of clarity about the intended scope and technological context of the claim. This also implies that such statements may not fully reflect the current state of cutting-edge technology.

In the context of translator training, which is the primary focus of this book, there may not be as strong a backlash against AI as in parts of the industry. For instance, while some industry stakeholders see AI as a tool to enhance efficiency and provide opportunities for post-editing work, many independent professional translators express significant concerns. Fears about the indiscriminate use of AI, especially by untrained users, persist due to its potential to cause quality issues and undermine the value and compensation for human translators. According to ELIS (2025), around 64% of students in translation programmes report using GenAI at least occasionally, with 19% using it regularly. University staff estimate MT use at 63%, compared to 58% among students. These figures mark a steep rise from the limited adoption reported in 2024 and highlight the contrast between the comparatively cautious professional industry and the rapid integration of GenAI in translator training.

This chapter proposes leveraging the accumulated knowledge assets of professional translators and translation researchers to actively interact with LLMs through basic prompts such as Chain of Thought (CoT) prompting and few-shot prompts. For example, translation memories, which are repositories of past human translations paired with their source texts, can be effectively used for translator training in conjunction with few-shot prompts. Additionally, the concepts of translation briefs and work instructions, traditionally used in professional translation, can be directly applied as prompts. Furthermore, the classical assets of translation research that describe translation strategies can be adapted for use as CoT prompting.

The concepts studied in translation research, which explain the act of translation, can be collectively referred to as the metalanguages of translation (Miyata et al. 2023). These metalanguages can be effectively applied to prompt creation. These attempts are not merely an engineering perspective to enhance the accuracy of translation outputs from LLMs but are proposed as significant considerations for translator training in the AI era.

2 Literature review

Previous studies on prompts for translation and LLMs have primarily been published in the field of natural language processing (NLP). Most research has fo-

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cused on using LLMs as replacements for existing MT systems or as tools to address the shortcomings of MT. Specifically, these studies often investigate how translation accuracy compares between LLM-generated translations and traditional MT when using simple prompts, such as zero-shot prompts that merely instruct the model to translate from one language to another. For example, several studies (such as [Hendy et al. 2023](#), [Jiao et al. 2023](#), [Wang et al. 2023](#)) have examined how accurately LLMs can translate using zero-shot prompts and found that, for high-resource languages—languages with abundant digital resources and training data such as English, Spanish, or Chinese—LLMs can produce translations with accuracy comparable to or exceeding that of traditional MT ([Way 2025](#)).

Other studies have explored using LLMs and prompts to enhance translation quality, particularly for low-resource languages. For instance, [Jiao et al. \(2023\)](#) analysed ChatGPT’s MT capabilities and found that while it competes with commercial systems for high-resource European languages, it struggles with low-resource and distant languages. [Gao et al. \(2023\)](#) developed a new method for translation prompts that includes task information, contextual domain information, and part-of-speech tags, which significantly improved ChatGPT’s performance, surpassing commercial systems in multiple translation directions. [Zhang et al. \(2023\)](#) provided a comprehensive summary of prompt strategies used to date and attempted to overcome the shortcomings in translation for low-resource language combinations and other challenging scenarios.

Recent experiments have also focused on the translation of high-context and multi-modal materials such as Japanese manga using LLMs. [Yang et al. \(2024\)](#) conducted experiments demonstrating that applying appropriate prompts can significantly improve translation accuracy in such contexts. However, practical translation organisations have raised concerns about the feasibility of applying these methods to manga, particularly in capturing its nuanced cultural and contextual layers, which are integral to the genre ([Japan Association of Translators 2024](#)).

While these studies originate from the NLP field and tend to focus on engineering improvements, there appears to be a difference in the understanding of translation between NLP researchers and Translation Studies (TS) scholars. In NLP, translation is often approached—particularly in evaluation—as if there were a single correct outcome. At the system level, however, MT models may generate multiple possible translations and then output the one judged to be the “most likely,” which does not necessarily align with what a human expert would consider the “best.” By contrast, TS generally recognises that translation may vary according to context and purpose. This variability makes it difficult

to define what constitutes a “good” or “quality” translation. For instance, classical translation theories such as Skopos theory suggest that translation strategies (e.g. domestication vs. foreignisation (Venuti 1995), covert vs. overt translation (House 1981)) might depend on the purpose of the translation. Over time, TS has developed insights into these variations.

The following sections attempt to address the gap left by NLP researchers by incorporating the metalanguages of TS, namely the concepts of translation, into LLM prompts and examining the resulting changes in translation output.

Yamada (2023), for instance, investigated the impact of incorporating translation purposes and target audiences into prompts on ChatGPT’s translation output. This study focused on the pre-production phase of the translation process, drawing on previous translation research, industry practices, and ISO standards. By including concepts and terms from TS as prompts, the research explored the potential for achieving flexible translations that traditional MT systems have struggled to produce. The study evaluated changes in translation output using subjective qualitative assessments and cosine similarity, incorporating concepts such as dynamic equivalence (Nida & Taber 1969/2003).

Further, He (2024) explored using translation research concepts to design prompts for LLMs to improve translation quality. The study discussed the effectiveness of incorporating conceptual tools and personas of translators and authors into ChatGPT’s translation task prompts. Although the small-scale experiments indicated limited effectiveness in improving translation quality, the paper highlighted the need for further research on the impact of TS concepts on LLMs translation tasks.

Building on the ideas from these two papers, this chapter presents the study finding evaluating the effectiveness and potential of incorporating TS concepts as prompts for translator training, rather than assessing improvements in LLM-generated translation quality. It further discusses the implications of this approach for translation education.

3 Aims and pedagogical potential of this chapter

As observed in the comparative analysis of previous research, prompts in NLP often tend to prioritise the engineering aspect of achieving equivalence between source and target texts, typically measured by standardised translation metrics such as BLEU (Papineni et al. 2002) and COMET (Rei et al. 2020). This approach generally focuses on how closely the translation aligns with the source text. In contrast, TS often emphasises the importance of how the translation is received by the target audience, considering the context and situational variables.

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This philosophical challenge in evaluating translation quality involves numerous variables and may not be solely based on source-target text equivalence. By incorporating the concepts and metalanguages of TS, it may be possible to create prompts for LLMs that better address these complex considerations.

This chapter aims to serve as a starting point for designing pedagogical activities that leverage concepts and terminologies (metalanguages) from TS to create effective and unique prompts for LLMs. While not attempting to exhaustively evaluate all metalanguages as LLM prompts for educational use, the chapter presents several specific examples to inspire translation educators and instructors in their practice. By doing so, it highlights the potential of integrating TS knowledge into LLM-assisted translation education.

To achieve this, the chapter first provides an overview of the foundational concepts of prompt engineering for LLMs, specifically few-shot prompts and CoT prompting. It then explores how these concepts might be integrated with existing TS knowledge. Through these examples and discussions, the chapter aims to offer practical suggestions and potential directions for incorporating TS concepts into LLM-assisted translation education, rather than presenting definitive solutions.

4 Few-Shot prompts and CoT prompting

The concepts of few-shot prompts and CoT prompting are explained in the Prompt Engineering Guidelines.¹ Applying these concepts in this chapter requires some modifications and expanded interpretations, with concrete examples provided in Sections 5 and 6.

Few-shot prompting is a prompt design method for LLMs that includes a few examples of input-output pairs along with task instructions. This approach allows the model to infer the task's intent and output format from the provided examples, potentially leading to more accurate and desirable outputs. Few-shot prompting is particularly effective for complex tasks or when zero-shot prompting (instructions without examples) may be insufficient.

Consider a scenario similar to a translation memory where portions of source and target texts are provided to the LLM. The model may learn implicit patterns from these examples and apply these learned patterns to generate translations. This method parallels actual practices in the translation industry, where translation service providers often provide translation memories to human translators. Translators typically learn from these past translations, observing how style guide rules are concretely applied. Similarly, translation coordinators or project

¹URL: <https://www.promptingguide.ai/>

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managers provide instructions to human translators. This chapter aims to investigate how comparable instructions, when used as few-shot prompts for LLMs, might affect translation output.

CoT prompting involves verbalising the thought process as a prompt. Just as humans follow a step-by-step reasoning process to solve problems, CoT prompting encourages LLMs to explicitly follow a CoT. This leads to more accurate and interpretable responses. In the context of translation, few-shot prompts provide examples of source and target texts without explaining the intermediate process. In contrast, CoT prompting could describe the translation process, referencing translation process research literature. For instance, a translator might read the source text, produce a literal translation, consider the context, monitor the translation, and then revise it to make sense for the target audience, culture, and readers (e.g., monitor model (Tirkkonen-Condit 2005)). This step-by-step description of the translation process can be given as a CoT prompting.

Additionally, as suggested by Yamada (2023), CoT prompting can be interpreted as detailed translation briefs. He (2024) demonstrated that setting translator personas (profiles) can also be considered a variant of CoT prompting. In professional translation settings, companies typically select suitable translators for specific tasks and provide detailed translation briefs. This chapter interprets CoT prompting as analogous to the language used for describing translation processes (Miyata et al. 2023) and investigates how such prompts might influence translation output, as well as their potential educational implications.

5 Example of a few-shot prompt

Firstly, we present an example of a few-shot prompt. This example aims to examine how the output changes when a portion of a translation memory is input into the LLM. However, from an educational perspective, it is valuable to consider how learners, when faced with their own translation tasks, can study and emulate various linguistic aspects from the given corpus and incorporate them into their subsequent translations.

In this case, we deliberately focused on linguistic tone and manner. A highly distinctive translation corpus was prepared. As indicated in the prompt below, the original English text was contextualised as a monologue reflection of an elderly person on their life. In the corresponding Japanese translation, we used expressions characterised by sentence endings typical of the linguistic style associated with Japanese elderly individuals, creating an exaggerated translation that exemplifies their manner of speaking. Although this example is quite ex-

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treme and may lack realism, it can be likened to the specific style and voice used in a particular company's user manuals in a real-world context.

The purpose of this approach is twofold: to observe how the LLM adapts its output based on the provided translation memory, and to encourage learners to critically analyse how they can learn from and apply specific linguistic features, tones, and styles in their own translation work. This exercise, while using an exaggerated example, aims to heighten awareness of the importance of adapting language to specific contexts and audiences in translation practice.

This prepared sample set was then given as a few-shot prompt to the LLM, instructing it to translate a new source text following the corpus. The LLM used was Claude 3.5 Sonnet. To provide a comparison, the new source text was also translated using the MT service DeepL. Subsequently, the LLM translations and the MT translations were compared against the original corpus. Additionally, evaluations were conducted using the automatic evaluation tool COMET². The results of these comparisons and evaluations are presented below.

LLM Prompt 5.1:

Below are provided the English text [source text], in which an elderly individual living in contemporary America reflects on their life, and its Japanese translation [target text]. Based on this corpus, please translate the [new source text] into Japanese.

[source text] "When I was young, life was so different. We didn't have all these gadgets and technology. We had to work hard for everything we had. But, those were good times too. I remember walking miles to school, playing outside until dusk, and everyone knew each other in the community."

[target text] 「若い頃はもう、生活はまったく違ってたんじゃ。今みたいなガジェットやテクノロジーはなかったんじゃ。何でも手に入れるために一生懸命働かなければならなかったんじゃ。でも、それもまた良い時代だったんじゃよ。学校まで何マイルも歩いて、夕暮れまで外で遊んで、みんながコミュニティの中でお互いを知ってたんじゃ。」

²We used WMT20-COMET for this evaluation.

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[new source text] “I miss those simple days. We didn’t have much, but we were happy. I spent my summers helping my father on the farm and learning about life from him. Sundays were always special, with family gatherings and big dinners. As I grew older, the world changed rapidly, and sometimes it’s hard to keep up. But I cherish those memories, they keep me grounded.”

Source	Reference	Claude	Comet score	DeepL	Comet score
I miss those simple days.	懐かしいのう、あのシンプルな日々が。	あの素朴な日々が懐かしいのう。	0.996	あのシンプルな日々が懐かしい。	1.052
We didn't have much, but we were happy.	あまり多くはなかったけれど、幸せだったんじや。	あんまり持っとらんかったが、幸せじゃったよ。	0.361	何もなかったけれど、幸せでした。	0.663
I spent my summers helping my father on the farm and learning about life from him.	夏には父の農場を手伝って、いろんなことを学んだもんじや。	夏はいつも父ちゃんの農場を手伝って、父ちゃんから人生について学んだもんじや。	0.578	夏は父の農場を手伝い、父から人生について学びました。	0.693
Sundays were always special, with family gatherings and big dinners.	日曜日はいつも特別で、家族が集まって大きな夕食を楽しんだんじや。	日曜日はいつも特別じゃった。家族が集まって、たくさん食べて。	0.557	日曜日はいつも特別で、家族が集まり、大きな夕食がありました。	0.691
As I grew older, the world changed rapidly, and sometimes it's hard to keep up.	年を取るにつれて、世の中は急速に変わっていったが、時々ついていくのが大変じや。	年を取るにつれて、世の中はどんどん変わってしもうて、ついていくのが大変なこともあるのう。	0.718	大人になるにつれ、世界は急速に変化し、ついていけないこともありました。	0.633
But I cherish those memories, they keep me grounded.	でも、あの思い出があるからこそ、今のわしがあるんじや。	でもわしは、あの頃の思い出を大切にしよう。そいつがわしの心の支えじゃよ。	-0.213	でも、私はその思い出を大切にしています。	0.022
Document average score			0.499		0.626

Figure 1: Claude vs. DeepL.

According to the COMET scores in Figure 1, Claude’s scores range from 0.360 to 0.995, while DeepL’s scores range from 0.633 to 1.051. These scores indicate generally high-quality translations, as scores closer to 1.0 typically reflect strong alignment with the reference text. On average, DeepL achieves higher scores at 0.626 compared to Claude at 0.499.

However, when evaluated by a human translator, a stark stylistic difference becomes evident between the two. Claude Translation skilfully captures the distinctive elderly speech patterns found in the reference translations. Using a small corpus, it creates the impression that the same elderly person is continuing the conversation seamlessly. In contrast, DeepL Translation employs a completely standard Japanese tone, making it appear as though a different person is speaking, thereby disrupting the flow of the monologue. This discrepancy highlights a limitation of the COMET scoring system: it fails to account for stylistic features or cultural nuances such as those present in Japanese “elderly speech patterns.”

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In an educational context, it is crucial for human translators to first anticipate the linguistic features that can be learned from the corpus provided in the prompt, and then compare these expectations with the LLM's response. Additionally, comparing with MT allows reflection on how these predictions differ from actual outputs. When COMET scores are available, it may be beneficial to critically analyse whether these scores accurately reflect our predictions. As mentioned above, in this case, it appeared that COMET did not adequately reflect the situation, so we decided to question an LLM (in this instance, ChatGPT) about why this might be the case. The response we received is as follows:

LLM Response 5.1:

Your observation about the tone and manner of speech in translations highlights an important aspect of translation quality that COMET might not fully capture. Here's why:

1. **Literal Accuracy vs. Stylistic Fidelity:** COMET is primarily designed to evaluate the literal accuracy and fluency of translations. It does well in judging how close the MT is to the reference in terms of meaning and linguistic quality.

2. **Stylistic Nuance:** While COMET can capture some aspects of fluency and naturalness, it might not fully appreciate stylistic nuances such as the specific tone, voice, or manner of speech that are present in the reference translation. For example, an elderly man's distinctive speech patterns might not be fully valued in COMET's scoring if the reference doesn't explicitly indicate the importance of these characteristics.

3. **Contextual and Cultural Sensitivity:** Human evaluators can recognise and appreciate subtleties such as tone, cultural references, and stylistic consistency. Automated metrics like COMET are still limited in their ability to assess these elements comprehensively.

(It should be noted that such categories are analytical constructs used by human evaluators; the LLM itself does not "understand" or deliberately apply them.)

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This type of activity can provide a novel translator training experience. First, learners understand the relationship between few-shot prompts and parallel corpora, learning that LLMs can generate similar translations from them. While the above example focuses on tone and manner, it could also be interesting to investigate whether LLMs can replicate aspects such as terminology and grammatical structures. As noted in related work, similar attempts have been made in the context of metaphor (Dorst 2024). To achieve this, it might be necessary not only to show parallel corpora but also to add specific prompts that supplement what aspects of the language should be learned from them. Here, the practical application of TS concepts and professional translation considerations can be implemented in the form of prompts.

Moreover, this approach encourages critical thinking about translation quality assessment. By comparing human evaluations with automated metrics such as COMET, learners can develop a deeper understanding of the complexities involved in translation evaluation. This specific exercise highlights the importance of considering both quantitative metrics and qualitative aspects such as style, tone, and cultural nuances in assessing translation quality. It also underscores the current limitations of automated evaluation tools and the continued relevance of human expertise in translation assessment. Furthermore, engaging with LLMs to analyse discrepancies between human and automated evaluations can provide valuable insights into the strengths and limitations of different evaluation methods, fostering a more comprehensive and nuanced approach to translation quality assessment in both educational and professional contexts.

6 CoT prompting and the translation process

In this section, a simple CoT prompt was used, employing a taxonomy of translation strategies. Rather than recreating the translation process through a CoT prompt, in other words, not a strict step-by-step prompt, it facilitated a relatively high-resolution cognitive process concerning translation. Specifically, the well-known procedures described by Vinay & Darbelnet (1958/1995) were provided to the LLM. The definitions were generated by ChatGPT-4o. Subsequently, the model was instructed to translate a Japanese sentence (previously used in an example) by applying all the procedures from the provided taxonomy. Let us first examine the following prompt:

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LLM Prompt 5.2:

Translate the English sentence “When I was young, life was so different” into Japanese, adhering to the provided taxonomy below.

Direct Translation Methods

1. Borrowing: Directly taking words from the source language into the target language without translation (e.g., “pizza”).
2. Calque: Translating a foreign word or phrase literally, creating a new expression in the target language (e.g., “skyscraper” as “gratte-ciel” in French).
3. Literal Translation: Word-for-word translation that adheres closely to the original syntax and meaning, appropriate when the languages share similar structures and cultural contexts.

Oblique Translation Methods

4. Transposition: Changing the grammatical category of a word without altering its meaning (e.g., transforming a noun into a verb).
5. Modulation: Changing the form of the message, introducing a shift in point of view or cognitive category to fit the natural expression in the target language (e.g., “It’s not difficult” instead of “It’s easy”).
6. Equivalence: Using an entirely different expression to convey the same situation, often applied in idiomatic expressions or proverbs (e.g., “Out of sight, out of mind” translated as “Loin des yeux, loin du cœur” in French).
7. Adaptation: Modifying the cultural reference when the source language situation is unknown or unrecognisable in the target culture (e.g., changing a reference to Thanksgiving in the U.S. to a relevant local holiday in the target culture).

While such taxonomies of translation strategies are often lectured on in introductory TS courses, it was not realistic to use classification tables for discussing translations in practical translation classes. Especially when teaching translation into Japanese, European-developed classifications often did not directly apply, necessitating the use of customised literature or adaptation for practical use. In essence, this strategy classification taxonomy exemplifies how categorisations and accumulations from TS research were seldom utilised in practical teaching classes. Frankly, it was cumbersome. However, this is not to say that human

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translators do not mentally engage in a process similar to this taxonomy, creating and selecting from multiple translation options. Nor is it to suggest that such taxonomies are entirely without merit. To reiterate, the primary issue was the time-consuming nature of the process.

Given this context, we provided the aforementioned prompt and requested a translation. The result is as follows. For the translation, Claude 3.5 Sonnet was utilised, consistent with the earlier example.

LLM Response 5.2:

Borrowing: ヤングの頃、ライフはとても違っていた。

Calque: 若かった時、生命は非常に異なっていた。

Literal Translation: 若かった頃、人生はとても違っていた。

Transposition: 若い時代には、人生は大きく異なっていた。

Modulation: 若かった頃は、今とは全く違う人生だった。

Equivalence: 若い頃は、今とは別世界だった。

Adaptation: 青春時代は、今とは天と地ほど違っていた。

As mentioned earlier, Vinay and Darbelnet's translation taxonomy is challenging to apply to Japanese translation. Therefore, I was impressed that the LLM could instantly 'understand' this taxonomy and create examples adapted to Japanese translations. However, understanding this classification in Japanese remains difficult, so I supplemented the prompt by asking for specific explanations on "which parts of the translation each strategy is applied to." Consequently, the following detailed explanations were obtained. Due to space constraints, only the first three results are presented here.

LLM Response 5.3:

1. Borrowing: "Yangu no koro, raifu wa totemo chigatte ita." (ヤングの頃、ライフはとても違っていた。) - The English words "young" and "life" are borrowed directly into Japanese as "yangu" and "raifu" respectively.

2. Calque: "Wakakatta toki, seimei wa hijou ni kotonatte ita." (若かった時、生命は非常に異なっていた。) - "Life" is translated as "seimei" (生命),

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which is a more literal translation of “life” as in “living being,” rather than the more common “jinsei” (人生) meaning “lifetime” or “course of life.” - “So different” is translated more literally as “hijou ni kotonatte ita” (非常に異なっていた).

3. Literal Translation: “Wakakatta koro, jinsei wa totemo chigatte ita.”(若かった頃、人生はとても違っていた。) - The sentence structure of the original is maintained, with each word translated to its Japanese equivalent.

...

In actual translation practice, these strategies would typically be combined to create a natural and effective target text. Some of these examples may appear unnatural as they are designed to highlight the specific features of each translation strategy.

While it may seem repetitive, I must emphasise how remarkably clear, accurate, and well-explained these examples are. The precision and correctness of both the instances and their descriptions are noteworthy.

This verification suggests the potential for LLMs to support human translators throughout their translation process using CoT prompting. However, more immediately apparent is how such activities could offer significant insights for translation education. The example presented here, applying a well-known taxonomy of translation strategies to create translation examples, is just one illustration of this potential.

Beyond this, we might also consider the application of other TS assets, such as learning about translation issue typologies. These exercises provide a tangible sense of how theoretical concepts from TS can be practically applied.

In essence, this approach not only demonstrates the impressive capabilities of LLMs in understanding and applying complex translation theories but also opens up new possibilities for translation education. By bridging the gap between theoretical knowledge and practical application, it offers a novel way to engage with TS concepts that have traditionally been challenging to implement in practical training settings.

7 Discussion and concluding remarks

Firstly, it is crucial to understand that prompt engineering is not solely aimed at improving automatic translation quality based on NLP technologies or unified value systems. We often perceive AI and LLMs as adversarial to human translators. While important, this perspective, which leads to a focus on cautious educational or practical use—such as copyright concerns or ethical considerations—may be overly conservative. To draw an extreme analogy, it is akin to focusing solely on avoiding accidents or problems when planning an enjoyable trip, rather than considering how to make the journey truly rewarding. This chapter has specifically explored the positive and proactive possibilities of utilising LLMs in translator training.

To elucidate these positive applications, we have examined the use of accumulated TS assets and metalanguages as prompts for LLMs. However, I must reiterate that the primary aim of this chapter was not to apply TS concepts to prompts for the simple goal of improving LLM-generated translation quality, as is often the focus in NLP. Instead, the objective was to demonstrate how translation metalanguages can be used to engage in dialogue with AI, potentially enhancing translation education.³

Specifically, we have presented concrete examples of learning scenarios using translation concepts within the frameworks of few-shot prompts and CoT prompting. The prompt utilising process taxonomy to generate various translation variations was particularly surprising and insightful. Critical analysis and AI consultation on this approach yielded highly accurate responses. While these examples are limited and the high accuracy may be partly attributed to the use of the well-known Vinay and Darbelnet taxonomy, they represent just the first step in exploring a vast potential. As an educator, I am both intrigued and obligated to further investigate these possibilities.

While these results are promising, it is important to note that this is just the beginning. Further research and experimentation could reveal even more ways in which LLMs can enhance translation education and support professional translation practices. Moreover, this method could potentially make the learning process more interactive and engaging for students, allowing them to explore various translation strategies and their implications in a more hands-on manner. It also provides a platform for discussing the nuances of translation choices, which is crucial for developing critical thinking skills in aspiring translators. This ap-

³It should be emphasised that such systems do not in fact “understand” translation concepts; rather, they generate output based on statistical predictions of likely word sequences.

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proach may offer opportunities to foster autonomous learning and develop essential qualities for independent translators.

It is also important to acknowledge some limitations. Not all responses from LLMs were accurate, and some prompts were less successful than others. For instance, while the LLM correctly explained why COMET could not provide a fair evaluation in the few-shot prompt example, it failed to give a reasonable answer when asked which translation (Claude or DeepL) was closer to the reference translation. Such errors and limitations of LLMs become more apparent with increased use. However, given that translator training inherently requires maintaining a critical perspective, I believe exploring the possibilities of using LLMs is as important as considering the risks.

In conclusion, this chapter has demonstrated concrete methods for exploring the potential of LLMs in translation education. By leveraging the concepts and metalanguages of TS in prompt engineering, we can create more engaging, interactive, and effective learning experiences for translation students. While challenges and limitations exist, the potential benefits of integrating LLMs into translator training are significant and warrant further investigation and development.

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Chapter 6

Teaching AI ethics for translation students

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A key part of teaching translation in the age of Generative AI (GenAI) is understanding when to use and when not to use GenAI. AI ethics is an applied subfield of Ethics that can help with decision-making, which may be useful to students after they graduate and as they move through their careers. In this chapter, we begin with some definitions of AI and ethics, introduce some ethical issues, along with some progress in addressing those issues. We finally consider some ways to introduce discussion and reflection in the classroom to maximise the impact of this teaching, focusing in particular on case studies and the use of mapping and images to understand our position within the complex interrelationships between individuals, organisations, and society regarding ethical issues pertaining to translation and GenAI.

1 Introduction

The topic of artificial intelligence (AI) ethics is very high on many agendas at present, as universities and schools grapple with how best to incorporate Generative AI (GenAI) into their curricula in a constructive manner. The intention is to teach AI literacy, defined by Long & Magerko (2020: 598) as the ability to critically evaluate, communicate and collaborate with, and use AI, equipping students with critical knowledge and skills that will stand to them throughout their careers, balanced with understanding of ethical issues regarding the use of AI.

AI ethics are important. There are lots of ethical issues regarding GenAI, as we will run through in this chapter. Our graduates will gain responsibilities as they

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continue through their careers, so it's important that they be conscious of the ethical repercussions of decisions and actions that they might take. The current hype about GenAI might mean pressure to use tools even if the circumstances are not appropriate – what a blogpost by the language service provision company Transperfect called the “push to implement”, when companies are “pressured by leadership to implement AI solutions” (Transperfect 2024). It's important that graduates are able to make well-founded arguments as to why using GenAI might be appropriate or inappropriate in any given situation.

In the classroom, questions of ethics tend to provoke discussion and allow for student input. This chimes with the call for critical and democratised education from authors such as Freire, or Giroux and McLaren, incorporating “forms of learning which serve to prepare students for responsible roles as transformative intellectuals, as community members, and as critically active citizens” (Giroux & McLaren 1997: 236). Currently, this call feels more urgent than ever. In Translation Studies, Abdallah (2011) presented a related three-step ‘ideology critique model for teaching’ that involves disassembly of the teachers’ own beliefs, critiquing and resisting unfairness, and finally fostering hope and encouraging agency. The aim of teaching about ethics in the translation classroom is to stimulate critical thinking so that students recognise ethical problems and can resist unfairness. However, consideration of ethics may present students with opportunities rather than only imposing limitations, and it is important for students to graduate with a positive sense of their own agency in ameliorating ethical issues. Equally, as educators and students, we should be aware of and take responsibility for our choices and decisions. With that in mind, this chapter borrows Bryan's (2022) ‘pedagogy of the implicated’ to describe and understand our position within the complex interrelated systems linking us as individuals to our surrounding political, cultural, social and economic systems.

This chapter will begin with some definitions of terms pertaining to AI and ethics, moving on to explaining some of the ethical issues. We have found it useful to create model use cases for in-class discussion and will suggest a couple of these. Thereafter we look at possible routes forward regarding GenAI and ethics and the recent progression on these. Finally, we look at ethical issues for researchers in translation and interpreting that might be relevant for both educators and students.

2 Definitions

The terms ‘artificial’ and ‘machine intelligence’ were first popularised in the 1950s, usually regarding the ability of machines to take on human tasks, but

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are rarely clearly defined. More recent definitions of AI are still sometimes related to thinking or behaving like humans or thinking or behaving in a rational way. Russell (1999: 12), for example, defines AI as “the capacity to generate maximally successful behaviour given the available information and computational resources”. Up until the 1980s, these resources might have involved sets of rules built by human experts – so-called ‘expert systems’. This was the common paradigm of rule-based machine translation (MT), before the advent of statistical MT (Brown et al. 1988).

Both rule-based and statistical MT used human-readable data or rules in what is sometimes called Symbolic AI, but once technologies such as MT, speech recognition, and recommendation systems moved to neural networks or machine learning, we moved to Subsymbolic AI, in which data and processes are transformed into numbers and no longer comprehensible by humans. Mitchell (2020: 12) defines subsymbolic AI as a “stack of equations – a thicket of often hard-to-interpret operations on numbers”.

In recent times, AI is often taken to mean machine learning using transformer-based neural networks, despite the term’s relatively long history. Machine learning is defined by Kelleher et al. (2015: 3) as “an automated process that extracts patterns from data”, and can be supervised, whereby the system learns how to carry out a repeated task based on training data, as with NMT, or self-supervised (i.e. by the system itself), where patterns are inferred from training data without explicit external instruction, as is the case with large language models (LLMs) – GenAI based on natural language processing.

Currently, much of the focus for very large (sometimes called ‘frontier’) LLMs is scale: bigger language models (LMs) with more data, as a key finding for LLMs was the emergence of unexpected abilities at scale (Wei et al. 2022). This does not mean that smaller LMs cannot be useful (see Moorkens et al. (2025) for more on local, small LMs), but the tendency towards larger LMs brings particular ethical concerns that we will return to in the following section.

So, what do we mean by ethics? Ethics is a branch of philosophy that helps us to decide if an action is right or wrong (see Moorkens 2024). Normative ethics describes theories to help us with that decision, such as consequentialism or deontology, and applied ethics involves the application of those theories to particular scenarios, for example in business ethics, data ethics, or AI ethics. The field of AI ethics looks to balance the benefits of AI with negative effects (which we will come to in Section 3), often by establishing principles for providers to adhere to at their discretion or by establishing governance rules within an organisation or in national (or international) law.

There's a growing sense that technology affects our ability to live a good life. Technologies have affordances, actions that are made available to us as users, and these can encourage positive or negative, healthy or unhealthy behaviours. Machine learning is enabled by powerful computers, built using rare metals and electrically powered, and also by huge amounts of proprietary and/or public data. The terms 'crawling', 'harvesting' or 'scraping' are often used for the collection of data via the internet, as if data is naturally-occurring rather than the product of human effort. So there are lots of ethical issues pertaining to AI and data that relate to translation. We'll discuss many of these in Section 3.

Discussion point: How might translation technologies or GenAI help us to live a better life? And might they also have negative effects? If so, what might these be?

3 Ethical issues pertaining to GenAI and translation

As noted previously, the availability of vast amounts of data facilitates GenAI. Translators who work with computer-aided translation tools are usually expected to include bilingual translation memories with delivery of translation jobs, either due to contracts or precedent. This draws from the Berne Convention, first enacted in 1886, in which translations are considered to be derivative works that "shall be protected as original works without prejudice to the copyright in the original work" (World Intellectual Property Organisation 1979: art. 2). There have been arguments that translators may have reasonable claims for copyright (see Moorkens & Lewis 2020), but this is not usually respected, and in many cases, digital platforms do not permit translators to even access these memories, nor to control how they are repurposed. MT and LLMs are trained using monolingual data from organisations' repositories and data 'harvested' from the internet using crawlers. The legal basis for this differs in different jurisdictions and many copyright owners for text crawled or harvested have taken legal action. For now, there does not appear to be any major restriction to using crawled data, even if it's copyrighted, perhaps because authorities do not want to be seen to stifle innovation and few content owners feel able to challenge powerful technology companies. For translators and writers whose data is used, this presents a problem in that their work may be put towards purposes that they do not agree with, without any further compensation.

Workers in translation and other professions may choose to use GenAI or MT (see Rivas Ginel & Moorkens 2024), but many have these technologies imposed on them without any choice. Firat (2024) and others demonstrate the relationship between the use of technology to reduce labour costs, particularly within digital

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platforms, and reduced access to what the International Labor Organisation term ‘decent work’, relating to pay, representation, time, stability, equality and management (Ferrero et al. 2015). Ruokonen & Svahn (2024) summarise numerous studies that show a relationship between increased technologisation and reduced job satisfaction and motivation, which threaten the sustainability of the translation industry. Beyond translation, teams of (ghost-)workers, little-documented and often with poor pay and conditions, work to mitigate biased and offensive output from GenAI as part of the process of reinforcement learning using human feedback (RLHF) or to moderate potentially offensive training data (Rowe 2023), with associated repercussions on workers’ mental health.

Some years after NMT became popular, researchers such as Vanmassenhove et al. (2018) found patterns of bias in NMT outputs. This output tended to be less lexically diverse than human translations and contained gender bias, with certain words tending to be associated with males or females. There is also likely to be a cultural or linguistic bias towards English, the language of most training data (Bender 2011, Moorkens et al. 2025). The uneven support for languages in the digital domain is likely to exacerbate existing digital divides. Researchers working with LLMs also found biased output regarding race, gender, and sexuality, and a tendency to overrepresent hegemonic views. This makes sense, considering that a lot of training data comes from the internet. In order to rebalance this and to prevent embarrassing culturally inappropriate or offensive output, developers of LLMs began to use RLHF (Ouyang et al. 2022). However, this intervention also offers LLM developers the opportunity to suppress output due to geopolitical concerns. According to the New York Times, the output of the Chinese GenAI tool Deepseek “will largely reflect the worldview of the Chinese Communist Party” (Myers 2025: B1) and US-based GenAI Grok censored “unflattering facts about President Donald Trump” and its owner, Elon Musk on its initial launch (Wiggers 2025), one of a string of controversies that appears to relate to adjustments of the RLHF guardrails behind the scenes. Cooperation between big tech companies and autocratic regimes has until now mostly involved removing content or search results (see Boyle 2025), but RLHF offers another opportunity to intervene and manipulate output based on political demands.

Translation and MT have long been closely related to power (see Tymoczko 2007 and Paullada 2020) with the patron of translation or MT research having some sway in production. The amount of data, money, and computing power required to train LLMs mean that production capacity is in the hands of very few wealthy organisations, with many produced by US big tech companies or their close relations. Users are further reliant on these companies’ cloud services to train and fine-tune, and later deploy their LLMs. Services may be changed or

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withdrawn with little warning due to a change of circumstances (or geopolitical expediencies). At the time of writing, there is a push for LLM sovereignty, with international organizations like the European Union, national governments and some companies investing in supercomputers and building their own LLMs. Of course, not all LMs have to be huge, but the focus for US big tech providers continues to be scale, creating larger and larger models that take longer to train, with knock-on effects regarding environmental sustainability.

The pretraining and deployment of foundational LLMs is a resource-intensive process that requires significant capital investment for acquiring the necessary hardware, training data and human talent as well as operating costs, and each large-scale pre-training cycle has significant environmental impacts due to high electricity and water consumption. Combined, these factors allow only a few companies to have very powerful LLMs. Since many national governments, large corporations and public organizations consider LLMs and AI in general as strategic enablers for their future, they are increasing their investment in this area as well and setting big targets for broader AI use. However, the impact of this AI ‘arms race’ on the environment is likely to be catastrophic on our planet if every entity tries to develop their own proprietary LLM. In recent years, AI systems have in general become more efficient, but the trend for scale means quality improvement is based on more data and parameters, requiring more training time. [Luccioni et al. \(2025\)](#) argue that increased efficiency in AI training and in time saved by using AI will tend to spur more use of AI, continuing an upward trend not only in emissions, but in water use for data centres, rare earth metals used for ICT, and more harmful electronic waste.

These environmental harms and the impacts directly from AI are difficult to measure exactly. Emission levels may differ depending on whether energy comes from fossil fuel or renewable sources ([Shterionov & Vanmassenhove 2022](#)) or depending on the time of day and data centre location ([Dodge et al. 2022](#)). Big tech firms are looking to buy up renewables and nuclear power sources as they come online, in order to maximise their net-zero credentials. Water footprints are also likely to differ depending on time and region ([Li et al. 2025](#)). Presently, GenAI does not require a large proportion of resources, but projections, such as those from the head of the UK National Grid predicting an AI-driven six-fold increase in power requirements for data centres in the next decade ([BBC 2024](#)), are worrying. For now, energy and water use attributable to GenAI are very small in comparison to the huge requirements for watching streaming media or joining a Zoom meeting ([Mytton et al. 2024](#)).

4 Possible solutions

Positioned within the international AI ‘arms race’, with politically-motivated support for some developers leading to preferred companies receiving preferential access to governments and cutting-edge technologies, there are also movements to use ‘AI for Good’. One platform by that name is led by the International Telecommunication Union of the United Nations (UN) to use AI to achieve UN sustainable development goals, and there are smaller initiatives such as the Distributed Artificial Intelligence Research Institute that looks to push back against the influence of big tech on AI research, development and deployment. Many people seek to use (Gen) AI to reduce harms and inequality. There have been proposed uses of AI to route power use to maximise the use of renewable energy and to improve energy efficiency in the design, building, and use of commercial buildings (Ding et al. 2024).

Van Wynsberghe (2021) feels that there are two motivations behind ‘AI for sustainability’ and ‘sustainability of AI’ that ought to be combined. The former seeks to do good, yet might entail negative environmental impacts, whereas the latter acknowledges that for AI to be sustainable, there needs to be lower environmental costs for AI training, tuning and inference. She defines sustainable AI as a necessary movement to “foster change in the entire lifecycle of AI products (i.e. idea generation, training, re-tuning, implementation, governance) towards greater ecological integrity and social justice” (Van Wynsberghe 2021: 217).

National and international legislation, most notably the EU AI Act, seek to limit harmful uses of AI. The AI Act defines a typology of tiered uses of AI based on risk, with ‘unacceptable risk’ uses forbidden and high-risk uses, such as recruitment decision-making and job allocation, subject to special regulation. This renders algorithmic job allocation in the EU illegal, although it’s likely that the recommendation of an automated project management tool will still be followed.

In a position paper by Moorkens et al. (2024) we borrowed the idea of a triple bottom line from Business Ethics (Elkington 1997) to propose that translation technologies and LLMs be evaluated not just focusing on performance, but rather giving equal weight to people, planet and performance. This is necessarily a heuristic rather than an exact metric, but follows on from criticisms of a focus purely on performance pushing AI development in the wrong direction by Schwartz et al. (2020) and others. For people, we might consider how the LLM impacts annotators, translators, platform workers, and those who have been dispossessed of their data, balancing these against benefits to people. For the planet, we might look at energy costs/CO₂, efficient models, and ICT cost

and disposal. Finally, for performance, we should use task-appropriate and comprehensive standards.

A previous suggestion for translation data from [Moorkens & Lewis \(2019\)](#) was a community-owned and managed digital commons, following the ideas of [Ostrom \(2011\)](#), with tiered access available for a cost. We argued that this would help to “sustain the occupation of translation and to minimise the potential risks and harms to translators and the public” ([Moorkens & Lewis 2019: 17](#)). This idea seems to be similar to the intended implementation of a European Language Data Space as a decentralised marketplace for text, video and audio data in different languages ([Rehm et al. 2024](#)).

The advice from some researchers, such as [Rudin \(2019\)](#), is to entirely avoid black-box, subsymbolic systems for high-risks uses. This is because opacity is largely baked into subsymbolic AI systems, as described previously, although Rudin and others believe that, in many cases, comparable results may be achieved with more transparent, hybrid systems.

For very large, closed source systems, we do not know what training data has been used or what the RLHF guardrails are. However, not all LLMs are closed source – or even that large. We mentioned the alternative options for small-scale LMs in Section 2. These can be useful for narrowly-defined tasks, with benefits of low cost, low environmental impact, and customisability. For technically confident students, guidelines for building a custom small LM are provided by [Moorkens et al. \(2025\)](#).

In academic research ethics, the credo has moved on from ‘do no harm’ to the need to actually benefit research participants. Relatedly, best practice for engaged research involves a participatory approach, working cooperatively, particularly with marginalised groups, as co-creators of knowledge rather than imposing narratives or putting words into their mouths. [Birhane et al. \(2022\)](#) propose this approach for building AI systems, offering examples of participatory approaches that aim to lessen existing imbalances of power. In this way, developers “acknowledge that the communities and publics beyond technical designers have knowledge, expertise and interests that are essential to the development of AI that aims to strengthen justice and prosperity” ([Birhane et al. 2022: 7](#)).

5 Translation and AI ethics in the classroom

5.1 Classroom discussion activities

As student users of translation technology – and most likely users of related AI tools and cloud services more broadly – students in a translation classroom

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are already part of the interconnected web of ethical issues from the previous section. They may not have given them much thought, as so much of the hype about GenAI focuses on its positive potential rather than ethical issues. GenAI is also entangled with what Brand & Wissen (2021) call the ‘imperial mode of living’, through which public and private organisational strategies and individual lifestyles and practices in the global north rely on the unlimited appropriation of resources, a disproportionate claim to global and local ecosystems, and cheap labour, ideally from distant locations. Bryan (2022: 330) conceptualises individual relations with the climate crisis as a “form of ‘difficult knowledge’, particularly as it relates to learners’ self-implication in the conditions that are being addressed”. We can reasonably broaden this to AI ethics for trainee translators and educators. Transmission-based lectures alone seem inappropriate for this topic, as difficult knowledge may raise sensitivities; nobody likes to be hectored or to feel that their personal ethics are in question.

In this section, we introduce two methods for stimulating discussion and reflection about translation and AI ethics in the classroom. The first uses scenarios or case studies, placing ethical dilemmas into familiar contexts for discussion. The second draws from Bryan’s (2022) ‘pedagogy of the implicated’, which seeks to prompt critical reflection about our own positioning as ‘implicated subjects’ and to foster agency for change. Bryan uses the notion of the implicated subject from Rothberg (2019) to look beyond dichotomies of individual versus institutional responsibility for injustices to a discussion of how we are enmeshed with systems in many ways across historical (diachronic) and contemporary social-structure (synchronic) lines.

5.2 Case studies

According to Benbunan-Fich (1998), a combination of lectures and discussion are complementary ways of introducing ethical issues in the classroom. To begin with, lectures about “ethical concepts can lay the theoretical foundations” so that students can “practice ethical analyses” thereafter using case studies (Benbunan-Fich 1998: 20). Case studies have proved to be a useful tool, particularly in business schools, for many years. According to Barnes et al. (1994), case studies extend learning beyond each class, stimulating deeper insights that link across classes and modules. Led by instructors with appropriate case studies, students will engage and can develop and articulate critical insights. These come through four particular factors: situational analysis, active student involvement, a non-traditional instructor role and the need to relate analysis and action. Situational analysis means that ethical issues are applied in situ rather than in

theory. Students are active, engaged and learning without direct teaching, energised and challenged by discussions. This puts the instructor in a non-traditional role of facilitator without dominating the classroom. Rather than only building knowledge, case studies combine analysis and action: the “importance of action influences the entire case discussion, which focuses on the practical and doable” (Barnes et al. 1994: 48).

The case study chosen must be appropriate – Zhou (2022: 400), for example, recommends finding “representative translation-ethical cases”. There are two that may be used or adapted to GenAI from Moorkens (2022) in the open access book *Machine Translation for Everyone*. Case studies may be accompanied with a general question to motivate discussion (e.g. ‘What ethical issues do you see in this scenario?’) or a case worksheet featuring a number of questions or tasks. Some examples that could be included are:

- Describe the key ethical issues and the stakeholders involved.
- What ethical theories might provide guidance in this scenario?
- Can you think of guidelines from a translator code of ethics that could be useful in this scenario?
- Are there any legal issues involved in this scenario?
- With the above questions in mind, are there any actions that you propose to resolve this scenario in an ethical manner?

Students may address these questions in groups or as part of a plenary discussion moderated by the instructor. A final recap, making note of any new or unexpected issues that were brought up by students, can effectively summarise a class and reinforce key lessons.

5.3 Pedagogy of the implicated

The approach proposed by Bryan (2022) in her article on ‘pedagogy of the implicated’ could work in combination with or instead of case studies. As mentioned, Bryan draws on Rothberg (2019) for her description of (most of) us as “implicated subjects”, who can be “beneficiaries and *perpetuators* of systems that are not of [our] own making or that [we] have no direct ancestral attachment to” (Bryan 2022: 338, italics in original). A preamble explaining this complex entanglement of implication is a valuable precursor to discussion in the classroom. The prompt

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for this discussion could come from carefully chosen images or a mapping exercise to visualise these entangled relationships.

Bryan's (2022: 341) article is focused on climate change education, for which there is a tendency towards what she calls an "apocalyptic sublime aesthetic". This aesthetic, she argues, "positions viewers as mere voyeurs or passive spectators – rather than active agents or implicated subjects – in the unfolding chaos". Images of AI are perhaps generally less apocalyptic, but are similarly uninvolved. These science-fiction-inspired images will be familiar to those who have seen many visualisations of AI featuring stylised robots with fingers to their chins among gigantic circuit boards, concealing any human involvement, ethical dilemmas, or even any question about the smooth transition towards wise and sentient intelligent robots.

Dihal & Duarte (2023) summarise how typical stock images of AI are misleading for non-experts, in that they hide societal and environmental impacts, promote unrealistic expectations of AI capabilities, hide human accountability, and often promote stereotypical assumptions about gender, ethnicity and religion. This led them to start the website Better Images of AI¹ featuring artist-created images, collages and illustrations that foreground the issues hidden in regular stock images. These images are highly considered and thus make an excellent starting point for classroom discussion and reflection.

Bryan proposes a mapping exercise as part of a discussion of self-implication in a complex interconnected consumer-focused system, such as that behind GenAI, to prompt critical self-reflection. The main example that she uses is what she calls the Social Ecology of Responsibility Framework (SERF), inspired by Bronfenbrenner's (2009) theory of the ecology of human development. Bronfenbrenner was "concerned with the mechanisms, processes and conditions that shape individuals' development and devised a model that theorises the reciprocal interactions that occur within and between different nested environments (or systems) which in turn affect developmental outcomes" (Bryan 2022: 338).

The SERF can be a useful tool to visualise the interconnections between the individual actions of the 'implicated subject' within their immediate context and larger social, national and international systems. These are represented by concentric circles, expanding from the individual's microsystem (our immediate context) to our macrosystem of societal, political, and cultural norms and context. Within these micro- and macrosystems are the mesosystem, describing points of connection between different levels, and the exosystem of institutions and organisations that exert influence without any direct contact with us as individuals.

¹<https://betterimagesofai.org/>

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Navarro & Tudge (2023) suggest that our microsystems should be subdivided into physical and virtual categories, as our day-to-day activities, social roles, and interpersonal relations may differ (or blur) between physical and digitally-mediated domains. The mesosystem groups our multiple microsystems together. We might consider the remarkably consistent messaging about technology and GenAI coming from the exo- and macrosystems, why that might be, and the potential impacts on our mesosystem.

Figure 1 features an adaptation of Bryan’s (2022) SERF, using a layered funnel design and bringing in the idea of the triple bottom line. This could form a useful starting point for classroom discussions or an example for a mapping exercise. The intention is not to blame individuals – the chronosystem part of the SERF makes clear the historical context for current systems – but to empower the individuals at the centre, which of course include the educators as elements and perpetrators within the overall system.

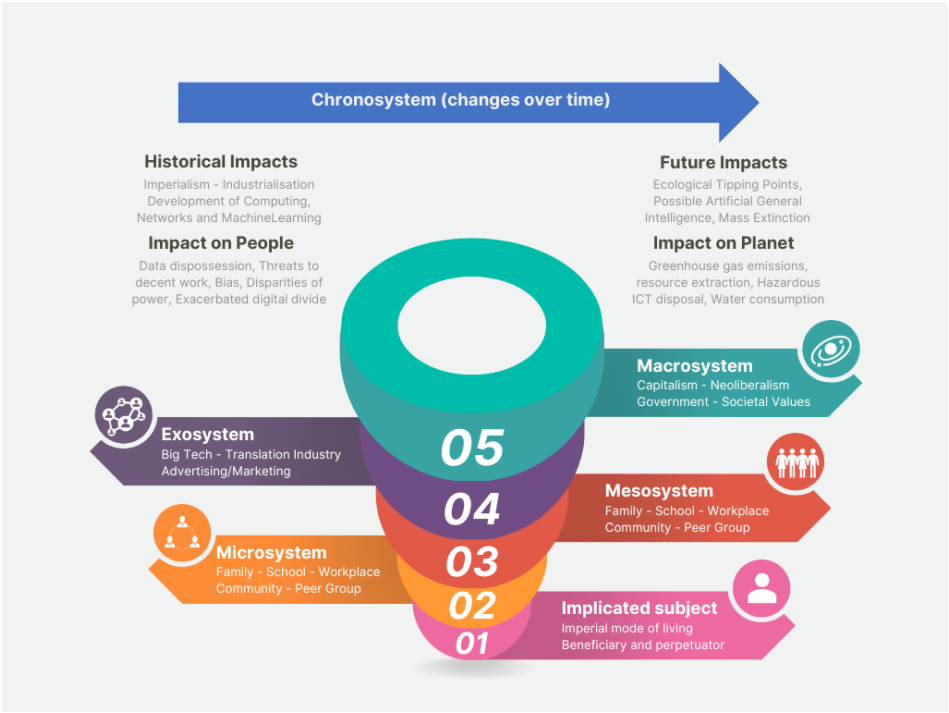


Figure 1: An adapted version of Bryan’s (2022) Social Ecology of Responsibility Framework with some ideas to use for classroom discussion

Another useful image intended to stimulate reflection and discussion of the

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translator's responsibilities, with the potentially conflicting pressures from ethical codes, business ethics, personal interests, and social responsibility comes from Joseph Lambert's (2023) book *Translation Ethics*. The section on responsibilities and related questions place many of the points from this chapter within the context of translation work.

6 Conclusion

Many curricula now incorporate AI literacy as an important competence. Chan & Colloton (2024) recommend a fine-grained interpretation of AI literacy that is tailored to subject or professional areas, and a major part of this for translation students is AI ethics in translation. In this chapter, we briefly introduce some of the issues and debates regarding AI ethics, along with some positive initiatives that are ongoing.

We propose interactive, social constructivist methods for teaching students about AI ethics for translation, highlighting two potential methods in particular. The first of these is the use of carefully chosen case studies with accompanying support material to help students talk through ethical issues. Benbunan-Fich (1998: 21) feels that it is "crucial for students to practice ethical decision making, so when they have ethical decisions to make in the real business world they have a framework to follow". The second suggested method is an adaptation of Bryan's (2022) pedagogy of the implicated, using images and mindmapping to visualise and understand our own place within the tangle of beneficiaries and perpetrators of the system within which GenAI is developed and marketed. A disingenuous tendency to highlight individual responsibility for issues of ethics and sustainability lets large organisations off the hook, but these methods show how individual and collective responsibilities are interlinked. Our status as implicated subjects means that we must also take responsibility for our actions and ethical decisions. We hope that these methods will be effective beyond the individual class, leading to deeper knowledge in students that can come to their own individual ethical position on the debates and issues in AI ethics for translation.

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Part III

New paradigm: New teaching approaches

Chapter 7

Re-positioning the human translator as the future expert of GenAI translation in the translation classroom: The results of a collaborative study

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With the launch of ChatGPT in 2022, which allowed the production of translation through a prompt-based *dialogue* between human and generative artificial intelligence (GenAI), the early characterisations of translation as “human-computer interaction” (O’Brien 2012) and a “dance of agency” (Olohan 2011) have become realities in the fullest sense. In previous research, it has been suggested that educators and students can collaborate to understand how to create “human added value” (Massey 2021) and raise the students’ “self-concept” (Haro-Soler & Kiraly 2019) as human translators in emergent MT tasks (Öner Bulut & Alimen 2023). Translator educators should continue to explore collaboratively with students this human-GenAI *dialogue*. For this aim, a research experiment was conducted where graduate level students and a translator educator collaborated to explore the dynamics of prompt engineering (Yamada 2023) as an emerging job role in the field of GenAI translation. The results showed that the experiment helped the students diagnose not only the strengths and weaknesses of GenAI but also their own, acknowledging the need for human intervention, editing and evaluation to turn raw GenAI output into a functioning translation. The students also re-evaluated their human added value and self-concept as educated and experienced human translators, highlighting the sense of empowerment and responsibility they felt as human decision-makers in the prompt engineering process while addressing areas of self-improvement such as developing skills to communicate with GenAI, which are distinct from communicating with a human being. The results also indicated that the established, holistic skill set of a trained professional human translator was regarded a prerequisite for effective prompt engineering. This needs to be considered

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by translator educators who now have the task of educating students so that they can cautiously but confidently communicate and dance with GenAI.

1 Introduction

The impact of machine translation (MT) on the translation profession and translator education has been profound in the last decade, as has the research into this impact. The literature has focused prominently on the tense relationship between the human translator and the machine/technology, which Pickering (2008) conceptualised as a “dance of agency”. Olohan (2011: 354) drew attention to the “decentring” of the human agent in this context. O’Brien (2012: 119-120) further characterised translation as “human-computer interaction”, defining what was happening then as a “shift in paradigm” and pointing out the potential for “dialogue” between human translators and technology.

Ruokonen & Koskinen (2017: 321) further elaborated on the human-machine dynamic, with a focus on the emotional dimension inherent in this relationship. They found that the human agents were both “willing” and “reluctant” in terms of assigning agency to the machine and that the relationship between the human and the machine is too “complex” to be reduced to “simplified man versus machine conceptions”. In a similar vein, Cadwell et al. (2018: 312) observed that human translators held both positive and negative perceptions of MT. Positive perceptions related to reduced effort and increased inspiration while the negative were linked to concerns about the negative influence of MT on human creativity. The authors also argued that the dance of agency between the human and the machine was a symbiotic one, as they mutually feed off and shape each other; the “distinction between the human and the material agent” is blurred (Cadwell et al. 2018: 303).

The practical consequences of the pervasive integration of MT technologies into the global translation industry and their economic impact have also been explored. Moorkens (2017), for instance, noted the negative consequences of the freelance employment model, which has been increasingly adopted in the global translation industry, in terms of prices and isolation. According to him, the focus should be on how the human translator actually works with MT. Similarly, Vieira (2018: 15) found that, rather than feeling an “intrinsic negativity to the technology”, the perceived threat from technology among translators appears to stem from business practices in the industry.

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Another focus in the related literature has been on the need to redefine the profile of human translators. Sakamoto (2019: 68-69) underlined the necessity of the “redefinition of the concept of ‘translator’, and consequently ‘translation’ per se” on the grounds that “the notion of ‘translation’ is now being challenged by the growth of technologies”. The author also noted that attention should be paid to the uncertainties related to the legal requirements in MT use, pricing for post-editing, and post-editors’ profiles and skills (Sakamoto 2019: 63-66). Based on a comprehensive analysis of the emerging needs and requirements in the translation sector, which was conducted with the aim of addressing the “distance between teaching and professional practice” in the context of MT competences, Gaspari et al. (2015: 333-334) reported “a strong need for an improvement in quality assessment methods, tools and training”, underlining the “growing uptake of MT and the perceived increase of its prevalence in future workflows”.

There has been extensive and substantial research into embedding in translator education the new skills, abilities and competencies that human translators should possess in the age of MT, as well as into the concomitant need to re-profile translator education. For instance, Doherty & Kenny (2014: 296-297) designed an SMT syllabus, which foregrounds the “empowerment” of the human translator by enabling him/her to intervene in and add value to the SMT workflows when facing legal, ethical and technical obstacles (Kenny & Doherty 2014: 288). This “empowering” approach has been further extended by Mellinger (2017: 284), who proposes a holistic, “cross-module or cross-curricular integration” of post-editing and MT “to avoid compartmentalisation of various competences and skillsets”. Mellinger also suggests the inclusion of “controlled authoring, terminology management, engine tuning, and post-editing” in translation practice courses, as these are “representative of many of the skills mentioned in recent works on machine translation pedagogy” (Mellinger 2017: 284). Pym (2019) drew attention to the identification of “automation-resistant skills” and their integration into translator education.

The measures to be taken by translator educators to empower students and help them become “aware of their usefulness in order to maximise their agency as translators” has also been the focus in a study by Moorkens (2018: 375-376), who designed a translation evaluation exercise to enable students to “demystify NMT output”. Based on a reconceptualisation of translator competence in the age of MT, Öner Bulut (2019: 3) designed a learning practice to help students “raise their awareness of their professional self-concept as human translators” and suggested the consideration of human translator competence and human translation meta competence. Nitzke et al. (2019: 248) proposed a novel “post-editing competence model”, which included the core competences of risk assessment competence,

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strategic competence, consulting competence and service competence. Öner & Öner Bulut (2021: 100) investigated translation students' perceived difficulties and benefits in the context of "post-editing oriented neural machine translation error annotation and quality evaluation".

Exploring the "dynamics of the human-machine dance in the translation classroom", Öner Bulut & Alimen (2023: 377) designed a learning experiment to enable the students to carry out emergent MT-related tasks of post-editing, pre-editing and error annotation. The authors reported that the experiment helped the majority of participating students "raise awareness of their self-concept as human agents and of the human added value they can create while dancing with the machine" (Öner Bulut & Alimen 2023: 387).

These examples of past studies focusing on the impact of MT on the translation profession and translator education show that translation studies (TS) research has indeed come a long way to come to terms with and adapt to the paradigm shift (O'Brien 2012) induced by the increasingly all-pervasive technologization of translation, especially since the launch of NMT in 2016.

Yet, the year 2022 presented both translators and translator educators with another, unprecedented challenge. With the launch of ChatGPT in 2022 (OpenAI), which allows for the production of translation through a prompt-based dialogue between human and generative artificial intelligence (GenAI), the early characterisations of translation as human-computer interaction (O'Brien 2012) and dance of agency (Olohan 2011) have become realities in the fullest sense. This poses new questions and opening new areas of research for the TS community, especially for educators and researchers of translator education.

In their above-cited research, Öner Bulut & Alimen (2023: 389) noted that, in the face of the rapid advancements in MT technologies and the uncertainties concerning their integration into education, the re-positioning of translator educators is as necessary as that of human translators. It has also been argued that such re-positioning involves an "awareness of the fact that the translator educator has the primary task of diagnosing and promoting emergent areas of human added value creation in the MT age", and that this can only be achieved by restructuring the translation classroom as a "platform of collaborative learning, where all involved can work together to discover what remains and will remain human in the MT age" (Öner Bulut & Alimen 2023: 389).

In the face of the paradigm shift induced by the introduction of ChatGPT and other GenAI tools, translator educators should once again work in collaboration with students to explore the dynamics of integrating these tools into their workflow. Motivated by this very need, the present chapter presents the results of a collaborative research experiment designed to explore the dynamics of the

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human-GenAI dialogue and find the pathways to be followed in incorporating the insights gained into translator education.

The remainder of this chapter is structured as follows. First, the unprecedented practical and theoretical challenges posed by GenAI technology in terms of the very *raison d'être* of the human translation are explored. Second, information on the design and implementation of the experiment is given. Third, the data obtained from the experiment is analysed. Fourth, a discussion of the results is presented together with conclusions and suggestions for future research.

2 Human added value and the human translator's self-concept in the age of GenAI: Re-visiting the interpretive task and responsibility of the human translator

ChatGPT is a GenAI tool powered by a large language model (LLM) and has not been developed specifically for interlingual translation. However, the revolution it has brought to translation is of great significance as the tool takes prompts from human agents and hence allows them to produce a translation by providing the tool with contextual information and guidance, i.e. to perform prompt engineering (Yamada 2023 and Chapter 5 of this book). This was certainly not the case with MT, which makes GenAI technology a real breakthrough that now requires translator educators to ask and try to answer new questions about human-technology interaction.

In my opinion, in this endeavour we should not attempt to reinvent the wheel. Instead, we should revisit the relevant existing insights into MT, especially those that deal with the integration of MT into translator education, in a way that empowers human translators to create human added value, as outlined in the previous section (see Introduction). This should also involve trying to answer the challenging question posed by GenAI: *what is human in translation?* This is because, unlike MT, GenAI creates the illusion of having an actual dialogue with a non-human agent which pretends to understand, interpret and produce a linguistic message.

This new form of challenge should urge us to revisit the interpretive task and responsibility of the human translator. Accordingly, the approach presented here is performing a backward reading. Rather than giving priority to the investigation of the performance, potential and/or limitations of GenAI, what the present study proposes is the instrumentalisation of the human-GenAI *dialogue* in order

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to try to reconsider *what is/will remain human* in translation in a new, brighter light and find ways to incorporate it into translator education in the age of GenAI.

In a seminal contribution by Massey & Kiraly (2019) titled “The Future of Translator Education: A Dialogue”, Kiraly states that “transcoding (the *mechanical replacement* of linguistic units from a list with corresponding units from a parallel list) is not at the *heart* of translation at all” (16, emphasis mine) and that, although in the future the translation profession will be “different”, it is “certainly not going to disappear – unless at some point in our evolution we no longer need to *interpret* texts” (16, emphasis mine).

It is no coincidence that Kiraly highlights the act of interpreting as the *raison d'être* of the human translator, which echoes the hermeneutical approach applied to translation by Schleiermacher (1813/1977). According to Hermans (2015: 101-102), Schleiermacher’s believed that the sole option for a translator is “to act as the hermeneuticist does: to work to attain the best possible understanding of the foreign text which nevertheless remains foreign, and present to the reader [...] exactly that understanding”. Accordingly, the meaning of a text is not encoded in a text to be decoded by the translator, rather it should be interpreted by the translator to attain the best possible understanding, which constitutes the hermeneutical task of the translator. The translator is then required to present this understanding to the reader, which, according to Herman’s reading of Schleiermacher, “exceeds hermeneutics” (Hermans 2015: 99). In a similar vein, Şerban & Larisa (2016: 295) underline that in Schleiermacher’s approach, the meeting between the author and the reader is “mediated and orchestrated by the translator” and the translator assumes “the most active mediating role”.

Although the hermeneutical approach is of critical importance in that it allows the translator room for active mediation, a specific conceptualization by Arrojo (1997: 18) of the “inevitable interference” of the translator in the context of translation ethics could further the attempt to reconsider *what is human in translation*. Arrojo views translation ethics in terms of the relationship between the “authorial power” of the translator and his/her “ethical responsibility” (Arrojo 1997: 18). Based on a postmodern view of “language and subject”, the author underlines that “[as] no reading can ever aspire to repeat or protect someone else’s text, translators necessarily have the right to exercise their authorial power ‘as long as their game is played up front’” (Berman 1995: 93, cited in Simon 1996: 36). In the view of the author, “[s]uch ‘right’ implies, however, an ethical responsibility which parallels that of ‘original’ authors” (Arrojo 1997: 18).

In the wake of the emerging paradigm brought about by GenAI, this line of thought can be followed by researchers and translator educators to foreground the human translator’s interpretive task and the accompanying responsibility

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with the aim of exploring human added value (Massey 2021) and human translators' self-concept (Kiraly 1990; 2000, Haro-Soler & Kiraly 2019 and Massey & Kiraly 2019) in the age of GenAI.

Drawing careful attention to the human aspect of added value, Massey (2021: 39) characterises human added value as being “manifest in translational decision-making and problem-solving on a conceptual level that transcends the surface lexical realisations by which meaning is conveyed in source and target texts”. He also observes that “the added value that human translators can and do bring to bear resides in their socio-cultural, socio-technical 4EA cognition, as opposed to disembodied, de-contextualised artificial intelligence” (Massey 2021: 39). In doing so, the author revisits Venuti's (2019) advocacy of the “hermeneutical model of understanding translation not as ‘the reproduction or transfer of an invariant that is contained in or caused by the source text’ but as ‘an interpretive act that inevitably varies source-text form, meaning, and effect’” and Pym's (2003) “minimalist definition of translation competence as ‘the ability to generate a series of more than one viable target text (TT1, TT2 ... TTn) for a pertinent source text (ST); the ability to select only one viable TT from this series, quickly and with justified confidence’” (Massey 2021: 51).

The delineation by Massey (2021: 39) of the added value of the human translator vis-à-vis AI strongly captures the conceptualisation of the human translator as an agent who has the hermeneutical task and power of interpreting the text and who bears the responsibility of their interpretive act. This is something that remains quite impossible for the “disembodied, de-contextualised artificial intelligence”. In the age of GenAI, we must therefore pay unprecedented attention to the translator's “self-concept” (Haro-Soler & Kiraly 2019), which includes “the image of the translator's social role”, “the translator's appraisal of his or her competency for translating a particular text” and “an understanding of *responsibility* towards the other actors in the translation context of situation” (261, emphasis mine). Consequently, we believe that the constructs of human added value and of (human) translator's self-concept should constitute the privileged foci in exploring the human-GenAI *dialogue* and incorporating the gained insights into translator education in a way that helps students raise their self-concepts (Haro-Soler & Kiraly 2019: 261-262) as human translators whose *raison d'être* is to perform the hermeneutical task of interpreting and bearing responsibility. It is with this focus in mind that the following collaborative research experiment was conducted.

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3 Design and implementation of the collaborative research experiment

The research experiment, the results of which are reported in the present study, was designed following the tenets of the collaborative research model, implemented by Haro-Soler & Kiraly (2019). It follows the social constructivist and emergentist approach to translator education (Kiraly 2000: 256), whereby the educator-researcher works collaboratively with translation students “to investigate topics in the domain of Translation Psychology with the goal of having teacher-researchers learn about the translator’s psychological ‘self’ right along with their students”. Accordingly, the research experiment presented here was collaboratively performed by graduate level translation students, who participated in the research on a voluntary basis, and a translator educator (also the author of the present chapter). The aim was to explore the dynamics of human-GenAI interaction as human translators work with GenAI tools as prompt engineers. The student-researcher group consisted of five students enrolled in the PhD programme in translation studies at Yildiz Technical University, Turkey. All had previous translation education and experience and near-native English proficiency. The educator-researcher was an associate professor of translation studies at the same university.

In the research process, the student-researchers and the educator-researcher collectively decided on the design of the task of prompt engineering in three sessions held online in June 2024. After the design was agreed upon, the student-researchers performed the task individually and reported on their learning/research processes through self-reflexive accounts of their experiences.

The initial decisions made in the recorded online discussions concerned the selection of the source texts and of the GenAI tool to be used to translate the texts through crafting and, when necessary, curating prompts (i.e. prompt engineering (Yamada 2023)) so as to pre-process, initiate, produce, edit and evaluate translations that achieved the intended quality level and functions. It was collectively decided to use the free version of OpenAI’s ChatGPT. As for the source texts, two English texts, an essay and a book description, were selected on the grounds that both texts were of mixed forms between expressive and operative text types, while also displaying the features of an informative text type (Reiss 1981: 124-125). Further, the translation of both texts demanded meticulous attention to cultural nuances and context, idiomatic expressions, tone, intent, and linguistic style. The first source text was an essay by Anna Quindlen, published in her “Life in the 30s” column in *The New York Times* (Quindlen 1987). The second source text was the book description of *The Time Regulation Institute*, the English translation of

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Ahmet Hamdi Tanpınar's Turkish novel titled *Saatleri Ayarlama Enstitüsü*, translated by Alexander Dawe and Maureen Freely and published in 2014 (Tanpınar 2014).

Next, decisions had to be made concerning the guidelines, namely the two translation briefs and the general principles of prompt engineering. First, the educator-researcher provided the student-researchers with the existing (albeit limited, due to the fact that is a relatively new topic) literature on prompt engineering. This led to an exchange of opinions in the first two online sessions. Drawing on those discussions, the educator-researcher then drafted the guidelines for the task and sent them to the student-researchers for evaluation. In the third session, the draft guidelines were discussed and finalised collectively.

The educator-researcher and the student-researchers then had to decide whether to use the finalised translation briefs as the first prompt in the dialogue with the GenAI tool. Taking into consideration the results of the experiments already conducted in other studies on prompt engineering (e.g. He 2024, Peng et al. 2023, Yamada 2023), it was decided not to include the translation briefs in prompt engineering and to use them, instead, as guidance of the human translators-prompt engineers only. In this way, the human translators were entrusted with the responsibility of the translation act despite performing this act through prompt engineering. The translation briefs for the first and second source texts were as follows:

(1) Translation Brief 1:

"The Name is Mine" is an essay by Anna Quindlen published in her "Life in the 30s" column in *The New York Times* on March 4, 1987. An online magazine website editor in Turkey needs a Turkish translation of the essay and asks you to translate it into Turkish to publish it on the website. The intended audience for the translation is primarily adult readers who would be interested in the works of Anna Quindlen, who writes primarily on feminism and family life and is a Pulitzer Prize-winning columnist, journalist and author. The editor asks you to translate the second and ninth paragraphs of the essay as a test.

(2) Translation Brief 2:

The source text is the book description of the novel *The Time Regulation Institute*, which is the English translation of the prominent Turkish novelist and poet Ahmet Hamdi Tanpınar's Turkish novel titled *Saatleri Ayarlama Enstitüsü* (1961), translated by Alexander Dawe and Maureen Freely and published by Penguin Classics in 2014. A Turkish literary

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scholar is asking you to translate the English book description into Turkish in order to examine how the English translation of the book (*The Time Regulation Institute*) is marketed to the English-speaking audience by Penguin Classics.

As for the general principles of prompting, “persona prompt” (He 2024) was selected as prompting method. This meant that prompting began with the statement “You are a professional translator” without specifying any specific domain. It was also decided that prompt engineering would include the following steps: 1) prompting the tool to analyse the source text for the purpose of translation, 2) prompting the tool to translate the source text, 3) prompting the tool to edit its translation through curating prompts where deemed necessary, and 4) prompting the tool to evaluate the quality of, and comment on, its translation. In order to foreground the responsibility of the human translator, a final step, not included within the scope of the prompt engineering task, was added: post-editing the obtained translation output through prompt engineering to produce the translation end-product. Other than these, no specific prompt statements were drafted in order to allow the human translators-prompt engineers some freedom and creativity in their individual dialogue with the GenAI tool. As a result, each source text generated five versions of prompt engineering and five translation end-products.

After completing the task, each student-researcher wrote a report on their prompt engineering experience, the analysis of which is presented below.

4 Analysis of the results: Towards deeper awareness of the power of the human translator vis-à-vis GenAI

After the experiment was completed, the student-researchers wrote retrospective process reports on their experiences working with GenAI by answering the questionnaire drafted by the educator-researcher (see Appendix A). Globally, their responses indicated that the collaborative research had allowed them to both evaluate the potential impact of GenAI on the translation profession and re-affirm their role as human agents and experts in translation.

The answers to Question 1 (“What is your general assessment of the experiment?”), indicated that most of the participants came to recognise that human intervention is still needed in several respects. One participant (P4), for instance, noted that during the experiment she “came to the realisation that human translators still have the final say”. Likewise, another participant (P3) stated that despite the “general conception” that LLMs “are already adequate for daily use or

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even technical translation at some point”, “they would never be able to replace human translators” in terms of “literary texts” and “mixed forms”. Another participant (P1) stated that human translators’ post-editing is “a necessity rather than a preference”, especially for “restructuring expressive language” and “transferring the style”. Participant 2 highlighted her frustration at the AI tool needing “continuous prompting” to edit its translation “until a final translation, which still need[ed] post-editing” was obtained. Another participant (P5) characterised the experiment as being “educating”, noting that although she had used “ChatGPT on various occasions”, the experiment was the first time that she “actually sat down and tried to generate a translation through prompting”.

The answers to Question 2 (“Upon completing the experiment, have you faced any emotional challenges as a human agent collaborating with GenAI to produce a translation? If yes, what were these challenges?”) showed that most of the participants faced emotional challenges while working with GenAI. A feeling of frustration was raised by two participants. Participant 1 stated that he got “frustrated” when the LLM did not “directly implement” his suggestions and he needed to “provide prompts again”. Another participant (P5) also stated that she “found talking to ChatGPT quite frustrating” and that she realised that she didn’t “have the required patience to guide it through everything”. One participant (P2) noted that it was “emotionally exhausting to prompt AI for non-literal translation”, especially “if a word is used figuratively” as it was “challenging to explain the function of the word and design the prompt so as to reach an acceptable translation”. Only one participant (P3) described his feeling as “constant puzzlement” rather than “frustration or anger” as the tool required “an in-depth thought process on how to write the prompts” and that he “felt like the LLM wouldn’t be able to complete the tasks” the way he “wanted it to do”. The emotional challenge expressed by one specific participant (P4) differed from the challenges experienced by the other participants in that the “feeling of stress” experienced by this participant stemmed from her underestimation of the “depth of AI’s analytical capabilities” and its “speed”. The participant (P4) also stated she “experienced a lack of confidence while providing prompts” in the translation of the first source text and “became more comfortable” in the translation of the second source text as she familiarised herself with the tool.

The answers to Question 3 (“Upon completing the experiment, what have you discovered about your power vis-à-vis AI and your self-concept as a human translator?”) revealed that the participants addressed various aspects of the power of the human translator on the basis of their experience working with GenAI. One participant (P3) described his experience by saying that it felt like being “[a] supervisor trying to teach a very intelligent new employee how to conduct the job

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and how it can improve itself in certain areas”. According to him (P3), the tool can “easily analyse the text” and “find out what kind of strategies could be needed to translate the text for the [target] audience”. However, it fails to “produce translation strategies to overcome translation problems” as it remains “too ‘faithful’ to the source text” unless the human user “specifically underlines the problems, provides strategies and solutions”. Similarly, Participant 1 emphasised the significance of the “review and intervention” of the human translator for “preserving naturalness and fluency”. According to another participant (P2), her ability “to make a hermeneutical analysis of the texts and guide AI in that respect” fostered her “self-concept as human translator” as she was the one deciding “what kind of a translation is required and when it is acceptable”. The answer by another participant (P4) highlighted the “sense of empowerment and authority” she had while working with the tool in terms of responsibility. This particular participant stated that she had asked the AI bot to take the responsibility for its translation but that it had refused to do so, stating: “[I] strive to provide accurate and high-quality translations based on the text and context provided. However, the final responsibility for assessing and using the translations ultimately rests with the user or the person overseeing the translation project”. Another participant (P5) also underlined how she had felt empowered by saying that “AI is clever and quick but it didn’t go through the life [sic] the way I did”. She also stated that AI “knows everything about the world but has never actually seen it”, which she found obvious in the translation of the essay “The Name is Mine”. It did not show any “sense of true-life experience”.

The answers to Question 4 (“Upon completing the experiment, how would you assess the impact of your previous training and experience on the way you collaborate with GenAI with a focus on your self-efficacy beliefs concerning your ability to prompt and guide GenAI to produce translations of an intended function?”), showed that most of the participants established a clear relationship between their previous training and experiences and the way they worked with GenAI. One participant (P3) stated that he believed that his previous experience and training was “vital during the experiment” and that if he had not “received any education in the field” it would be impossible for him “to guide the LLM”. Another participant (P1) stated that “providing the necessary prompt to the AI” requires “the holistic set of skills that constitute human translator competence” that can only be acquired by “translator training”, while another participant (P2) emphasised the significance of the knowledge of “text types and purpose of a translation” in effective prompt engineering. Another participant (P5) stated that thanks to her previous training and experiences, especially in working with and evaluating “automatic translation”, she “was able to spot what does not sound

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right as the translator and categorise these improvable aspects”. One participant (P4), on the other hand, stated that she had “come to understand that interacting with an AI is distinct from engaging with a human being”. It required skills other than the “social skills” needed to interact with fellow human beings.

The answers to Question 5 (“Upon completing the experiment, do you have a clearer idea of the strengths and weaknesses of GenAI tools in terms of producing translations of intended function?”) revealed that speed was the most appreciated feature of the tool. Yet, despite this, participants highlighted the tool’s “weakness in terms of fluency” (P1) and its “inability to provide adequate translation outputs without feedback from a professional” (P3). One participant (P5) stated that the tool is able to provide “usable output in many ways when guided accurately” but that “it is not completely reliable” given that it is not fully possible to make the tool “achieve what one has in mind”. Another participant (P2) specifically referred to the text analysis ability of LLMs as a strength. Similarly, another still (P4) described the ability of the tool to “deliver rapid and deep analysis” as its “most notable strength”; however she also added, referring to the essay “The Name is Mine”, that the LLM was “unable to convey the feminist perspective” in its translation no matter how hard she tried to prompt the tool in that direction.

The answers to Question 6 (“Upon completing the experiment, how would you assess the contributive potential (if any) of GenAI tools for professional translation workflows?”) revealed that the participants mostly highlighted the time-saving benefits of GenAI. One participant (P1) argued that “refining translations produced by GenAI through post-editing can save time” while another participant (P2) drew attention to the ability of the GenAI tools to yield “faster translation outputs compared to a fully human translation”. Another participant (P3) described GenAI as a “great tool” as it is able to “efficiently analyse a text in detail”. The role GenAI can play in the translation pre-process was also underscored by another participant (P4), who stated that “AI can enhance the first stage in the translation process by offering comprehensive analysis to facilitate translation” and that it can assist the human translator “in the decision-making process by supplying necessary information”. One participant (P5), on the other hand, expressed her reservations. According to her, GenAI “won’t be able to be professionally integrated into workflows until it is tailored to only translation projects”.

The answers to Question 7 (“Upon completing the experiment, how would you define prompt engineering as an emergent job position for the human translator and the skills it requires?”) showed that the participants had gained a clear understanding of the requirements of prompt engineering as an emerging job position. Participant 5 stated that “the understanding of the translation process”

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is required to “excel in prompt engineering” together with “translation, language and technology competence” and “AI literacy”. Another participant (P3) stated that he believed that many human translators will become prompt engineers as “most LLMs can be customised according to different needs” and that such people will be necessary for “customising” and “upgrading” LLMs. Prompt engineering was regarded as “an essential and emerging role for human translators” by Participant 4, who also noted the importance of the re-consideration of “the effectiveness of communication skills in the context of prompt engineering” as giving prompts is “directly linked to the communicative abilities” of a human translator. One participant (P1) stated that without the human translator’s skills in “reading, understanding and analysing a text for translation purposes” and the skills in “reformulating a text in the target language in a way that it is appropriate for the purpose”, it would not be possible “to design the necessary prompts to guide the AI effectively”. The significance of prompt design was also highlighted by a fellow participant (P2), who argued that apart from his/her existing skills, the human translator is required to develop his/her skills in prompting through experience.

The answers to Question 8 (“Upon completing the experiment, what factors do you believe should be taken into consideration when integrating GenAI tools into translator education?”) displayed various considerations. Participant 1 highlighted the importance of “instilling in students the awareness that the importance and necessity of the human translator’s added value remain undiminished” as the human translator “retains the role as the expert and bears ultimate responsibility for the target text”. Another participant (P2) highlighted the necessity to “equip translation students with necessary qualifications to make efficient use of GenAI tools” and to encourage them “to test the full potential of GenAI tools to participate in the technology development phase as well”. Another still (P3) focused on the issues of “accuracy” and “bias”, which are frequently associated with LLMs, and stated that translation students “need to be able to critically evaluate AI-generated content and to efficiently conduct fact-checks, cultural appropriations, fitness for scope, etc.” as LLMs “can generate outputs with factual errors, cultural insensitivity, or biases reflecting the data they were trained on”. A similar concern can be found in the answer of another participant (P5), who stated that the bot’s “strengths and weaknesses and limitations (potential biases in training data, the possibility of factual errors, and the inability to capture cultural nuances) should be taken into account” when integrating its use into translator education. Another participant (P4) stated that, based on her experience during this experiment, “it is advisable to incorporate AI into translation-oriented text analysis courses”. She also added that students need to be informed about the

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“potential disadvantages of relying only on AI, since it basically tends to produce ‘literal’ translations”.

The answers to Question 9 (“Upon completing the experiment, what competences do you believe translation students should have to become experts of GenAI translation tasks?”) indicated that all participants agreed on the relevance of the already established translator competencies and skillsets for human translators to become experts of GenAI translation tasks. Additionally, one participant (P2) drew attention to the emotional aspect involved in working with GenAI and stated that it is necessary for students to learn to “control their emotions especially in the dialogues with the AI tools” because “by giving agency to the AI tool, the user attributes human features, which makes it even harder to control especially negative emotions”. Likewise, another participant (P4) stated that “the competence to communicate with the AI”, “emotional competence” and “post-editing competence” are fundamental and that “the quality of the translation can be unavoidably influenced when there is a communication gap between a human translator and the AI”. Another participant (P3) stated that students should be familiar with LLMs so that “they comfortably use different LLMs for translation, be aware of LLM-related problems such as LLM hallucination or limitations of each LLM”. According to another participant (P5), “critical-thinking skills” and “creative skills” are needed in addition to “technological competence”, “editing competence”, “cultural competence”, and “research competence”.

The answers to Question 10 (“Upon completing the experiment, what competences do you believe translator educators should have to educate students to become experts of GenAI translation tasks?”) showed that participants saw a parallel between the competences of required of students and those required of the educators (P4 and P5) concerning the integration of GenAI into translator education. According to one participant (P1), translator educators need to engage in a “learning process themselves” by “closely following the developments in the field of AI”. Other participants stated that educators should be “users of the state-of-the-art translation technologies” (P2) and should be “able to design courses that combine traditional translation practices with AI-assisted translation workflows” (P3).

Finally, only one participant (P4) provided an answer to Question 11 (“Other comments”), in which she stated that the experiment helped her “identify” her “strengths and weaknesses as a human translator” as well as those of GenAI. She also stated that “communicating and cooperating with the AI” was more “challenging” than she had anticipated and that “upon completing the experiment”, she identified herself “as a *beginner but a quick learner* when it comes to dancing with the AI” (P4, original emphasis).

5 Discussion and conclusion

On the basis of the above analysis of the participants' answers to the questions, it can be claimed that the experiment provided the student-researchers with the opportunity to work with GenAI to produce a functioning target text according to a given translation brief and, in doing so, to evaluate certain aspects of the human-GenAI dance through an experiment-based diagnosis of the strengths and weaknesses of GenAI.

In the accounts of the participants, the main strength associated with GenAI was, rather expectedly, speed. Apart from this, participants placed special emphasis on the text analysis abilities of GenAI, which were considered to be a significant factor that can facilitate and improve the translation process. Notable weaknesses were GenAI's inability to produce a stylistically adequate and fluent translation, its need for continuous prompting from a human translator (i.e., its failure to implement text-specific translation strategies without guidance from a human translator), its potential to generate biased, cultural insensitivity and/or factually inaccurate output which can only be fixed by the careful monitoring and evaluation of a human translator. Last but not least, GenAI's lack of a true-life experience was also mentioned.

Further, participants' accounts also highlight that, during the course of the experiment, the student-researchers re-evaluated their human added value and self-concept as trained and experienced translators. Despite their familiarity with, and experience of, the dynamics of human-machine interaction during translation, they recognised the necessity of engaging in a new form of collaboration with GenAI. Indeed, the participants' accounts of their own strengths and weaknesses were closely related to the strengths and weaknesses they had identified for LLMs. The need for a professional human translator to intervene (edit and evaluate) the raw output of GenAI tools to it into a functioning translation was emphasized by all participants. This echoes the findings of recent studies on the necessity of a human translator's intervention due to the limitations of GenAI (e.g. [Alimen 2023](#), [Öner Bulut & Alimen 2023](#)). Thus, the process by which the students adjusted the LLM output through prompt engineering mirrors the hermeneutic role of the human translator, who interprets the 'meaning' of a text and reframes it appropriately for the target audience> This underscores that GenAI-assisted translation is not merely mechanical but requires deep cultural adjustments and human judgment.

Participants also highlighted the sense of empowerment and authority they felt as decision-makers, who must make decisions concerning translation strategies and prompt the GenAI tool accordingly. This sense of responsibility brought

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them power as the ones responsible for the final translation output. This aligns with the ethical and hermeneutic roles of the human translator. That being said, some of the participants also addressed areas of self-improvement such as developing skills to communicate with GenAI, which is distinct from communicating with a human being, and designing and curating effective prompts to guide AI through the translation process without becoming impatient and stressed. This gives hints about the possible emotional challenges human translators can face while working with GenAI, evidencing the need for “introducing an emotional intelligence dimension into translator training” (Penet & Fernandez-Parra 2023:349). This is something we believe should be addressed in future research on the integration of GenAI into translator education.

The analysis of the results also provided insights into other factors that need to be considered towards this. All student-researchers acknowledged the significance of their previous education and experience in performing the emerging task of prompt engineering. They also agreed that the holistic set of skills of a trained professional human translator is a prerequisite for effective prompt engineering. This had already been suggested in a recent study on the role of human competencies in the “tech-driven language services industry” (Öner & Bengi 2024:92). Yet most of the participants noted that, in addition to this prerequisite, translation students should learn to communicate effectively with GenAI (i.e., prompt it), to critically evaluate and edit its output and to develop an awareness of the fact that human translators are still the ultimate authority with the right to the final say by virtue of the trust and responsibility invested in them. This particular perspective underscores the importance of acknowledging translators’ active role in reconstructing cultural value and meaning within the interactive translation processes facilitated by GenAI. Further, this also points to the significant potential of hermeneutics-based education in fostering a deeper understanding of the human translator’s role in GenAI-assisted translation.

This, in turn, places greater responsibility on translation educators, who, according to the participants, need to become dedicated learners and users of state-of-the-art GenAI technology so that their translation classrooms become a safe space where translation students learn to communicate and dance more confidently with GenAI.

These experience-based insights, which were gained through a collaborative research experiment, shed light on the emerging opportunities, as well as challenges, for re-positioning the human translator as the future expert of GenAI translation in the classroom. They also show a possible path to the fulfilment of the translator educator-researcher’s - also emerging - task of re-evaluating *what*

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is/will remain human in translation and incorporating it into education in the age of GenAI.

Appendix A Instructions for process report writing

Please write your report by answering the below questions.

1. What is your general assessment of the experiment?
2. Upon completing the experiment, have you faced any emotional challenges as a human agent collaborating with GenAI to produce a translation?
3. Upon completing the experiment, what have you discovered about your power vis-à-vis the AI and your self-concept as a human translator?
4. Upon completing the experiment, how would you assess the impact of your previous training and experience on the way you collaborate with GenAI with a focus on your self-efficacy beliefs concerning your ability to prompt and guide GenAI to produce translations of intended function?
5. Upon completing the experiment, do you have a clearer idea of the strengths and weaknesses of GenAI tools in terms of producing translations of intended function?
6. Upon completing the experiment, how would you assess the contributive potential (if any) of GenAI tools for professional translation workflows?
7. Upon completing the experiment, how would you define prompt engineering as an emergent job position for the human translator and the skills it requires?
8. Upon completing the experiment, what factors do you believe should be taken into consideration when integrating GenAI tools to translator education?
9. Upon completing the experiment, what competences do you believe translation students should have to become experts of GenAI translation tasks?
10. Upon completing the experiment, what competences do you believe translator educators should have to educate students to become experts of GenAI translation tasks?
11. Other comments

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Chapter 8

Computer-assisted language mediation in teaching human-centred augmented translation

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In the evolving landscape of translation education, the integration of computer-assisted language mediation has emerged as a key strategy for teaching human-centred augmented translation. This study explores the implementation of generative AI (GenAI) models, in particular Large Language Models (LLMs), in translation workflows to enhance translator competence while maintaining ethical standards. Using custom AI systems that adapt and evolve with new translations, students can be trained to critically evaluate AI-generated content and manage hybrid translation workflows that combine AI output with human expertise. The research emphasises the importance of understanding the limitations and potential biases of AI, and advocates a balanced approach in which AI augments rather than replaces human judgement. Practical examples and exercises demonstrate the strengths and weaknesses of using AI in information processing and multilingual text generation, with the ultimate aim of training a new generation of translators to use AI technologies responsibly.

1 Introduction

1.1 GenAI in translation education

In translation education, integrating generative AI (GenAI) models into translation processes presents challenges in maintaining translator autonomy while embracing technology (O'Brien 2024). Ethical concerns, including privacy, algorithmic bias, and responsible use of machine translation and linguistic data,

Maria Zimina-Poirot. 2025. Computer-assisted language mediation in teaching human-centred augmented translation. In JC Penet, Joss Moorkens & Masaru Yamada (eds.), *Teaching translation in the age of generative AI: New paradigm, new learning?*, 147–165. Berlin: Language Science Press. DOI: ??

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must be carefully considered as translation education undergoes a paradigm shift (Ramírez-Polo & Vargas-Sierra 2023, Slimi & Carballido 2023, Tavares et al. 2023).

Developing computer-assisted language mediation in translation education requires successfully integrating data analysis and AI assistants into pedagogical workflows. In this respect, AI is one of the building blocks of translation technologies, along with computer-assisted translation environments, project management software, ontologies, expert involvement and data management facilities (Garcia 2023, He et al. 2023, Mitchell-Schuitevoerder 2020).

1.2 Computer-assisted language mediation

Used as part of a computer-assisted dynamic mediation framework (Hu et al. 2024), GenAI tools allow translation students not only to produce multilingual content, but also to compare and fine-tune the generated output, fostering a deeper contextual understanding of the strengths and weaknesses of different translation models and enabling future translation professionals to become more proficient reviewers of automatically generated content (Zhuang et al. 2024). This process requires new translation skills and up-to-date knowledge of AI assistants built on underlying AI models, such as Large Language Models (LLMs) (Douglas 2023).

As LLM technology is introduced to the field of translation, enabling trainees to navigate language variations within adaptive pathways, the concept of a linguistic norm emerges as an increasingly crucial analytical tool, facilitating efficient translation across diverse contexts (Fabricius 2022, Schulz & Ollig 2023, Sinner 2020). From this perspective, computer-assisted language mediation can be taught as a set of new translation competences, including the ability to create and use a language assistant, and potentially to understand how to employ LLM-enhanced linguistic data responsibly (Raza et al. 2025).

From a technical standpoint, building personalised AI assistants from scratch requires substantial resources, emphasising the efficiency of leveraging pre-trained models (Shichkina & Krinkin 2022). This process can be part of training programs specifically tailored to prepare data scientists specialised in translation. Few-shot learning (FSL),¹ particularly in-context learning (ICL),² empowers

¹Few-Shot Learning (FSL) is a machine learning approach that enables models to adapt to new tasks with only a small amount of labelled data (Mosbach et al. 2023). For more information: <https://medium.com/ubiai-nlp/step-by-step-guide-to-mastering-few-shot-learning-a673054167a0>

² In-context learning, as popularised in LLMs such as GPT-3 and GPT-4, involves a model's capacity to comprehend and execute tasks based on contextual information embedded within the input sequence, without modifying the model's parameters (Dong et al. 2024).

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translators with limited natural language processing (NLP) expertise. This approach allows LLMs to rapidly acquire new capabilities, thereby facilitating seamless integration of AI into a range of translation workflows. In this regard, prompt engineering³ facilitates the development of a versatile approach that can be applied to a range of tasks, including multilingual translation, fine-tuning of translation models (Doğru & Moorkens 2024), information extraction, and text summarisation. Training future translators to effectively prompt LLMs for these tasks in digital communication is essential. Our analysis in Section 3 will provide specific examples that illustrate these approaches in teaching.

1.3 Investigating biases and inconsistencies of GenAI

An important area of research in teaching GenAI technologies in translation courses is to make students aware of the potential biases and inconsistencies they may encounter when using this new generation of tools. The shortcomings of AI-generated content become apparent when the translation task is fully mastered. Unfortunately, if students are not fully proficient in the target language and lack specialised knowledge and translation practice, they may find it difficult to effectively assess the quality of the output (Tavares et al. 2023). Understanding both the strengths and limitations of GenAI through a series of practical examples is vital for efficient use of translation tools (Farrell 2023).

By exposing students to real-life scenarios and practical exercises, it is possible to develop a better understanding of the qualities and shortcomings of automatically generated content in the context of multilingual communication. This hands-on approach allows trainee translators to identify potential biases, inconsistencies and errors that may arise from relying solely on AI assistance. In addition, by analysing and discussing these practical examples, students can learn to critically evaluate the output of AI assistants and identify when human intervention or additional expertise is required to produce accurate and contextually relevant content. This knowledge is also essential for developing a new generation of translation-revision workflows that successfully combine GenAI and human expertise.

³The field of prompt engineering is concerned with the design and optimisation of prompts to help a model perform better on new tasks with limited data. For prompts to be effective, they must clearly define the task at hand, include context to help the model use its existing knowledge, and gradually introduce new information or tasks.

2 Contextual understanding and automation

Artificial intelligence is now capable of processing a wide variety of information, including text, images, speech, facial expressions, and, to a certain extent, body movements. These capabilities can be used to develop a large panel of AI assistants (Yang et al. 2024). One of the most intriguing aspects of these technologies is their handling of context, which differs significantly from human understanding (O'Brien 2024, Zhu et al. 2024).

Human understanding involves a combination of experience, emotion, cultural background and situational awareness. This allows people to naturally interpret nuances, implicit meanings and subtleties in communication. Assessing human understanding of context through written expression has long been a focus of research, with methodologies evolving to encompass cognitive, linguistic, and technological perspectives (O'Brien 2024). Researchers in cognitive psychology and linguistics have devoted considerable time and effort to studying the mechanisms of language processing, with a particular focus on the role of lexical access, syntactic parsing and discourse coherence in comprehension. In addition, discourse analysis has provided valuable insights into how contextual cues shape interpretation, revealing complex interplay between language and cognition in human comprehension (Van Dijk 2006, Pleyer & Winters 2015, Gledhill & Pecman 2018).

In contrast, GenAI models process context through the identification of patterns within the data they learned during the training phase (Zhuang et al. 2024). In contrast to a true understanding of context, these models rely on vast quantities of data to identify and generate responses based on statistical correlations. To illustrate, when presented with a sentence, an AI assistant analyses the sequence of words and predicts the most probable continuation based on the patterns it has learned. Currently, this approach can effectively imitate human-like responses, but it lacks genuine comprehension and the capacity to grasp more profound meanings or unexpected cultural nuances. Additionally, it tends to accentuate recognised patterns in the generated output.

This difference in contextual understanding has significant implications for translation. While AI may be good at generating coherent and contextually appropriate text, it may struggle with tasks that require cultural sensitivity or complex problem-solving that requires nuanced human judgement. Furthermore, AI's understanding of context is limited to the data on which it has been trained. If training data sets lack diversity or are biased, AI-generated content will reflect these limitations (Martínez et al. 2023). Understanding these differences is

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crucial when integrating GenAI assistants into translation workflows. While AI-based technologies can significantly increase productivity and efficiency by performing routine tasks and providing quick information, they should complement rather than replace human judgement and expertise in translating certain types of content (Denning & Arquilla 2022).

Balancing the strengths of AI in processing and generating text with advanced human skills used to understand and interpret complex contexts can lead to more efficient and nuanced applications of translation technology for translation project management, known as *human-centred augmented translation* (O'Brien 2024).

In practical terms, what specific tasks can help students understand GenAI weaknesses as reflected in the warning “ChatGPT can make mistakes. Check important info.” displayed by ChatGPT (<https://chatgpt.com>)? In what types of situations are these tools helpful, and what are the potential pitfalls? The following section explores the point of studying biases and inconsistencies in GenAI operation through practical examples of classroom activities.

3 Exploring AI in practice: delving into knowledge-rich texts

3.1 Text summarisation and interactive learning with AI assistants

In this section, we consider how practical tasks can help students gain a deeper understanding of GenAI. One of the easiest ways to address these complex processes in translation education is to use examples of knowledge-rich texts (Gledhill & Kübler 2016). In this context, information synthesis is one of the key capabilities that AI brings to streamline information processing, with tools such as SciSpace and PDFgear being able to summarise key information almost instantly for efficient retrieval.⁴ In this respect, linguistic standards, specifications and style guides provide an excellent basis for practice (Gledhill & Zimina-Poirot 2023). These well-structured documents precisely outline the standards that should be applied in written communication. At the same time, the length and complexity of these documents often require detailed human analysis and extensive practice to fully master the rules.

To illustrate how AI can be used in teaching, let us consider a specific example. In a Master's level 2 course on technical translation and controlled languages, stu-

⁴SciSpace is an advanced platform designed for researchers that uses natural language processing and machine learning to facilitate efficient literature review and analysis in scientific fields.

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dents are expected to familiarise themselves with ASD-STE100 (Simplified Technical English).⁵ ASD-STE100 is an international specification for writing technical documentation in a controlled natural language.⁶ This comprehensive document of over four hundred pages requires considerable effort to master the principles of STE for writing procedures, descriptive technical texts, warnings and cautions. In addition, there is an extensive “core dictionary” containing a wide range of lexical items that students must learn to use in context. The dictionary is accompanied by illustrative examples and contrasting cases (APPROVED: STE/ not approved: non-STE).

Integrating AI into ASD-STE teaching can potentially improve the learning experience. The following examples highlight several ways in which this can be achieved.

- Use case: Use tools such as PDFGear or SciSpace to quickly summarise lengthy sections of ASD-STE100.
- Benefit: By using a concise format in classroom settings with limited teaching time, students can focus on understanding and practical applications of the STE rules rather than being overwhelmed by the volume of text.

In this context, AI assistants can act as interactive tutors, answering students’ questions about specific language policies and providing immediate feedback. Prompt clarification and reinforcement can accelerate learning. In addition, AI assistants can quickly locate specific sections or rules within lengthy documents through enhanced research and reference, saving time and improving efficiency. The main benefit is that students can spend more time applying the rules and less time searching for them.

At the same time, through this type of exercise, students will realise that both PDFGear and SciSpace copilots are quite effective at extracting relevant knowledge and providing a comprehensive list of essential rules to be applied in writing. Nevertheless, this knowledge extraction is based on patterns in the text and may be superficial when it comes to applying these rules in writing and rewriting

⁵See curriculum for Master 2 ILTS, Université Paris: <https://odf.u-paris.fr/fr/offre-de-formation/master-XB/arts-lettres-langues-ALL/traduction-interpretation-K6JMSAFS/master-traduction-interpretation-parcours-industrie-de-la-langue-et-traduction-specialisee-JRQNFV5Z.html>

⁶ASD-STE100 (<https://www.asd-ste100.org>) is a set of writing rules and a controlled vocabulary for producing clear and simplified technical documentation in English. It was developed in the 1980s by the aerospace industry associations (AECMA/ASD and AIA) to make aircraft maintenance documentation more readable for people with limited English proficiency.

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texts. Students can rapidly become aware of these limitations and gain insight into the technical aspects of producing content with GenAI. This dual awareness can help them to understand both the potential and the limitations of using AI assistants in the context of technical communication and translation:

- Use case: Use PDFgear and SciSpace copilots to generate examples and counter-examples based on the core dictionary and ASD-STE100 rules, then critically evaluate the AI's output.
- Benefits: Firstly, this type of exercise encourages active learning and critical thinking as students compare AI-generated content with examples of human rewriting. Secondly, it provides insight into the limitations of automatic text generation in the practical application of writing rules.

The following section will demonstrate how previous exercises can be further developed into a more conclusive experiment, using tools such as ChatGPT, Perplexity AI, or Claude.ai.⁷ In this case, a prompt language model (input in Technical English and output in Simplified Technical English) would be employed to demonstrate the outcomes of few-shot learning on a specific data set. In addition, prompts can be used to highlight inconsistencies in the generated output, thereby directing attention towards potential enhancements (such as the avoidance of noun clusters and the use of articles encouraged by STE).

3.2 AI output refinement using prompt engineering and few-shot learning

From a technical perspective, building LLMs from scratch requires significant resources, which highlights the efficiency of using pre-trained models. This process can be integrated into teaching programs specifically designed to train data scientists specialising in translation. Few-shot learning, especially in-context learning, enables translators with limited NLP expertise to quickly acquire new skills and seamlessly integrate AI into translation workflows across multiple applications.

In the context of few-shot learning, a model is presented with a limited number of labelled examples corresponding to a desired input-output scenario. Transfer learning principles are embedded in the model to enable it to use these examples

⁷ChatGPT is a conversational AI model developed by OpenAI based on the GPT (Generative Pre-trained Transformer) architecture. Based on OpenAI's model, and a standalone LLM with NLP capabilities, Perplexity AI is a conversational search engine that answers queries using natural language predictive text, using sources from the web. Claude.ai is trained by Anthropic using Constitutional AI (Bai et al. 2022).

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to adjust its parameters and fine-tune its representations to enhance the presence of specific patterns. The main advantage of few-shot learning is its ability to generalise from limited examples to unseen data, due to the rich abstract representations acquired during pre-training. However, even in this scenario, the results may not be entirely satisfactory and may require human refinement.

- Use case: Create a prompt language model with labelled examples of “Input text in Technical English” and “Output text in STE”, demonstrating the results of few-shot learning on a given data sample.
- Benefit: This hands-on exercise allows students to interact directly with the technology and gain practical experience using prompt language models for text generation tasks.

In addition, this approach provides an opportunity for students to observe the limitations of AI in fully capturing complex linguistic nuances, highlighting the importance of human analysis and editing to refine the output for satisfactory results.

3.3 Terminology extraction with LLMs: understanding model limitations

In parallel, other initiatives are exploiting the capabilities of combining AI and NLP to facilitate access to specialised knowledge. Teaching terminology extraction from multimodal sources using LLMs is one of the areas which can reinforce learners’ augmented reasoning in translation.

Before using LLMs with multimodal capabilities (such as Perplexity AI playground: <https://labs.perplexity.ai> or <https://gemini.google.com>) for the analysis and extraction of terms from multiple sources, it is beneficial for students to contrast the outcomes generated by LLMs with those produced by conventional term extraction, text-mining, and text annotation tools. This comparison enables learners to become aware of potential biases and hallucinations. For example, the frequencies of candidate terms identified by GenAI are not always 100% accurate compared to traditional text segmentation and term extraction methods. AI models can propagate errors from misinterpretations in the text, resulting in inaccurate term frequencies. Misreading sentence structure can lead to incorrect counts of certain terms (Uchida 2024).

In addition, AI models trained on diverse and large datasets may inappropriately apply patterns learned in one context to another. For example, if a model

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has frequently seen “climate change” discussed alongside terms such as “policy” and “regulation”, it might overestimate the frequency of these terms in any document mentioning “climate change”, even if the specific text focuses on scientific data rather than policy.

These errors highlight a key limitation of AI models: their reliance on patterns and associations learned from vast amounts of data, which can sometimes lead to misinterpretations if the specific context of a new text differs from those patterns. The generated output exaggerates the learned patterns. As a result, the frequency of terms identified by GenAI may not always accurately reflect the actual text and its context, especially in cases where a nuanced understanding of language is required. Therefore, comparing AI-generated results with those of traditional text segmentation and term extraction tools, which often use straightforward statistical methods, can help identify and correct such inaccuracies.

- Use case: Use LLMs to extract terms from different types of documents. Check the extracted terms for accuracy and relevance. Use text-mining and navigation tools (such as Voyant Tools: <https://voyant-tools.org>) to critically evaluate the results produced by AI assistants.
- Benefit: Learn to use quantitative methods to evaluate the quality of AI-generated text analysis.

Note that ready-made prompts for text analysis can be found on many websites.⁸ Such models can be adapted, for example, for term extraction (works for Perplexity AI):

- Use case (sample prompt): Extract the most relevant terms with contexts from the attached document. Present the results in a markdown table with the following columns: term, frequency, contexts, definition.⁹

3.4 Towards “augmented reasoning” in translation: linguistic data and LLMs

Many other experiments with AI tools are possible in the classroom, both to demonstrate the role that AI models can play in different application settings,

⁸For example, Keywords Everywhere ChatGPT Prompt Templates: <https://keywordseverywhere.com/chatgpt-prompt-templates.html> (also available for Claude.ai).

⁹More examples are available in #2024TEF -AI-powered terminology extraction: A hands-on guide for translators by Josh Goldsmith: <https://youtu.be/5Y5PhzyeMGI?list=PLLqIRaiVCGCR80tysPOQ2AHJ30HEyItXX>

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such as text simplification, paraphrasing, answering questions, and generating synthetic data to automate translation tasks. Furthermore, students can conduct experiments to explore and address biases in word embeddings,¹⁰ which play a pivotal role in the functioning of LLMs.

In addition, corpus linguistic tools and methods can be used to identify recurrent characteristic patterns in source language texts and comparable target language texts. This approach enhances the performance of LLM-based translation assistants by leveraging targeted datasets containing these characteristic elements. For example, translation students can be trained to identify comparable sentences in multilingual annotated corpora (treebanks) and learn how to prompt LLMs to translate these sentences correctly within a given linguistic context.

Moreover, in this scenario, human-centred information retrieval becomes essential for identifying comparable texts in the target language. To support this, corpus linguistic tools and methods, such as co-occurrence network analysis and syntactic dependency annotation, can be employed to extract specific patterns that capture terminological, phraseological, and syntactic features. These patterns are then integrated into LLM training for translation purposes, potentially through the implementation of a translation bot, such as one designed to write bilingual definitions.

Developing such skills requires creativity and non-standard thinking, offering many potential applications in the language industry, particularly in managing data intelligence platforms.¹¹ Consequently, GenAI can become a crucial component of translation technology. However, it must be carefully monitored by experts and combined with reliable data management facilities.

Finally, a promising area of educational research lies in the development of augmented human translation in translation pedagogy. This approach aims to foster a new competence called ‘augmented reasoning’ in translation, which combines human intuition with robust data support to optimise translation processes (EMT 2022).

¹⁰An embedding is a vector representation of data designed to capture and preserve aspects of its content or meaning. For more information on bias detection in word embeddings and LLMs for European languages: https://www.biasproject.eu/scientific_publications/the-bias-detection-framework-bias-detection-in-word-embeddings-and-language-models-for-european-languages

¹¹Databricks: <https://www.databricks.com>

4 Hybrid translation workflows: enhancing human expertise with GenAI

4.1 Augmented cognition: challenges in translation education

In translation education, the integration of LLMs into translation poses challenges in terms of maintaining learner autonomy and developing translation competence while taking advantage of technological advances. Maintaining learner autonomy means emphasising that GenAI assistants should be seen as tools, rather than as a substitute for human translators. Learners should retain control over the translation process and engage in critical evaluation of the output generated by automated translation systems, thereby developing skills in assessing quality and appropriateness (Gledhill & Zimina-Poirot 2023). Furthermore, learners must learn to provide clear instructions to LLMs, which is crucial for effective human-machine collaboration.

As translation education undergoes a paradigm shift, it is becoming increasingly important to encourage responsible use of machine translation and linguistic data, with careful consideration of ethical concerns such as privacy, data education and algorithmic bias. To ensure effective use of LLMs in translation processes, it is essential to adopt a comprehensive approach, including seamless integration of technology into pedagogical workflows through computer-assisted language mediation.

In the past, efficient interaction with machines and data navigation techniques also relied on human intuition backed up by data, but implied a clear distinction between quantitative information, which can be measured and counted, and human judgement (qualitative analysis) (Zimina & Fleury 2016). The realisation of augmented cognition requires a stronger connection between the user and the system, which should progressively learn to understand human functioning and respond effectively to situational demands. From this perspective, personalised AI plays an important role in facilitating this integration.

The convergence of human-machine relationships is a major challenge for translation professionals, spanning academic disciplines such as translation studies and the translation industry (Raus et al. 2023). The emphasis on augmenting intelligence rather than replacing humans marks a shift from emulation to empowerment, which is a progressive way forward.

In the industry, the foundations of this approach lie in hybrid translation workflows that seamlessly combine human expertise with AI capabilities, rather than relying on a distinct separation between qualitative analysis and automatically generated content, which is then edited and revised by a human expert. In the

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coming years, acquiring expertise in the design of advanced translation workflows that integrate innovative methods and technologies will become a key competence in Master's level translation education. In the context of industrial applications, tools that facilitate such workflows, such as those available in the RWS Language Cloud,¹² will be enhanced with upcoming GenAI features. For example, RWS's linguistic AI solution EVOLVE aims at "integrating human and artificial intelligence, revolutionising translation processes and the time it takes to achieve high quality results".¹³

The following section offers a brief overview of the principles behind a hybrid translation workflow that integrates elements of Dynamic Translation Memory (DTM) and leverages dynamic human-AI interaction in project-based translation.

4.2 Dynamic Translation Memory (DTM) in project-based translation

Unlike traditional Translation Memory (TM),¹⁴ a Dynamic Translation Memory (DTM) is part of custom AI systems that not only store translations, but also adapt and evolve, learning from new translations and adapting to different contexts within a project-oriented translation workflow (Hao & Pym 2023, Kim et al. 2022, Sin-wai 2023, Zimina 2015, Zimina & Fleury 2016, Zimina 2019).

Recent developments in LLMs have demonstrated their capacity to outperform traditional neural machine translation models in certain contexts and language combinations. For example, a Lionbridge assessment (2023)¹⁵ showed that certain GenAI models can perform better than neural machine translation engines, challenging the established paradigm in this sector. At the same time, combining neural machine translation with neural fuzzy adaptation (NFA) is a promising way to achieve flexible control over translation resources while avoiding potential biases inherent in general purpose LLMs.¹⁶

Given these technical advances, a DTM system based on AI technologies differs from traditional TM systems in that it not only stores translated text, but also adapts and evolves in response to new translations and changing contexts within an AI-enabled translation workflow. The development of new translation management skills is therefore essential in this evolving landscape.

¹²<https://www.rws.com/localization/language-cloud>

¹³<https://www.rws.com/fr/about/news/2024/evolve-by-RWS-transforms-global-translation-practices>

¹⁴<https://www.trados.com/learning/topic/translation-memory>

¹⁵<https://www.lionbridge.com/fr/blog/translation-localization/machine-translation-a-generative-ai-model-outperformed-a-neural-machine-translation-engine>

¹⁶<https://www.systransoft.com/translation-products/translation-memories>

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To illustrate some practical applications of this approach in teaching project-based translation at the Master's level, a number of specific examples are required. Let's consider the task of implementing a dynamic workflow where both TM and LLM interact. The following examples illustrate the range of related processes and tools/services that are currently available on the market for designing a dynamic workflow:

- Leveraging TM in a dynamic workflow (segment analysis). For each segment of the source text, the TM software checks for matches. If an exact or high confidence fuzzy match is found, it is such as user dictionaries and TM (available, for example, on the SYSTRAN Translation Server) (for practical examples: [Zimina-Poirot & Gledhill 2023](#)).
- If a fine-tuned translation model is not available, it is possible to use LLM integration with Adaptive Generative Translation (AGT).¹⁷ For segments without customised NMT output, LLM integration allows a translation to be generated. This can be done in real-time or batch mode. For example, memoQ AGT uses the in-context learning capability of LLMs to generate more consistent translations using in-context resources (TM and glossary), similarly to [Moslem et al. \(2023\)](#).¹⁸
- Translation consistency checks. LLMs can be used to ensure consistency across multiple documents or versions. In this case, translated texts (bi-texts) can be fed into AI assistants, such as Poe (<https://poe.com>), to analyse terminology, phrasing, and style consistency. Prompt example: Identify and resolve inconsistencies to ensure consistency across translations. Produce a report highlighting areas of inconsistency and providing recommendations for improvement.
- Human post-editing and revision. Translators review, post-edit and revise the combined output to ensure consistency and quality. At this stage, an assessment report quantifies the revisions and highlights the strengths and weaknesses of the translated content.¹⁹
- Overall translation quality assessment and evaluation. Various assessment techniques are employed in the industry for translation evaluation, and these methods are increasingly enhanced with AI-based features.²⁰

¹⁷<https://agt.memoq.com>

¹⁸<https://www.memoq.com/resources/webinars/memoq-agt>

¹⁹<https://www.rws.com/localization/services/language-quality-assurance>

²⁰ A good starting point for an analytic summary can be found here: <https://videotap.com/blog/7-translation-quality-assessment-techniques>

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These industrial practices are key building blocks of a hybrid workflow that leverages AI-suggested output to enhance translation project consistency and efficiency. However, the true success of such project workflows lies in seamless integration of machine-driven efficiency and human oversight. In this respect, revision emerges as a crucial phase in ensuring semantic integrity and contextual relevance. While AI systems can flag inconsistencies, suggest improvements, and support adherence to glossaries and style guides, human intervention is essential to complement these tools. This evolving landscape highlights the importance of equipping trainee translators with both revision skills and the ability to design and implement workflows that ensure high-quality output at every stage.

4.3 Leveraging revision history in AI-assisted translation training

Trainees must actively participate in project management, navigating the intricate cycles of post-editing and revision while leveraging AI tools effectively. In a structured translation/revision workflow, the revision process is ‘evolutive’, meaning that modifications should not be random. With the exception of occasional edits, changes should result from a complex interplay of various editing constraints (for example, writing in Plain Language). This underscores an ‘evolutive’ perspective on the translation project: each stage of translation and revision must be carefully planned in advance, in collaboration with both clients and students (Gledhill & Zimina-Poirot 2023).

From this perspective, a revision history archive becomes a valuable resource in AI-assisted translation training, offering a comprehensive record of the edits and decisions made throughout the translation process. The analysis of revisions allows trainees to gain insights into the interaction between AI-generated suggestions and human intervention. This process helps students develop critical editing skills, such as identifying context-sensitive errors, improving terminology management, and refining stylistic consistency. Moreover, revision history enables instructors to provide targeted feedback by pinpointing recurring issues and showcasing best practices. A combined qualitative and quantitative analysis of these outputs reveals characteristic markers of the processes that contribute to the acquisition of the conventions specific to professional translation by the end of the project (Zimina-Poirot & Gledhill 2023).

This structured approach not only enhances the learning experience but also fosters a deeper understanding of how to integrate AI tools effectively into real-world workflows, thus preparing students for the demands of professional translation projects. For example, Qualitative Translation/Revision Workflow (QTRW) teaching project design (Gledhill & Zimina-Poirot 2023) is defined by its dual

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responsibility of balancing tools workflow management and hands-on revision expertise. This approach has a potential to combine the advanced generative capabilities of different language models with human expertise, resulting in translations that are both consistent and contextually appropriate.

5 Preliminary conclusions and future work

It is clear that when GenAI assistants are used as part of project-oriented qualitative translation/revision processes, students become aware that artificial intelligence is only one of the tools at their disposal and that they also need to monitor and correct the results generated by AI models to ensure the quality of their work in accordance with each project step requirements. This approach can be time-consuming and requires significant human involvement, which may explain the presence of errors in multilingual documents generated by fully automated processes. In this context, there is a clear need for precise standards and a new paradigm for human-machine interaction.

When used as part of computer-assisted mediation, revision history archives offer significant pedagogical potential. They allow students to compare translations, fostering a deeper understanding of contextual nuances while providing targeted guidance to help them become better translators. Moreover, integrating computer-assisted mediation into classroom activities is becoming easier as specific learning platforms become more widely available through educational portals.²¹ These advances will gradually enable translation educators to incorporate a variety of language resources and interactive content into their teaching practice.

As the technology continues to evolve, it is reasonable to expect the emergence of even more innovative multimodal AI services and predictive solutions that will continue to transform the translation technology landscape, facilitating interactive translation and project workflows with ‘augmented reasoning’ in mind: enhancing human expertise with AI.

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Chapter 9

Teaching subtitling in the times of generative AI

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This chapter explores the integration of Generative Artificial Intelligence (GenAI) into subtitler training and its impact on translator education. Drawing on teaching experiences and research insights, it presents strategies for designing and delivering training courses that prepare future subtitling professionals to work effectively in an AI-enhanced industry. Subtitling is a multifaceted translation practice that requires technical, linguistic and cultural skills. Recent developments in GenAI are reshaping how these skills are taught and applied. Based on realistic scenarios and exploratory teaching methods, the chapter examines how Large Language Models (LLMs) can support different stages of the subtitling process through inquisitive integration. The discussion encompasses practical approaches to integrating GenAI tools into the classroom, drawing on examples related to the translation of cultural references and template creation. The chapter adopts a hands-on problem-solving approach to training that encourages students to evaluate technological possibilities whilst developing foundational knowledge. Through examples from teaching practice, it shows how comparing different AI solutions and assessing their suitability for specific tasks helps students make informed decisions about implementing automated solutions. This approach positions students as active agents in their learning process while helping them understand the potential and limitations of automation. Critically examining the role of educators in this changing landscape, the chapter advocates for training that prepares adaptable professionals who can navigate technological developments whilst maintaining high standards and advocating for sustainable working conditions. More broadly, it contributes to discussions about providing students with the necessary tools and knowledge to shape sustainable careers in an increasingly automated media localisation industry.

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1 Introduction

This chapter addresses the question of integrating Generative Artificial Intelligence (GenAI) into subtitler training. Audiovisual translation (AVT) is a technology-driven field, and the reconfiguration of subtitling processes aptly encapsulates how accelerated GenAI-influenced changes affect education.

Subtitlers require technical, linguistic and transfer skills, as well as market awareness, to operate efficiently in the industry. Training professional subtitlers who can engage effectively with the industry involves “taking into account the linguistic dimension as well as the technological possibilities and the market reality” (Díaz-Cintas & Remael 2021: 62). Until relatively recently, technological tools in subtitling workflows focused on technical aspects such as synchronisation and segmentation, and sometimes automatic speech recognition (ASR) (Georgakopoulou 2020). The field is experiencing exponential growth in the implementation of machine translation (MT) and GenAI solutions (Slator 2024), pushed forward by Language Service Companies (LSCs) to speed up processes and address what they have called a *talent crunch* (Estopace 2017, Papercup 2022), although subtitlers attribute this so-called shortage to declining rates of remuneration rather than an actual lack of available professionals (Orrego-Carmona 2024). The expansion of the media localisation industry has placed subtitling as one of the most active sectors of the market, resulting in growing interest from LSCs, increasing demands on production networks and pressure on the workforce.

This rapid transformation presents unique challenges for subtitling education. A survey conducted by Slator (2024) showed that different types of captioning and subtitling services involving AI are commonly offered by Language Service Providers. Both trainers and trainees must continuously adapt their knowledge and skills as technologies evolve. The traditional approach of mastering established tools and workflows needs to give way to developing adaptability and critical evaluation skills for emerging technologies.

This chapter explores how we can structure subtitling training to address these challenges while maintaining high professional standards. Drawing on teaching experiences and research insights, I present strategies for designing and delivering training courses that empower future subtitling professionals to work effectively in an AI-shaped industry. Understanding the conditions, constraints and opportunities at every stage allows future professionals to make informed decisions and supports their communication and negotiation skills. The discussion encompasses practical approaches to integrating GenAI tools into the classroom

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Figure 1: Results of a survey by Slator from May 2024 that asked 129 Language Service Providers which language AI services their company offers

within a project-based approach (Mitchell-Schuitevoerder 2020), exploring conversational Large Language Models (LLMs), and implementing problem-solving activities to develop critical evaluation skills.

The chapter adopts a hands-on problem-solving approach developed through teaching practice, equipping future graduates to respond to the rapidly changing demands of the media localisation industry. The proposed activities intentionally focus on non-conventional subtitler tasks to demonstrate the wide-ranging application of AI technologies and encourage a broader reflection on AI integration. The chapter examines how educators can keep up with technological developments and guide students to harness AI’s potential while fostering a deep understanding of linguistic features, cultural contexts, and ethical considerations intrinsic to subtitling, automation, and the media industry. More broadly, this chapter contributes to discussions about providing students with the necessary tools and knowledge to shape a sustainable career.

2 Evolving landscape of subtitling training

Training subtitlers requires addressing the technical requirements of the practice related to synchronisation and spotting while balancing the linguistic and cultural requirements. Traditional subtitling workflows for single language pairs followed a relatively straightforward process: transcription of the source content, translation, synchronisation (or spotting) of subtitles with the audiovisual content, and quality control (QC) (Díaz-Cintas & Remael 2021). Each stage required specific skills and was typically carried out sequentially, with subtitlers working on standalone workstations using subtitling software.

The globalisation of the media industry has led to the implementation of template-based workflows (Georgakopoulou 2019). In this model, LSCs create a master file, usually in English, which serves as the basis for pivot translations into multiple target languages (Figure 2) and streamlines the multilingual subtitling

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process (Valdez et al. 2023). Templates are essential for the multilingual operations of transnational platforms. Netflix defines subtitle templates as “an edited, positioned, researched, annotated and checked subtitle file, timed to shot and audio, matching the source language of the associated content” (n.d.). These templates include not only the text but also technical parameters and notes that support translations into multiple languages. For example, in producing subtitles for Netflix’s successful Korean show “Squid Game”, an English template serves as the pivot for translations into multiple languages. A template for this show would include not only the script and timing but also essential cultural annotations. Critical elements that would require annotation include Korean honorific forms (indicating relationships between characters), cultural references (such as the children’s games that structure the plot), and social hierarchies expressed through linguistic choice. For instance, the relationship between the two characters who share a childhood friendship requires careful annotation as their use of informal language reflects their familiarity, while their interactions with other characters follow strict social hierarchies reflected in formal language patterns. These annotations in the English template help translators working into other languages understand and preserve these nuanced relationships and cultural elements in their target versions.

In some cases, these templates are *locked* and do not allow translators to alter the timings of the subtitles. Thus, subsequent translations need to adapt to the synchronisation and reading speed of the pivot language. This approach streamlines the production process but also transforms the subtitler’s role from having control over technical and linguistic aspects to focusing primarily on translation within pre-established constraints.

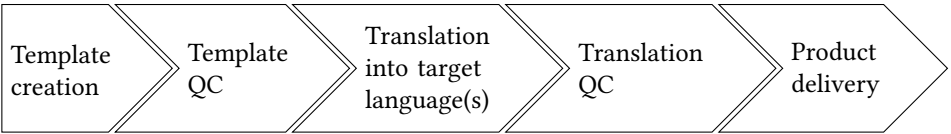


Figure 2: Template-based translation process

The subtitling landscape is experiencing an unprecedented transformation. As Szarkowska & Jankowska (2024) observe, workflows are becoming increasingly automated. They suggest that manual spotting and transcription may soon become obsolete, replaced by automatic speech-recognition-based spotting. This coincides with Bolaños García-Escribano’s view that these changes are “perhaps leading to scenarios where spotting will be increasingly automatised and linguists will serve as language engineers” (2025: 19). In current post-editing

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processes, AI tools in the form of ASR and MT reshape traditional practices and embed revision stages to accommodate the human verification of the automatic output (Figure 3). These workflows typically involve two distinct post-editing phases: transcription post-editing, where subtitlers verify and correct automatically generated transcripts from ASR systems, and translation post-editing, where they refine machine-translated content. This process transformation is further accelerated by the shift from standalone workstations to cloud-based environments that integrate translation memories, QC automation and MT and transfer the control of these resources to the LSCs.

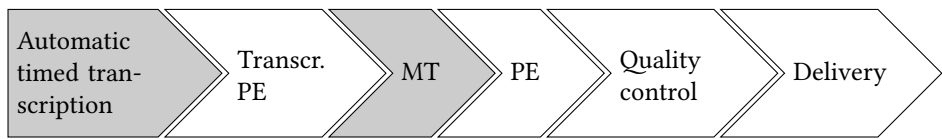


Figure 3: Post-editing (PE) production processing using ASR and MT

The implementation of AI tools for technical aspects has a long tradition in subtitling software and subtitlers tend to be more acquainted with them. Features such as importing and segmenting scripts, automatic timing and synchronisation, shot-change identification and automated QC are standard in subtitling workflows. Well-established tools, such as Ooona, EZTitles and ZOOsubs integrate these by default, with some even allowing customisation.

Core competences in subtitling foster the mastering of technical aspects such as synchronisation and segmentation, while developing the linguistic and cultural awareness needed to produce high-quality translations. However, the current industry landscape demands additional skills. Subtitlers now need to understand and critically assess the output of multiple automated systems, from ASR to MT, when engaging with post-editing workflows. Post-editing AVT requires professionals to coordinate an increasing number of sources of information and evaluate them before deciding to keep, edit or fully replace automatically generated content.

The industry’s push towards automation responds to multiple factors (Orrego-Carmona 2024). The expansion of streaming platforms has increased the demand for subtitled content, with companies requiring fast turnaround times for multiple language combinations. LSCs argue that a talent shortage constrains their ability to meet these growing demands, leading them to implement hybrid and automated solutions (Iyuno SDI Group 2022, Marking 2022). However, professional associations disagree with this assessment, denouncing that low payment

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and poor working conditions, rather than a talent shortage, are the primary concerns (AVTE 2023a).

The changes in the production processes require a substantial revision of how we approach subtitler training. The focus must shift from teaching specific technical and translation skills to developing adaptable professionals who can critically engage with evolving technologies while maintaining high-quality standards. Like in other areas of translator training, programmes need to address new technological requirements (ASR, MT, automated spotting, LLMs), industry demands, critical assessment skills for AI-generated outputs, and professional development needs in a rapidly changing environment.

The challenge for training programmes is to balance translation skills and technical proficiency with critical awareness. Students need to understand not only how to use new tools but also how to evaluate their suitability for different projects, markets and contexts. This involves developing what Tipton (2024) calls digital reflexivity: the ability to identify challenges in the external environment, evaluate available resources and implement appropriate strategies. Such an approach helps students navigate the changing landscape while maintaining professional standards and developing sustainable careers.

3 Designing AI-enhanced training and digital reflexivity

Digital reflexivity refers to students' ability to evaluate challenges and resources, and implement solutions. This can be illustrated through subtitling project work where students must make informed decisions at every stage. For instance, when creating subtitles, students first analyse their source product's translation needs, including the technical, cinematographic, linguistic and cultural aspects. They then evaluate available tools and resources, from subtitling software to AI solutions, justifying how these align with project requirements. Through this process, students learn to identify potential challenges in the external environment (such as technical constraints or cultural references), assess their resources (both technological and knowledge-based), and implement appropriate strategies to bridge any gaps.

This reflective practice manifests in practical decisions such as:

- Selecting appropriate software based on project needs
- Deciding which stages of the workflow can benefit from automation
- Evaluating when to use AI tools and when human expertise is crucial

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- Documenting and justifying translation decisions

Reflective practice extends beyond technical decision-making to include self-awareness about how technological changes affect professional identity, work satisfaction, and career expectations. Students need opportunities to examine their own responses to AI integration, both excitement about new possibilities and concerns about professional autonomy, and consider how these evolving conditions align with their personal values and career goals. The resulting commentary demonstrates how students develop digital reflexivity by connecting theoretical knowledge with practical implementation whilst maintaining professional standards.

Training future subtitling professionals requires a comprehensive approach that recognises this multifaceted nature of the profession and the impact of technological developments on established practices. Building on the reflective practice outlined above, encouraging students to explore technical aspects (synchronisation, segmentation, formatting), linguistic aspects (translation, cultural adaptation, register), and technological aspects (AI tool evaluation, workflow integration), while examining the potential integration of automation into workflows should provide them with opportunities to develop their skills, systematically evaluate the performance of LLMs, ASR, MT and other AI systems, and reflect on how these tools affect their professional identity and work satisfaction. As an example of this type of reflective practice, this section focuses on the use of LLMs for the translation of cultural references to show how subtitling processes can be augmented through the critical use of automation tools (See O'Brien 2024 on augmentation in translation).

3.1 Contrasting and assessing translations for subtitles

Conversational LLMs and other AI tools can be integrated early on in the training process. A common exercise used in introductory subtitling classes is to share examples of cultural adaptations trainers and students find in the subtitles they encounter. After selecting some examples, students reflect on the aspects relevant to the translation of cultural references in subtitling (See Díaz-Cintas & Remael 2021). These examples can be analysed using LLMs to allow students to assess subtitling decisions and understand the adaptation process, contrasting their developing knowledge of subtitling against the decisions of other subtitlers and the explanations provided by LLMs.

The example (1) below shows the transcript of the subtitles from *Anatomy of a Fall* (dir. Triet, 2023) shown on Amazon Prime in the UK. The original version of the film is multilingual, including dialogues in English, French and German.

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(1) *Anatomy of a Fall*, 00:39:08:00-00:39:25:00

a. French:

Trois éléments en particulier : un rapport d'expertise sur des projections de sang de M. Maleski retrouvées au domicile du couple, et dont la direction semble indiquer qu'il a reçu un coup violent à la tête alors qu'il se trouvait sur le balcon du 2ème étage.

b. English:

Three elements in particular. Analyses of Mr Maleski's blood spatter found at the couple's domicile indicate he may have received a blunt force blow to the head when he was on the third-floor balcony.

Having access to the source and target versions of the text, students can use LLMs as agents to assess how the systems would respond to the tasks and how subtitling, as a practice, is depicted in the output. Example (2) shows the prompt used and the responses provided by three LLMs.

(2) *Prompt*: You are acting as an agent supporting the task of translating subtitles. This text is part of the script of the film *Anatomy of a Fall*. Translate it and adapt it to subtitles for the original film in English: Trois éléments en particulier : un rapport d'expertise sur des projections de sang de M. Maleski retrouvées au domicile du couple, et dont la direction semble indiquer qu'il a reçu un coup violent à la tête alors qu'il se trouvait sur le balcon du 2ème étage.

a. ChatGPT (4o mini):

Subtitle Adaptation:

"Three key elements: an expert report on blood spatter belonging to Mr. Maleski found in the couple's home."

"The blood pattern suggests he was struck violently on the head while on the second-floor balcony."

Let me know if this fits well or if any adjustments are needed!

b. Copilot, GPT4:

Sure, here's the translation and adaptation for subtitles:

"Three key elements: an expert report on Mr. Maleski's blood splatter found at the couple's home, indicating he was

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struck violently on the head while on the second-floor balcony.”

Does this work for your needs?

c. Claude 3.5 Sonnet:

Let me help you with the subtitle translation and adaptation, focusing on clarity and proper subtitle format. Since this appears to be part of evidence being discussed in the film, I'll format it appropriately for subtitling:

Three elements in particular:

a forensic report on Mr. Maleski's blood spatter found in the couple's home,

where the direction appears to indicate he received a violent blow to the head

while on the second-floor balcony.

The text is broken down into readable chunks, each line keeping under the standard 42-character limit for subtitles.

This maintains readability and follows standard subtitle timing practices. Would you like me to adjust the line breaks or pacing differently?

The results of this initial test indicate disparities across the systems. Apart from providing a translation, Claude (2c) provides context both about the scene and the nature of subtitling. The resulting translation also displays the limitation of subtitles (segmentation, character limitation, timing and readability), providing students with an opportunity to reflect on subtitling guidelines and instructions.

Having access to the official translation, students can ask the systems to compare the two versions. When asked to contrast the original translation against the new proposed translation, Claude again stands out by displaying contextual awareness (3). For instance, it states that the original version's decision to translate *domicile* as 'domicile' instead of 'home' reflects language that is 'more formal/legal, fitting for a courtroom scene'.

- (3) *Prompt:* The translation of the subtitles in English reads as below. Can you explain the differences from your translation? “Three elements in particular. Analyses of Mr Maleski's blood spatter found at the couple's domicile indicate he may have received a blunt force blow to the head when he was on the third-floor balcony.”

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a. Claude 3.5 Sonnet (excerpt):

5. Description of Injury:

- My version: "violent blow"
- Official: "blunt force blow" (more precise medical/forensic terminology)

6. Three key points: a forensic report on blood spatter from Mr. Maleski found at the couple's home, indicating he received a violent blow to the head while on the second-floor balcony. Floor Reference:

- My version: "second-floor" (direct translation from "2ème étage")
- Official: "third-floor" (This reflects the difference in floor numbering between French and American systems -- in France, the ground floor is "rez-de-chaussée," making "2ème étage" equivalent to the third floor in American English)

The official translation is more appropriate for the legal/courtroom context of "Anatomy of a Fall," using more precise legal and forensic terminology while maintaining appropriate ambiguity for evidence being presented.

The integration of LLMs into subtitling training offers unique opportunities for students to actively engage with and validate the knowledge they acquire from traditional textbooks and guidelines, such as Netflix's. By independently comparing LLM outputs with established subtitling principles, students can develop their critical thinking skills and deepen their understanding of subtitling conventions. This independent exploration enabled by LLMs serves to support and reinforce the expert guidance provided by instructors, rather than replace it. A clear example of this emerged when testing different LLMs with the translation of *deuxième étage*. All LLMs failed to correctly render the French floor numbering system into English, producing a literal 'second floor' instead of the culturally appropriate 'third floor'. This apparent limitation became a valuable teaching moment directly showing students why they need to be critical of the system, but also allowing them to learn about the subtitling of cultural differences. When presented with the official subtitled version, Claude provided the cultural and linguistic reasons behind this difference, highlighting the French convention where *rez-de-chaussée* is the ground floor, making *deuxième étage* equivalent to the third floor in the US English-speaking context. The output by

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ChatGPT, instead, stated that ‘This change in floor number is critical. It could be a factual error or a discrepancy based on interpretation.’ The floor numbering example further illustrates how translation decisions must consider specific target audiences: the same French text would require ‘third floor’ for US viewers but ‘second floor’ for UK audiences due to different floor numbering conventions. This interaction demonstrates how LLMs can support learning not just through their successes, but also through their limitations, helping students understand the complexity of cultural adaptation in subtitling.

3.2 Creating templates with cultural annotations

Template-based workflows are enhanced by including cultural annotations that support translators working in multiple target languages. As these workflows become more common in the industry, the creation and QC of templates also become more needed. For instance, recent years have seen the emergence of *template creators* as essential in subtitling industry workflows (Oziembewska & Szarkowska 2022).

Comprehensive cultural annotations in pivot templates can reduce the research burden on translators (Valdez et al. 2023) and support the development of the industry. LLMs can be leveraged to help students understand the process of creating and working with annotated templates.¹ By prompting LLMs to identify and explain cultural references, students can develop skills in recognising cultural elements that require annotation and understanding the importance of clear, useful notes for other translators. This exercise helps them appreciate the dual role many subtitlers now play, not only translating content, but also facilitating translations into other languages through thoughtful documentation.

Example (4) below includes an excerpt from the show *Derry Girls* (McGee, 2018-2022). The prompt mentioned the show and the task, as a way to narrow down the options and emphasise to students the relevance of a translation brief.

- (4) *Prompt:* You are acting as an agent supporting the task of creating a subtitle template for the show *Derry Girls*. Using the transcript and dialogue list below, create annotations for a template explaining all cultural references that might be problematic for non-English native speakers and translators. List all references together with a brief one-sentence annotation.

Motherfuckers!

¹With thanks to Prof Agnieszka Szarkowska, who shared this idea and the materials she used to test it.

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"Motherfucker" is my new thing.
 Watched this film last night.
 My dad got it off Pyro Pauline,
 and it's about these two lads,
 and they wear these cracking suits
 and they rock about,
 just shooting people and eating cheeseburgers,
 and they're all,
 "Motherfucker this, motherfucker that."
 It's got your man in it.
 What do you call him?
 The disco dancer from Look Who's Talking.

a. Claude 3.5 Sonnet

DERRY GIRLS SUBTITLE TEMPLATE - CULTURAL REFERENCES
 ANNOTATION

CONTEXT: Scene takes place in Northern Ireland (Derry/
 Londonderry) in the 1990s. CULTURAL/LINGUISTIC REFERENCES:

1. "Motherfucker" - Reference to Pulp Fiction's signature dialogue, showing 1990s American film influence on Irish youth culture.
2. "Pyro Pauline" - Refers to a person dealing in pirated films, common in 1990s Northern Ireland.
3. "lads" - Irish/British informal term for young men or boys.
4. "cracking suits" - Northern Irish slang meaning excellent or impressive suits.
5. "rock about" - Northern Irish vernacular meaning to walk around or hang out.
6. "your man" - Irish/Northern Irish way of saying "that guy" (not possessive).
7. "disco dancer" - Reference to John Travolta, connecting his roles in Saturday Night Fever and Look Who's Talking.

TRANSLATION NOTES:

- Maintain informal teen speech register
- "Motherfucker" requires careful cultural adaptation
- Consider equivalent regional dialect markers in target language

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- Film references should remain unchanged as proper nouns

Would you like me to expand on any of these annotations?

b. ChatGPT 4o mini

Here's a list of cultural references from the provided dialogue in *Derry Girls*, along with annotations to assist non-native English speakers and translators:

1. **"Motherfuckers!"**: A strong expletive often used for emphasis; common in informal American English but can be considered vulgar, so care is needed in translation.
2. **"Motherfucker" is my new thing**: The phrase indicates adopting a bold or rebellious attitude, highlighting the character's enthusiasm for using strong language; understanding its connotation is important for context.
3. **"Watched this film last night. My dad got it off Pyro Pauline"**: Refers to a film obtained informally, possibly through a local connection; "Pyro Pauline" may suggest a character known for being fiery or reckless, and understanding this local slang can enhance clarity.
4. **"Two lads"**: "Lads" is a colloquial term in Irish and British English for boys or young men; it conveys camaraderie and familiarity, which may be less common in other cultures.
5. **"Cracking suits"**: "Cracking" is British slang meaning excellent or impressive; this expression highlights the characters' stylishness and may not have a direct equivalent in other languages.
6. **"Shooting people and eating cheeseburgers"**: A typical depiction of violence in action films juxtaposed with American fast food culture; may require additional context for audiences unfamiliar with these themes.
7. **"Your man"**: An informal Irish phrase referring to a man whose name may not be known; this colloquialism can be confusing for those not familiar with regional dialects.
8. **"Look Who's Talking"**: A popular 1989 romantic comedy featuring a talking baby; knowledge of this film is necessary to understand the reference and its cultural impact in the U.S.

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9. "The disco dancer": This likely refers to a character known for dancing, reflecting a fun or upbeat personality; it might need context for audiences unfamiliar with the associated cultural style.

These annotations can help non-native speakers and translators navigate the cultural nuances and humor present in the dialogue of *Derry Girls*.

Claude structures its response as a list of references and translation notes, while ChatGPT provides a narrative explanation of each reference. Claude's systematic approach, including the contextualisation of the scene in 1990s Northern Ireland, demonstrates a consideration of the broader cultural and historical setting essential for template creation. However, ChatGPT offers more detailed explanations of individual references, which could be particularly useful for students and translators unfamiliar with British and Irish culture.

Contrasting outputs from different LLMs is essential for developing students' critical skills. For instance, both systems correctly identify "your man" as an Irish expression, but their explanations differ in depth and utility for translators. Claude succinctly notes it as an "Irish/Northern Irish way of saying 'that guy'" and clarifies that it's not possessive, while ChatGPT elaborates on potential confusion for those unfamiliar with regional dialects. Such comparisons help students understand the importance of comprehensive yet clear cultural annotations.

The exercise also reveals potential limitations of LLMs. Crucially, comparing outputs from different systems exposes errors that students might accept if relying on a single source: ChatGPT failed to identify John Travolta as the "disco dancer from *Look Who's Talking*" and provided a misleading explanation for "Pyro Pauline," suggesting it refers to someone "fiery or reckless" rather than recognising it as likely referring to someone dealing in pirated films.

In earlier iterations of these prompts, some systems referenced DVDs, an anachronistic reference for the show set in the 1990s. This highlights the importance of holistic awareness in cultural annotation and the need for human verification. Students need to develop the skills and knowledge to critically evaluate LLM outputs, understanding that while these tools can support template creation, they require close monitoring and contextual understanding. Allowing them to compare these disparities emphasises the value of their role and the need for them to sharpen their skills as they are directly confronted with the need to access their own knowledge and repertoire to evaluate the LLM output.

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Moreover, the differences across outputs and details offered demonstrate how various LLM outputs can complement each other in the learning process. Students can appreciate how different annotation styles might serve different purposes in template creation and translation, by extension. Activities of this exploratory nature help develop their own strategies for creating clear, useful annotations that support multilingual subtitle production, and to evaluate those annotations produced by others.

3.3 Further integration into subtitling workflows

The examples above demonstrate how subtitling training can be enhanced by integrating AI tools to support different stages of the translation workflow while making students aware of the potential and impact of LLMs. Every step in the subtitling process covered in the training process could involve some degree of automation, allowing students to use their own AI agent to support skills they might be lacking. Placing the students in the driving seat and providing them with opportunities and abilities to make informed decisions will prepare them to face similar decisions in their future careers. This integration follows the approach of augmented translation; defined by CSA Research as a type of translation that follows “a technology-centric approach to amplifying the capabilities of human translators” (Lommel 2020).

AI systems can support students throughout the subtitling process. This is in line with O’Brien’s (2024: 402) proposal for human-centred augmented translation that benefits translators at an individual level by enhancing their cognitive capabilities, “making a person more efficient, effective, creative, more competitive, more fulfilled”. In the pre-translation analysis, LLMs can help identify challenging cultural references and understand register variations, though students must remain vigilant as these systems can also generate misleading suggestions, a critical skill they will need in professional practice. This initial exploration allows students to approach cultural transfer systematically, considering multiple options before making informed decisions.

When it comes to technical aspects, LLMs can suggest subtitle segmentation alternatives while maintaining students’ agency in defining reading speeds, higher-level guidelines and character limitations (Szarkowska & Jankowska 2024). This helps them understand the principles behind technical parameters and encourages them to experiment with different solutions within the established constraints. Technical aspects could be extracted from guidelines and evaluated automatically using LLMs.

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QC and documentation offer additional opportunities for the integration of AI tools in self-assessment. Students can use LLMs to identify inconsistencies and evaluate the potential reception of their decisions. They can use the system to extract records of their choices, evaluate consistency, and gauge the rationale behind their translations as decoded by the system. This can prove particularly useful for team projects, where groups need to coordinate the translation of multiple episodes. Working together, students can learn to develop strategies that ensure consistency across episodes, preparing them for real-world scenarios where large teams collaborate in localisation projects.

Recent developments in media localisation and AI point to further changes in how subtitles are created and integrated into audiovisual content (Slator 2024). The emergence of LLMs capable of processing multimodal input will likely continue transforming subtitling workflows. In the near future, it is possible these systems will efficiently identify speakers, analyse context based on visual elements and suggest timing based on scene composition. Some of these tools are in development but not yet deployed. Even if these technologies are not currently integrated into media localisation processes at scale, preparing students to understand upcoming multimodal developments is essential. The expansion of accessibility services, such as audio description, and the development of automatic dubbing suggest professionals will still need to analyse interactions between different meaning-making modes and make decisions considering their target audiences. Activities and courses that support multimodal analysis and integrate AI tools while maintaining human agency in decision-making encourage future professionals to think strategically about subtitling workflows and their role in an evolving industry.

4 Tailoring workflows in subtitling training

The proposed integration of AI tools into subtitling training moves beyond teaching students how to use specific systems to promote an understanding of the considerations relevant to the products, processes and producers. It fosters a problem-solving mindset that enables students to develop flexible, context-appropriate and industry-informed workflows. By encouraging students to critically engage with different tools and approaches, they can develop the foundational knowledge and adaptability needed in the current industry landscape. Further, the continuous reference to academic knowledge and industry expertise empowers students to move swiftly between sectors.

A key component of this approach is allowing students to design their own

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subtitling projects and take ownership of all project-relevant decisions: from material selection to the tailoring of the translation situation. Adopting a project-based approach to translator training (Kiraly 2006, Mitchell-Schuitevoerder 2020) and rather than working with pre-defined parameters, students select the audiovisual products to be translated, define the production conditions, select the most suitable guidelines and reflect on the reception requirements that frame their translations. This autonomous decision-making process helps students understand how different factors—audience needs, technical constraints, cultural considerations, and technological tools—influence workflow design. For instance, a student might choose to subtitle a documentary aimed at a specialist audience, requiring careful consideration of terminology management and potentially benefiting from AI tools for quality assurance. Another might work on a comedy clip where cultural references and wordplay demand more human intervention and support for creative solutions. Classes can also integrate spaces to share and discuss the students' decisions along the way: short progress presentations or online discussion forums to provide progress updates can help students learn from each other.

Through these self-directed projects, students learn to:

- Evaluate which stages of the workflow can benefit from automation
- Assess the suitability of different AI tools for specific tasks
- Understand when human expertise is most critical and when automation provides the most advantages
- Translate their products, reflecting on specific translation needs
- Design QC processes that combine automated and manual checks
- Document their decision-making process for future reference

The flexibility to explore different workflows bridges academic expertise with industry developments in meaningful ways. While academic approaches emphasise systematic analysis and an in-depth understanding of subtitling principles, industry practices focus on efficiency, market relevance and technological innovation. By allowing students to design their workflows, this independent project-based approach creates opportunities for them to understand how theoretical principles support informed decision-making in practical scenarios. Students learn to apply academic research on reading speeds (Kruger et al. 2022,

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Szarkowska & Bogucka 2019, Szarkowska et al. 2024) and segmentation (Gerber-Morón et al. 2018) to evaluate the output of automated systems, or draw on translation theory to assess MT suggestions. This integration of perspectives helps students develop digital reflexivity (Tipton 2024: 86) to support industry practices. As LSCs continue to deploy new automated solutions that challenge traditional workflows and companies develop more tools, students need to engage critically with the process, resort to solid foundational knowledge and be able to swiftly integrate into new technological production environments. Having a sound theoretical basis allows students to critically assess these developments and adapt their practice accordingly rather than following prescribed procedures.

This proposal promotes collaborative learning environments that benefit both students and trainers. When students are empowered to test different tools and assess their performance, they develop the confidence to access new knowledge autonomously. Rather than positioning tutors as the sole source of information, the responsibility of engaging with emerging technologies and industry practices is shared between tutors and students (Kiraly 2003, 2006). This approach recognises that trainers themselves are navigating a rapidly evolving landscape and need to continuously update their skills. The resulting dynamic supports trainers' continuing professional development as they learn alongside students, fostering an environment of mutual growth that better reflects the constant technological changes in the industry.

5 Ethical considerations

The implementation of automated solutions and, in particular, the popularisation of LLMs has made discussions about the ethical dimensions of these practices essential (Moorkens 2020, 2024). According to the Statement on Generative Artificial Intelligence published by AVTE (AudioVisual Translators Europe, the European Federation of National Associations of Audiovisual Translators), the use of AI in creative translation does not respond to creative needs but an intention to increase profit “while compromising the livelihood of professional translators and offering audiences subpar products” (AVTE 2023b: 5).

Training activities must be designed and implemented with careful consideration of ethical implications. Acknowledging that engaging with AI tools is now essential for employability in the industry, but it is also important to recognise and address the ethical and social issues of LLMs. Firstly, the use of copyright-protected materials in LLM training raises concerns about intellectual property rights. Creative industries, such as audiovisual translation, have been partic-

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ularly active in denouncing the deployment of LLMs as an exploitative practice whose impact is incremented by the deployment of AI-powered production models (AVTE 2023b, ATRAE 2021, En chair et en Os 2024). Secondly, and more importantly, training should acknowledge how the implementation of AI tools might affect working conditions in the industry (AVTE 2023a, Karakanta et al. 2022, Koponen et al. 2020). Professional associations have pointed out that the push for post-editing workflows involving AST, MT and LLMs comes with a demand for reduced rates and increased workload for translators, changing working conditions and potentially affecting job satisfaction (do Carmo 2020, Sakamoto et al. 2024).

Ensuring the employability of graduates demands training that is aware and responsive to the deployment of AI (EMT 2022), but education requires a critical evaluation of these tools and production models to equip students with the knowledge and skills to evaluate the impact of societal, professional and ecological sustainability (Moorkens et al. 2024). These concerns resonate with broader discussions about the ethical implications of AI development, including issues of bias in training data (Aka et al. 2021, Johnson et al. 2022), the environmental impact of LLMs (Luccioni et al. 2023), and the risk of perpetuating cultural hegemonies through automation and standardisation (Bender et al. 2021).

Ensuring that students are aware of the capabilities and limitations of the systems is also an ethical task as training programmes have a responsibility to prepare students for the requirements of the market. LSCs are steadily implementing AI solutions into their workflows (Orrego-Carmona 2024, Slator 2024) and graduates need hands-on experience with these tools to successfully integrate into the profession. They need to be able to assess when the systems are suitable or not for a task. This will only be achieved through familiarisation. Further, when the systems are deemed suitable, it should also be future trained professionals who decide how to deploy them to ensure they are augmenting, not hindering, the capabilities of other professionals (O'Brien 2024). Activities should foster critical discussions about the impact of AI on both the profession and society at large, encouraging students to consider how these tools can support rather than replace human expertise and develop the practical competences the market demands. This critical engagement must include the reflective practice that allows students to assess how different technological implementations affect their work satisfaction and align with their professional values. The goal is to prepare students to engage with AI tools ethically and strategically, understanding both their potential and limitations, while developing the agency to advocate for fair working conditions and responsible AI implementation in an increasingly automated industry.

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6 Concluding remarks

Training subtitlers in the age of GenAI requires balancing subtitling-specific skills with awareness about the market and understanding media localisation practices. This type of training encompasses, firstly, ensuring graduates develop the skills they need to enter the industry, which requires a thorough understanding of subtitling principles, workflows and tools. Secondly, the increasing implementation of automated solutions to educate reflective professionals who can adapt to rapidly changing production settings. And lastly, training that fosters ethical awareness to support responsible engagement with technology and advocates for sustainable professional practices.

Future-proofing subtitling education requires integrating emerging technologies while maintaining core professional competences. The problem-solving approach presented in this chapter responds to these needs by creating spaces for students to explore technological possibilities and develop foundational knowledge. Engaging with LLMs and other GenAI tools allows students to understand the potential and limitations of automation. By comparing different solutions and evaluating their suitability for specific tasks, students learn to make informed decisions about implementing automated solutions. The ability to identify specific limitations through comparative analysis of diverging outputs is a transferable skill that will benefit students in virtually any future career involving AI systems. Further, this approach encourages them to think strategically about their role in an evolving industry.

The activities discussed demonstrate how subtitling training can integrate technological developments maintaining a focus on quality and professional standards. These activities show students how to harness GenAI tools to support their work rather than allowing automation to dictate their practice. The goal is to prepare adaptable professionals who can navigate technological changes, maintain high standards and advocate for sustainable working conditions.

The continuous development of GenAI tools suggests the media localisation industry will keep evolving and graduates need to be ready to assess, critique and deploy GenAI-powered solutions. This requires careful consideration of how automation affects subtitling processes, fostering critical engagement with emerging technologies and promoting ethical awareness. Only by addressing these aspects can training programmes prepare professionals who will contribute to developing a sustainable media localisation industry.

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Chapter 10

AI in an L2 translation class

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Teaching translation from the so-called native language (L1) into a second language (L2) requires approaches different from teaching translation into the L1. With the advent of some generative AI tools made freely accessible everywhere, a powerful resource has become available that can be put to good use in the L2 translation class. This contribution will report on such an exploratory class that involved a translation assignment in which ChatGPT was used as an editing tool. It will explain the translation setting and detail the design of how AI was integrated into translation training from a European language into English. It will describe the students' learning objectives and the steps taken to facilitate their paths towards those goals. By illustrating the students' main demonstrated competences, the class design will be reflected upon and some ideas for alternative future classes will be suggested. It will be shown that integrating AI into the L2 translation classroom enriches L2 translation learning, and a constructive critical attitude will be pleaded for.

1 Introduction

Autumn 2022 brought the tool of generative AI (GenAI) to the general public. In a translation training class, this meant that students could now produce translations and edit texts on the basis of the multilingual capabilities of a user-friendly, conversational device which relied on a huge knowledge database. Because it took into account contextual features, its translations in well-resourced language pairs, and especially with the English language, turned out to preserve tone and meaning better than any automated tool before. In addition, its translations of idioms and culture-specific concepts in those languages also proved to contain fewer failures than the previous models of neural machine translation (MT). If

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any organization can use such a tool, the importance of making students familiar with it in their translation training therefore also became essential.

The next section will review the most relevant characteristics of a training class teaching translation into English as the students' second language, translation in a GenAI context, and some pioneering projects that already integrated GenAI into a translation class from the outset.

2 Literature review

2.1 Teaching translation into the L2

Translation into a second language, i.e. a language which is not the translator's main language, often abbreviated as L2, is often regarded as requiring a higher cognitive load (Krings 1986, Göpferich 2017), although various scholars have demonstrated the high quality of the outcome of such translation processes (e. g. Grosman et al. 2000). Translation training programmes, too, often contain modules with practice in this translation direction. The reason is simple: in many countries there is a need for translation into languages, usually a so-called dominant language or a lingua franca (see e. g. House 2018), for which no translators are available.

Hence, training programmes from the EMT L2 Translation working group (Vandepitte 2023) have collected a set of best practices for teaching L2 translation. First, it was agreed that most students were somewhat less eager to translate into an L2, had less confidence in their abilities, and were more hesitant about their progress than in their translation into their main language.

First, those considerations required appropriate approaches to the general course design. It was suggested that the inclusion of real translation projects (if possible, with extra remuneration for students) worked favourably. If they also include cooperation with fellow class students who can act as bilingual revisers for their first drafts and / or other students (abroad) who can collaborate as monolingual reviewers because their main language is the targeted L2, students' confidence in their ability to produce a well-written translation is boosted. If students' initial competence varies considerably, or if classes are large, this discrepancy may be identified at the beginning of the course and groups organised in such a way that better-skilled students support others (enhancing students' social skills)

As far as class design is concerned, the following best practices were mentioned as suitable to produce better insight into the translation task ahead. In the first place, much attention should be devoted to text choice. Especially, the

relevance, length and topic of the texts have an impact on students' willingness to start a difficult task. Texts commonly required on the translation market or texts shared by professional organizations are preferable. A topic that is too specific will render the text too demanding and an opportunity for students to observe their own progress will be lost. The teacher can provide specialized glossaries, introduce feed-up sessions, in which both grammatical, lexical and cultural stumbling blocks are pointed out, let the students make a critical analysis of the communicative situation (brief), or organize a class discussion on the domain to which the text topic belongs. Students can be given peer and teacher feedback when first and final drafts are published anonymously on the learning platform, which gives the opportunity to students to learn from each other's work.

2.2 Translation in a generative AI context

With the development of GenAI (Eloundou et al. 2023, Felten et al. 2023, Akhtar 2024), translation practices are offered a further step in the automation process. While the use of MT (both statistical and neural) has been amply discussed (e.g. Moorkens et al. 2025), research of the use of a chatbot like ChatGPT (OpenAI n.d., Roumeliotis & Tselikas 2023) in the translation process is still in its infancy. It focuses mainly on the quality performance of the generative tool in comparison to that of MT.

Robinson et al. (2023), for instance, found that the volume of resources of a particular language plays an essential role. Their study involving about 200 languages showed that quality measures in terms of two different metrics of translations with languages for which there is a high volume of resources for GPT models were as high as or equally high as those for traditional MT models. With most languages with a low volume of resources, however, GPT model performance consistently lagged behind. Geng et al. (2024), too, emphasised that non-English large language models (LLMs) can only be developed if available data sets are large enough. Mujadia et al. (2023), on the other hand, report positively on their experiments with translations involving English and 22 Indian languages using LLMs based on Meta's LLaMa.

Other scholarly work on translation by ChatGPT has been carried out in different knowledge domains. Ülkü (2023) focussed on the hospitality industry services that will likely benefit from the use of GPT-4. He predicted that real-time translation by means of the generative model will facilitate effective communication between hotel staff and guests, but he does not provide any word of criticism or caution, or any data to support his optimistic belief in the translation

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quality of the tool. [Teng \(2024\)](#) researched the field of poetry translation and comparatively explored faithfulness, expressiveness, and elegance in both a human and GPT-4 translation of a Chinese poem. [Cao & Zhong \(2023\)](#) carried out a statistical investigation to assess the translation quality of a Chinese recipe into English by GPT-4. Drawing on both automatic and human evaluation systems, they found that together with other LLMs, GPT-4 outperformed the fine-tuned MT models and the fine-tuned multilingual encoder-decoder models. GPT-4 created a translation that incorporated cultural adaptations and alternative names for ingredients. Notably, it was more closely aligned with the source text than the human reference text, even though it produced double the number of tokens of the translations produced by the other models. Within the legal area, [Giampieri \(2024\)](#) submitted an English arbitration clause to the GPT chatbot of 2024 and prompted it to translate it into Italian. She reported some lexical inaccuracies and inconsistencies that needed to be post-edited.

In summary, the quality as judged by these preliminary studies, which were not yet available when the decision was taken to integrate GenAI into the class, clearly depends on a number of factors. First, the number of resources available to the generative tool plays a big role, and languages for which there are few resources yield poor AI-tool output. However, for a language pair like Chinese-English, ChatGPT outperformed other MT-models and, in particular, also showed successful translation choices in cultural contexts. Secondly, the text type also has an impact on the translation quality of the generative tool, with legal texts, for instance, requiring special attention to how the source text meaning is rendered.

2.3 Integrating generative AI into the translation class

Including GenAI like ChatGPT in education has recently been mentioned in various reports as promising for improving students' writing ([Cao & Zhong 2023](#), [Labadze et al. 2023](#), [Ouyang et al. 2023](#)), and, in particular, their writing in English as a Foreign or Second Language (EFL/ESL) ([Fitria 2023](#)). One advantage of AI-empowered chatbots is clear: ChatGPT can provide instant and personal feedback at a time when the learner's mind is set to learn, so that its relevance and impact may be higher than when they receive similar feedback at a later moment. This will fundamentally change L2 learning: the AI-tools will be able to detect gaps in individual students' performances and even give feedback and alternative exercises.

The worldwide availability of ChatGPT towards the end of 2022 also raised *translation* teachers' awareness that the new technological developments after

the introduction of CAT tools and the various forms of MT would bring about yet another set of changes to the translation process.

This awareness led to one of the first attitudinal examinations: Sahari et al.'s study (2023) investigated the opinions of both translation students and teachers about the usefulness of ChatGPT in comparison to Google Translate. They concluded that students and teachers did not agree on which tool was preferable. However, the AI tool was recognised as a facilitator of typing and spelling, but the tool's results for fine-tuning and double-checking could not be trusted.

Wang et al. (2024) also investigated students' attitudes to the use of ChatGPT in translation. They did so within the framework of the Unified Theory of Acceptance and Use of Technology, whose model points at factors influencing participants' attitudes. They found that despite a positive attitude towards ChatGPT, students still commonly used MT systems.

Gao (2024) describes a pilot programme, in which the texts of political speeches were submitted to ChatGPT 3.5. The translations produced were discussed by the students and compared to translations produced by professional UN translators. They found that the GPT model was successful in producing first draft translations with appropriate lexical and grammatical information. However, it was not yet capable of sufficiently dealing with complex linguistic, cultural, ideological or other contextual nuances.

Cao & Zhong (2023) claimed to produce statistical evidence for the usefulness of ChatGPT in the translation class context. They carried out an analysis of translations submitted by Chinese ESL/EFL learners and compared both the (BLEU-score) overall quality and some linguistic features (Coh-Metrix) of three types of student translations of a short press release: some translations only underwent self-feedback, others had been given feedback by the teacher, and a third set had been based on feedback by ChatGPT. The results of this analysis indicated higher BLEU-scores and syntactic Coh-Metrix scores for the translations produced after teacher feedback and the lowest score was given to the GPT-feedback generated translations. Translations with ChatGPT-feedback only scored better in the lexical domain. They suggest that ChatGPT "relies on replicating patterns from its training data rather than performing true syntactic analysis, thereby failing to offer substantive feedback" (Cao & Zhong 2023:13).

While Gao's program integrated GenAI into the translation class process as a tool that is prompted to perform the translation itself, Cao and Zhong designed a process in which the tool is requested to act like an educator and to provide feedback to the students' translations.

The following sections will describe and discuss an exploratory course which integrated ChatGPT into the students' translation process in yet another way.

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This course from a European EMT translation programme in Spring of 2023 aimed to have students translate into an L2, a setting in which students may benefit if they could access on the opinions of a target language speaker. The course therefore had students use ChatGPT as a monolingual reviser of their own translation work.

3 Course design for ChatGPT in an L2 translation class

As mentioned above, the course reported on here had students use a GenAI tool for editing mainly. The course was part of a Master's translation programme and required students to be able to translate from their L1 into English as an L2. The use of the AI-tool at the revision/review stage of their translation process would therefore give them more confidence about the choices they had made in their L2 text production. The class included 28 students, and although English is not the only L2 in their programme, their level of English proficiency can be described at C1/C2.

At the time of submitting course descriptions, GenAI had not yet become publicly accessible and was therefore not yet included in the course description. Hence, it was one of the primary ethical and legal requirements of the course design to assess whether the set of learning objectives already identified in the course description contractually allowed for the introduction of AI into this final-year master's course. It turned out that the most general learning objective, *acting in an unpredictable and complex context*, and the learning objective including the appropriate methods applied during the student's individual cognitive translation process, *applying the appropriate digital (reference) tools*, readily made it possible to include an exercise with the use of GenAI.

At the start of the course, before some of the research reviewed above was published, few students had heard about GenAI tools, which they would be made more familiar with in their compulsory language technology module of the programme. In order to introduce them very quickly to the GenAI tools, the students were presented a straightforward list with direct AI website links from which they could choose one, including ChatGPT-3.5, Perplexity and Jasper. There was also a reference to future developments such as Meta's Llama and Google Gemini (previously Bard).

The student's use of a generative tool was integrated into their translation process as follows. First, students were asked to translate a passage from the source text and place it on the course collaboration platform. Second, they were asked to revise and review a team partner's translation independently. Third,

the original translator would submit the edited translation to an AI-tool, asking it to improve the text or giving it another prompt of their choice, and pasting the outcome underneath the edited translation. Finally, they would compare the AI tool version with the revised translation and indicate any differences with an annotation; they would put a + sign in the balloon if they agreed with the proposed change or write an alternative solution, if they did not agree with the AI tool version.

Similar to previous non-AI translation processes, students were reminded to adopt a mind frame that is similar to that of a reader of the target text (in this case, a foreign student, see below) and to ask themselves what expectations the typical target reader might have in terms of register and terminology. They were also reminded to interpret the source text as best as possible, and to be aware of spelling and layout, grammar and terminology.

As a result, students' work would look as follows. Figure 1 shows the first draft submission of a student and the two editing comments made by their peer, one of which has been complemented by a teacher comment.

Figure 2 illustrates the GPT-3.5 output of an editing prompt given by the original translating student and her own annotated comments with pluses or alternative solutions.

Further noteworthy details of the translation setting in this course are the following. First, a real-world client was involved: the university's legal office needed a translation of its University Codex from the students' main language into English. This source text meant that legal language was involved and their target audience consisted of foreign staff and students. Since this translation setting included a paying client, students would also be remunerated (with book tokens) for their extra translation work. In addition, the client was available for any questions that might arise relating to the translation and students would, hence, be able to practice client communication. For this purpose, a working document was prepared, called 'Questions to the client', in which students could write about their translation problems. This document was discussed in class and questions that were retained for referral to the client were collected and sent by the teacher. Each communication from the client was then also provided to the whole class in the same document. The students could further consult both a parallel and comparable corpus and glossaries, which had been compiled in previous classes for translations for the same client and the same target text.

Finally, as in other L2 translation classes, students were able to work on a student collaboration platform, in which each student's contribution was visible relatively anonymously to everyone.

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In case the preparatory documents and/or annexes contain a working document for which only an opinion by the Executive Board is requested for the time being, the chair can decide to exclusively send the document to members of the Executive Board, without the prejudice to the right of inspection provided in Chapter VIII.

CHAPTER VII. DISCLOSURE OF THE OVERVIEW OF DECISIONS AND THE REPORT FOR NON-MEMBERS

Article 28

No later than 5 working days after the relevant meeting, the overview of decisions is published by the secretariat of the Board of Governors on the intranet of Ghent University.

On the first working day after the approval of the draft report by the Board of Governors or the Executive Board, the final report is published by the secretariat of the Board of Governors on the intranet of Ghent University.

If necessary, the confidentiality of one or more of the decisions made by the Board of Governors or the Executive board is taken into account when disclosing the overview of decisions and the report for non-members.

CHAPTER VIII. RIGHT OF INSPECTION

Article 29

The members of the Board of Governors and the Executive Board are entitled to a reciprocal right of inspection in the confidential or non-confidential reports and preparatory documents. This right of inspection also relates to the files covered before the directorship of a member, when relevant for the assessment of the files presented to the [administrative body] during the term of office.

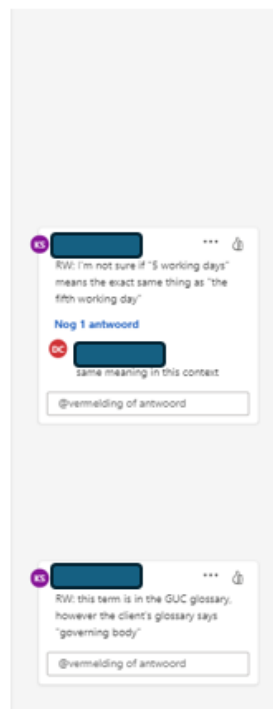


Figure 1: A student's first draft translation with peer and teacher feedback.

All materials provided by the students who participated in this project are under student consent for research purposes and allowed for the description of what the students were capable of in this pilot project.

4 Findings

First, students did not do much experimentation with a variety of AI tools but only applied ChatGPT and only used it in their revision process. It also turned out that some students confused Machine-Translated texts with AI-generated texts and referred to, for instance, Deep-L, which was already familiar to them. However, students did not prompt ChatGPT to produce a translation.

More importantly, students were able to identify differences between the translation version that had undergone monolingual and bilingual revision by their peers and the text as edited by the GenAI tool used. In particular, they were not only able to identify better formulations in the L2, but also unwelcome omissions,

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Figure 2: : The GPT-edited version of the peer-revised draft (from Figure 1) with the translating student's comments.

additions, and even – sometimes minor – differences of meaning between their own versions and the ones provided by ChatGPT.

The following set of examples, for instance, shows that students agreed with ChatGPT improvements. They involved the more idiomatic use of the Anglo-Saxon genitive (less frequently used in their own native language) (1), the avoidance of a split *to*-infinitive construction (2) and the use of the active rather than the passive voice (3):

- (1) a. Peer-revised draft
an opinion by the Executive Board is requested
- b. GPT-edited version

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*the **Executive Board**'s opinion is requested*

c. Student comment

+

(2) a. Peer-revised draft

*the chair can decide **to exclusively send** the document to members of the Executive Board*

b. GPT-edited version

*the chair can decide **to send the document exclusively** to members of the Executive Board*

c. Student comment

+

(3) a. Peer-revised draft

*the overview of decisions **is published** by the secretariat of the Board of Governors on the intranet*

b. GPT-edited version

*the secretariat of the Board of Governors **must publish** an overview of decisions on the intranet*

c. Student comment

+

The next set of examples (4-6) illustrates students' recognition of undesirable addition and omission as provided by ChatGPT:

(4) a. Peer-revised draft

The Internationalisation Council creates a policy plan regarding internationalisation (Integrated Policy Plan) evaluates and guides the implementation of this policy in cooperation with all parties involved. All themes regarding internationalization can be discussed. Regarding the educational and research related aspects within the themes, the Internationalisation Council shall seek advice from the Educational Council and Research Council, respectively.

b. GPT-edited version

INTERNATIONALISATION COUNCIL

The Internationalisation Council develops a plan for internationalisation (the Integrated Policy Plan) and ensures that it is implemented with input from all relevant parties. The Council addresses

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all aspects of internationalisation, and seeks input from the Educational Council and Research Council on related issues.

- c. Student comment
title is not necessary

- (5) a. Peer-revised draft
The Internationalisation Council creates a policy plan regarding internationalisation (Integrated Policy Plan) evaluates and guides the implementation of this policy in cooperation with all parties involved. All themes regarding internationalization can be discussed. Regarding the educational and research related aspects within the themes, the Internationalisation Council shall seek advice from the Educational Council and Research Council, respectively.
- b. GPT-edited version
The Internationalisation Council develops a plan for internationalisation (the Integrated Policy Plan) and ensures that it is implemented with input from all relevant parties. The Council addresses all aspects of internationalisation, and seeks input from the Educational Council and Research Council on related issues.
- c. Student comment
a lot of text is left out, which I wouldn't do in a text like this

- (6) a. Peer-revised draft
*The secretary registers the application, indicating the member and, if applicable, the reasons for the application, **retrieves** the file or the information and grants access to the applicant.*
- b. GPT-edited version
The secretary will record the request, including the name of the member and the reasons for the request, if applicable. The secretary will then grant access to the requested files or information.
- c. Student comment
'retrieves' is omitted

Notably, meaning differences, even minor ones, have been identified and rectified by students. Examples (7-8) show that the student is able to rely on the information from the co-text to interpret the source text as intended:

- (7) a. Peer-revised draft

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*The secretary registers the application, indicating **the member and, if applicable, the reasons for the application**, retrieves the file or information and grants access to the applicant.*

- b. GPT-edited version

*The secretary will record the request, including **the name of the member and the reasons for the request**, if applicable. The secretary will then grant access to the requested files or information.*

- c. Student comment

positioning of the original translation better as in this version [=GPT's] it could be interpreted as both the member and the reasons are only registered if applicable

- (8) a. Peer-revised draft

Members with an advisory vote :

- *the environmental coordinator*
- *the sustainability policy officer (secretary)*
- *experts in sustainable energy, mobility, purchasing, food, education, and research (present depending on the agenda)*

- b. GPT-edited version

*Members with an advisory vote **include** :*

Environmental coordinator

Sustainability policy officer (secretary)

Experts in sustainable energy, mobility, purchasing, food, education, and research (present depending on the agenda)

- c. Student comment

'include' implies that there are also other members with an advisory vote

Although example (9) shows that the student did not have the complete information about university faculties, which the teacher comment makes clear, they did see a meaningful difference with the first draft, which happened to render the situation as intended by the client:

- (9) a. Peer-revised draft

Three assistant academic staff representatives, of which one from the alpha, one from the beta, and one from the gamma faculties

- b. GPT-edited version

Three assistant academic staff representatives, one from each of the alpha, beta, and gamma faculties

c. Student comment

I would not adopt this phrasing as it seems like there are multiple alpha, beta and gamma faculties

d. Teacher comment

There are multiple alpha, beta and gamma faculties, but not each faculty is represented on the Council, which the ChatGPT version implies

Apart from identifying better or worse alternatives, students were also able to suggest better formulations, as, for instance in example (7) above or (10–11) below:

(10) a. Peer-revised draft

CHAPTER VII. DISCLOSURE OF THE OVERVIEW OF DECISIONS AND THE REPORT FOR NON-MEMBERS

Article 28

No later than 5 working days after the relevant meeting, the overview of decisions is

b. GPT-edited version

***Chapter VII outlines** the disclosure of decisions and reports for non-members. Within five working days of the relevant meeting, the secretariat*

c. Student comment

this is a title so a verb is not necessary

(11) a. Peer-revised draft

Composition

Chair: rector or vice-rector

Voting members, in addition to the chair:

- the chief logistics administrator*
- the Director of Administrative Affairs*
- the Director of Infrastructure and Facility*

b. GPT-edited version

Composition: The committee is chaired by the rector or vice-rector and consists of voting members and members with an advisory vote. Voting members include

- Chief logistics administrator*
- Director of Administrative Affairs*
- Director of Infrastructure and Facility*

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c. Student comment

I would keep the original structure as this wording is longer and not really necessary for clarity

5 Discussion

This descriptive exploration of an L2 translation course in which GenAI has been included in the editing stage of the translation process has shown that students are not only be able to identify differences, but also to assess them in terms of desirability. They did so at various language levels: grammar, sentence structure, layout and, importantly, also at textual level, which seemed to pose some difficulty to ChatGPT. Such behaviour testifies to their level of competence in the L2 language, their awareness of client-intended content matter from a legal domain and their ability to use GenAI as a revision tool. It was also shown that students are able to suggest better formulations to ChatGPT itself, and therefore improve the AI tool by providing feedback.

This does not mean, however, that all student comments or assessments were appropriate. Other observations revealed that students may wrongly identify ChatGPT suggestions as better alternatives, or, vice versa, wrongly determine that ChatGPT's different formulations were undesirable. The latter could be seen as cases of hyperrevision (Robert et al. 2018) and the undesirable imposition of one's own style onto the target text, as when a student changes *within 5 working days* to *no later than the fifth working day*. Unless such a change is made on the basis of explicit rules about consistency of certain phrases, it may be due to the higher frequency of a particular phrase in a student's mental lexicon.

In terms of the students' learning objectives, the course design integrating ChatGPT offered the students ample practice in exploring a changing environment and digital tool use. In addition, the GPT-input also provided alternative phrasings which may stimulate their creativity in the L2 and contribute to their command of English and language sensitivity. The presence of an additional touchstone for their own work may also boost their feeling of independence.

Students further practised the skill of underpinning their views on the desirability of translation solutions (an essential post-editing skill) and communicating about them, even to the client. In her final communication, the latter, for instance, repeated her words of gratitude for the student comments and the depth of reading that they had demonstrated, even across different regulations, which had led to better regulations for the university.

However, the course design did not give the students the opportunity to compare different generative writing tools or to practice various prompts as [Vidakovic \(2023\)](#) or [Giampieri \(2024\)](#) recorded. A comparative exercise of different translations and revisions by the same tool may already stimulate a valuable class discussion on the desirability of translation solutions, as the two translations by ChatGPT from March 2024 show in the following boxes (underlined phrases show passages where the two texts differ).

LLM Response 10.1:

Tom Lanoye receives prestigious Prize of Dutch Literature 2024 This autumn, the Flemish writer Tom Lanoye will be honored with the Prize of Dutch Literature. Flemish Minister of Culture Jan Jambon (N-VA) announces this on behalf of the Committee of Ministers of the Language Union. Lanoye is the first Flemish author since 2012 to receive the triennial prize. Following in the footsteps of renowned figures like Hugo Claus and Jeroen Brouwers, the Flemish writer Tom Lanoye is now also awarded the Prize of Dutch Literature. This is the most important literary prize in the Dutch-speaking area. The prize, which comes with a cash amount of 60,000 euros, is awarded every three years by a jury of critics, academics, and authors. The jury members always seek an author whose work holds a significant place in Dutch-language literature. “Tom Lanoye uses the theatricality of literature in all its forms to express himself about the real world,” says the jury in its report. “It is this baroque, language-virtuoso, yet equally serious interpretation of literary engagement that we are pleased to honor.” Who is Lanoye? Tom Lanoye was born in 1958 in Sint-Niklaas but currently lives and works alternately in Antwerp and Cape Town. As a novelist, poet, columnist, screenwriter, and playwright, he is one of the most read and praised authors in the Netherlands and Belgium. Giel Bosmans

Can you give an alternative translation for the same text?

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LLM Response 10.2:

Tom Lanoye to be awarded prestigious Dutch Literature Prize 2024. This fall, Flemish author Tom Lanoye will be honored with the Dutch Literature Prize, as announced by Flemish Minister of Culture Jan Jambon (N-VA) on behalf of the Committee of Ministers of the Language Union. Lanoye is the first Flemish writer since 2012 to receive this triennial award. Following in the footsteps of literary giants such as Hugo Claus and Jeroen Brouwers, Tom Lanoye, the Flemish author, is now set to receive the Dutch Literature Prize, the most esteemed literary award in the Dutch-speaking region. The prize, carrying a monetary reward of 60,000 euros, is bestowed every three years by a panel of critics, academics, and authors. The jury consistently seeks an author whose work occupies a significant position in Dutch-language literature. The jury's report states, "Tom Lanoye utilizes the theatricality of literature in all its facets to comment on the real world. It is this baroque, language-virtuoso, yet equally serious embodiment of literary commitment that we are delighted to recognize." Who is Lanoye? Born in 1958 in Sint-Niklaas, Tom Lanoye currently resides and works alternately in Antwerp and Cape Town. Renowned as a novelist, poet, columnist, screenwriter, and playwright, he stands as one of the most widely read and acclaimed authors in both the Netherlands and Belgium. Giel Bosmans

A comparative analysis of the output of different generative tools, too, will present an opportunity for an in-depth class discussion on translation processes. While GPT-4 (OpenAI 2024) only rendered a translation, Google's Gemini (2024), for instance, added comments to its translation of a passage from the students' source text:

- (12) The translation aims to accurately convey the meaning of the original Dutch text. However, nuances in language and cultural context might not be fully captured. For more specific or legal interpretations, it is always recommended to consult with a legal professional familiar with Belgian law and university governance.
[...]
"Bestuurders" has been translated as "governors" to reflect the role of these individuals in the university's governance.

“Raad van Bestuur” and “Bestuurscollege” have been translated as “Board of Governors” and “Executive Board” respectively, based on common English terms used in university governance.

The phrase “maatschappelijk geëngageerde” has been translated as “socially engaged” to convey the university’s commitment to societal issues.

(Google 2024)

Nor did the course design include any evaluation stage of the students’ individual use of ChatGPT, which is an important stage of the student learning process. Our complex and ever-changing world will need translators to be able to translate texts themselves, performing all the traditionally required cognitive writing processes, as well as to post-edit a GenAI-tool.

6 Conclusion

To conclude, integrating AI into the L2 translation classroom with a real-world task not only follows the call of leaders at all levels of government to introduce AI into teaching in order to maintain global competitiveness. It also facilitates the road to reaching basic learning objectives, and, therefore, enriches L2 translation learning in a way that enhances reflective thought. The latter may in itself increase intrinsic motivation, a focal point in much L2 translation training.

Considering the importance of the role of GenAI in education (Labadze et al. 2023), this exploratory project recommends the critical application of a GenAI-tool to be added as an additional learning objective in future L2 translation courses. It further shows teachers the direction in which the course design could be improved. In order to retrieve all the relevant information, students can be given a template for their assignments with space for their first draft of the translation, the monolingual and bilingual revision of that draft, a screenshot of students’ prompts showing the generative tool used, an AI chatbot output with students’ comments, and a final conclusion on the comparison between the revised version and the AI chatbot output (including the number and type of changes accepted/ignored and whether they would use the tool again). Devoting course time to explore different translation-generative tools and to discuss their output is also recommended.

While this qualitative inquiry has only focused on just one aspect of using AI in translation training, other teaching implications have been left undiscussed: ways in which a chatbot can help the teacher set up an examination text on a

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topic from texts seen in class or design a schedule for peer collaboration in large groups.

However, as Gao writes: “GPT is not, and will not be, an adequate substitute for the language expertise, cultural awareness, ideological sensitivity, and creative abilities to ‘re-write’ on the part of professional human translators” (2024:1). Therefore, the training of a constructive while at the same time critical attitude towards GenAI is essential, perhaps more strongly than ever before. This should keep the standards of academic integrity and increase awareness of local / international ethical and legal issues. Today, courts have decided that AI systems do not have any rights, nor do they have a legal personality (which is reserved for natural and juridical persons). Nor will AI systems ever show true intelligence (Chomsky 2023). The main problem with them derives from the fact that they are solely based on a volume of data and algorithms that apply to that volume and were provided by often unidentifiable humans. In spite of GPT’s recently implemented measures for bias reduction and content moderation, that volume may still contain bias, toxic language or other harmful content. Better training and more reliable data sources will not completely stop an AI-tool from yielding hallucinations or responses that do not mirror true facts, nor will protection measures be foolproof against malicious actors. In other words, quality control will remain an ongoing effort that users need to be aware of.

In order to build the necessary constructive critical attitude, various roads can be taken by translation trainers. They could provide for a creative skills module in the translation curriculum (Guerberof-Arenas & Asimakoulas 2023), they could make their students aware about sustainability issues in the evaluation of the technology they use (Moorkens et al. 2024), and/or they could attribute human-centred augmented intelligence (O’Brien 2024) a central place in their own course.

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Chapter 11

Gamification as a pedagogical instrument in interpreter training

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Translation & Interpreting (T&I) have only recently recognised the potential uses of simulated learning on market- and emotional-readiness of students. Educational simulation games have been found to contribute to increased performance and the motivation to learn. Utilizing games as a pedagogical instrument may not be novel, but the incorporation of gamification within the strategies of teaching and lesson planning is an innovative endeavour. There are research gaps that underline the need for a practice-based approach that supersedes the use of existing technology with specialist T&I-centred technology. With that in mind, this chapter presents a study that introduces Project “[Al]phra”, simulation software aiming to establish an interdisciplinary approach to interpreter education and training using gamification. The main goal is to develop skills before and after entering the professional market and to obtain a better understanding of the cognitive and emotional processes that take place during interpreting, especially in demanding contexts. [Al]phra consists of a virtual reality environment, simulating various scenarios where interpreters can practice in different settings and envisions an online platform, fostering connectivity and facilitating resource sharing including videos, articles and other interpreting-related materials.

1 Introduction

Simulated learning has developed as a powerful tool for preparing students for the job market, where they will inevitably encounter metalinguistic and emotional challenges. Many translation students graduate without understanding that the market demands a certain set of entrepreneurial skills upon which their

success will hinge. Graduates' discrepancies between expectations and realities in Translation and Interpreting (T&I) may stem from the misapprehension that their jobs are inherently isolated from the unpredictability of human interaction.

While T&I can be considered a discipline and industry eager to adopt technology and interdisciplinary research to create a hybrid environment for students to train and grow academically and professionally, research on how this might be initialised in the foundational years of translation education remains relatively scarce. Taking into consideration the many emerging conversations about the notion of artificial intelligence (AI) being trained by major corporations to replace human translators, the rising interest in utilising this technology to enrich students' learning experience seems to be a reasonable and timely response. There appears to be some reluctance to use AI in the classroom (He 2021: 1). The literature suggests that courses teaching machine-assisted translation or machine translation (MT) post-editing require swift adoption, as the developments in the field are moving quickly. In addition, as project-based T&I education is starting to pick up pace in the European Commission through projects like OPTIMALE and CEL/ELC, the T&I industry is entering a new era of consolidating pedagogy and professional outputs.

Simulated learning in T&I has been explored through projects such as The International Network of Simulated Translation Bureaus,¹ yet its systematic integration into curricula that bridge academia and the market remains limited. Research on simulations that can be used to teach the basics of T&I mostly in relates to case studies of simulating workflows from translation offices (Konttinen et al. 2020). Outside of the translation classroom, however, technologies and simulated environments such as the Metaverse have made the concept of utilising simulated environments highly appealing. Simulation has long been employed in fields such as aviation and the military, yet its appeal and feasibility have only recently been recognised in other domains. This growing attention can probably be attributed to the expected positive impact on student engagement and learning experiences. As reported by Nyström & Ahn (2024: 1), several meta-analyses of simulation-based learning demonstrate significant positive effects on learning and facilitation of complex skills across different fields.

Simulated learning, we argue, inherently involves the use of gamification in education. There are several definitions of simulation and gamification that, depending on their usage and context, allude to this connection to other fields. Within the context of education, the focus of this chapter, we rely on the definition of simulation from Alessi & Trollip (2000: 246) as:

¹<https://www.instb.eu/>

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a dynamic representation of a real or hypothetical system that allows learners to interact with a model of the system, explore its behaviors, manipulate variables, and observe the resulting outcomes without the risks, costs, or practical limitations of engaging with the actual system directly.

This definition accounts for variables that will change with different needs and requirements from a pedagogical perspective, such as student levels, different assessment types, and even expected learning outcomes.

With gamification, the core principle is to create a motivating environment for solving ordinary problems and to break down the ideas introduced by a new concept in a digestible manner. It also involves the gradual introduction of higher-level education according to student levels with a degree of flexibility (Park & Sangkyun 2021). Gamification as a term only entered the mainstream vocabulary around 2010 (Dichev & Dicheva 2017: 1). Simões et al. (2013: 345) define gamification as the use of video games or some of their elements in non-game applications or as learning tools, which they also refer to as Game-Based Learning (GBL). Kapp (2012), Wang (2021) and Zicherman (2011, quoted in Deterding et al. 2011) propose that gamification involves engaging people, motivating action, promoting learning, and solving problems using game thinking and game-based mechanics and aesthetics. Kiryakova et al. (2014: 679) state that gamification, in general, is the integration of game elements, game thinking, and game approaches in activities and contexts different from games. In their review, Dichev & Dicheva (2017: 2) suggest that gamification in education involves the introduction of elements of game design and gamified experiences in the design of learning processes.

A close relationship between games and learning has already been the subject of study in lower-level education, and in fields such as medicine, aviation, and architecture. However, it remains relatively new in T&I. The application of gamification in learning can create very promising opportunities, particularly in learning simulations. The stated research goals, along with an understanding of how game design features such as points, levels, challenges, and rewards can be applied in educational simulations, align with the principles of gamification, enabling learners to develop skills needed for the job market while supporting their integration into academic contexts. We propose a new way to use simulations as teaching tools, encouraging students to maintain a good level of operation by introducing game elements.

The authors propose adopting gamification in simulated learning in T&I as in Project [AI]Phra, a pilot project by the authors from 2023. In essence, it aims

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to confront the pressures in the field by utilizing AI to enrich the learning experience in a manner befitting the future of education in T&I and the future-readiness of the profession. It is, to an extent, an attempt to address the concerning reliance on machine-assisted translation, the changes occurring in the field of interpreting, and the technological necessities that are becoming an evident reality in teaching and training interpreters. It involves a human-made simulator model that hinges on the generation of arbitrary scenarios that are used to train interpreters in the Middle East and North Africa region (henceforth MENA for generalisation, with Arabic the predominant lingua franca). The languages are English and standard Arabic, at least for the duration of the pilot testing. The interpreting settings are mainly medical, legal, and public service. One of the main interests of the project is to address the issue of underqualification of interpreters in the aforementioned fields, on both academic and professional levels. On that note, this chapter aims to illustrate the set of realities that necessitate the project [AI]phra in its current form, and to illuminate the realities and expectations in the context of interpreting studies in Arabic-speaking regions.

2 Interpreting Studies: Realities and Expectations

Research in interpreting studies has progressively been growing in a manner that matches the advancements occurring on a professional and an academic level in the anglophone academic sphere. However, it remains a relatively young, under-regulated and under-researched field of study in the Arabic-speaking world. Interpreting in general is not given its due, as most translation programmes in the Arab world focus mainly or exclusively on translation. Even within translation, [Al-Batineh & Bilali \(2017: 198\)](#) note a mismatch of focus between these translator education programmes and the translation industry: “[L]iterary translation represents 25% of the field-specific courses offered at the graduate level while the required experience in literary translation represents 3% of market demand.” In addition, bachelor and master programmes for interpreters are notably absent, despite the urgent need for qualified professionals to meet the demands of a region that has witnessed a drastic influx of refugees and involuntary migration over the past few decades.

The need for trained professionals is under-acknowledged in public service settings in the MENA, and little has been done in terms of regulation, education, and certification ([Raddawi 2015](#), [Taibi 2011](#), quoted in [Taibi 2021: 73](#)). The most common settings for interpreting are conferences, hospitals, legal and immigration settings. Conference interpreting is, to an extent, a regulated field where

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qualifications and quality are expected. As a profession, it is what interpreting students aspire to do, and as a field of study, what educators aspire their students to be capable of performing. The same, however, cannot be said about the other aforementioned settings. According to a study conducted by Taibi (2021: 73), when attempting to uncover the nature of interpreting in the MENA region, there are issues related to training and qualifications. A survey conducted on medical interpreting in the United Arab Emirates and Saudi Arabia shows that "most of the language services offered at Al-Ain hospitals [United Arab Emirates] are now performed by ad hoc interpreters who are neither professional and well-trained, nor accredited or certified." (Taibi 2021: 73). As for interpreting in legal settings, reliance on ad hoc interpreters has been normalised in the context of less common languages. The number of interpreters in said settings is never quite sufficient to allow for choice in quality. For instance, in Saudi Arabia, this shortage has led defendants to interpret for other defendants (Taibi 2021, 73).

The means to solve these issues start from the classroom. The provision of adequate education hinges on several factors, including fitness for purpose, utilisation of technology, and most importantly, acknowledging the gaps that exist in the educational system in monitoring compliance and quality assurance. This field requires a high degree of emotional and situational awareness. Therefore, the education in question must also be at a level that enables the delivery and coverage of such skills. Adding to that, the context of the MENA requires a degree of acknowledging the variations in diglossia and dialectical issues that arise in the setting of public service interpreting (PSI), where there is a lack of training, as mentioned. All in all, the traditional modes of delivering interpreter education appear to be no longer sufficient nor fit for the realities of the job market in the region. Therefore, and as a way of acknowledging the changes and shifts in teaching paradigms in other regions in the world, the authors have devised a project that enables the use of generative AI in a manner that can make use of the arbitrary nature of scenarios used in different simulations, among other considerations such as providing a gamified approach to interpreting education, and use technology to provide access to real-life scenario-based situations. Moreover, the project seeks to foster a stimulating and engaging environment to increase motivation and engagement among novice and professional interpreters and to empower female interpreters and researchers who are relatively under-represented in interpreting and computing-related fields.

3 Training as pedagogy: An Entry Point into Simulations as Teaching Tools

3.1 Interpreter Competencies

As a starting point, the type of simulations that a project like [AI]phra entails are quite similar in *modus operandi* to the types of simulations utilised in the training of medical students. The principle relies on the creation of real-life scenarios that are expected to occur on a daily basis in an interpreting task. The generation of said tasks is mainly reliant on human-generated ideas that are extended and modified through generative AI. It is true that current models of generative AI that are commonly used can generate arbitrary scenarios that appear to be consistent with the pedagogical requirements of any interpreting curriculum. To break down that idea into practical pieces, it is important to divulge two things: first, what the pedagogical requirements are considering the developments in the field, and the context of the region, and second, the value of simulation as an illustration of practices that build metalinguistic and emotional capacities in interpreting students. A third, consequential aspect that may be considered is the sustainability of the model to be usable beyond the classroom, namely in the training of professional interpreters based on their varying degrees of experience in the field and in their respective interpreting settings (medical, legal and social services).

When designing the curriculum for interpreting courses at both undergraduate and postgraduate levels, there are well-recognised fundamental competences that must be addressed. Phelan et al. (2020: 25) categorise core competencies into several areas, as may be seen in Table 1.

An argument can be made that these competences are intertwined and complement each other from the perspective of an educator. However, for the purposes of this chapter, the connection can be made through the competence of technology. The training provided aims to create a sustainable opportunity to develop all competencies through one medium. Regardless of the type of setting, there are essential skills that are meant to be developed through the training of the competences. They include, but are not limited to, listening and comprehension skills, memory and note-taking abilities, public speaking skills, proficiency both in the context of language and in metalinguistic elements (body language, business skills), and gaining insight into real-life practices and intricacies of every interpreting task.

Additionally, Hale (2007), Wadensjö (2014) Tipton & Furmanek (2016) and Phelan et al. (2020) point to the need for an extensive understanding of the ethical

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Table 1: . Types of interpreter competencies

Competences	Details
Linguistic competences	grammar, specialised terminology, register, prosody.
Thematic	identifying relevant topics for interpreting assignments, researching and extracting relevant terminology from resources, demonstrating knowledge of various interpreting modes.
Interpersonal	interaction coordination, trust-building strategies, bias awareness.
Intercultural	understanding cultural differences and power asymmetries, knowledge of users’ cultural backgrounds.
Technological	terminology management, videoconferencing.
Business-related	client and assignment management, finance, membership of professional associations.
Developmental	ongoing learning, collaboration, flexibility, and change management.

and professional considerations of interpreting. This entails, for instance, understanding client confidentiality and the fact that an interpreting task starts before the linguistic exchange takes place. Another example is maintaining an acute awareness of the nuanced differences in discourse between different interpreting settings (hospital, solicitor’s office, court, police station, or mental health counselling session).

3.2 **Training as a social capital: equipping interpreters for real-life scenarios**

While considering the educational importance of training, literature has also stated that there is a social function for training, seeing as it provides a context for professional socialisation and contributes to advancing the interpreting profession and improving its status (Hale 2007: 167) especially in the case of PSI. It can be argued that one of the main aims of teaching interpreting is to fulfil a social responsibility towards the community. In the MENA region, the clients that benefit from PSI in particular and interpreting services in general are disempowered

minority groups, or those who may not be fortunate enough to have received a bilingual education. It also serves individuals who have been contracted from different countries around the world. There is an overlooked human rights aspect to interpreting that requires an advocate's approach in certain instances, which may prove difficult to grasp unless an interpreter is faced with it in a real-life scenario. Therefore, a simulation can be a low-risk instance where that interaction is enabled and navigated in the presence of a more experienced interpreter or educator.

3.3 What are we training when we train interpreters through simulation?

Simulation-based training, which involves replicating realistic situations in a controlled environment, offers opportunities to develop reflection-on-action skills, deliberate practice (entailing individualised training activities tailored to enhance an individual's current level of performance through repetition and successive refinement - see Chapter 2 of this book), and achieving and tracking technical, cognitive, and behavioural objectives (Anderson et al. 2008: 596). Additionally, simulation allows the modification of preconditions that are typically difficult to replicate for a specific learning element (Nyström & Ahn 2024: 138). Simulation-based learning can provide a safe environment for all types of learners, regardless of age, and be more accessible, especially for learners with disabilities.

Kincaid et al. (2003: 273) assert that simulation is significant to the field of education as it:

- Can be applied to learners of all levels and ages,
- Enables learners to view complex relationships that might otherwise involve costly equipment or risky experiments,
- Provides learners with new problem-solving methods and realistic training and skills, and
- Reduces risks to learners and is cost-effective.

Despite the prevalence of simulation in some fields, language-related fields, including interpreting, are still relatively far behind. Interpreting, however, is an excellent candidate for simulation due to the nature of the job. Interpreters often face a high level of unpredictability, especially in public service contexts, where

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they can be exposed to sensitive situations, traumatic events, physical and verbal abuse, health risks, highly specialised terminology, and cultural barriers. Interpreters may encounter all these challenges simultaneously or not at all, depending on their specific encounters. Without comprehensive and proper training, interpreters can struggle not only to provide adequate and quality interpretation, but also to manage emotional and psychological distress. Unfortunately, it can be quite difficult to replicate all possible scenarios in real-time within a safe environment that is accessible to all students and ensures their engagement.

A virtual simulation game application that facilitates the holistic comprehension of a subject is necessary to allow students to delineate the interrelationships between concepts, recognise the trade-offs and resource constraints, and comprehend their practical applicability. This would maximise the transferability of academic knowledge acquired by students to practical tasks in the industry (Loureiro-Koechlin & Córdoba-Pachón 2012).

Many studies have explored the potential benefits of simulation in different fields. Educational simulation games have been found to be a source of increased performance and motivation to learn (Randel et al. 1992, Terrell & Rendulic 1996, Prensky 2003, Schwabe & Göth 2005). In addition, simulators offer learners an opportunity to hone their skills in a risk-free and less stressful environment, make mistakes without consequences and thus increase their confidence (Alinier 2003, Salas et al. 2009, Mahboubian 2010, Lean et al. 2021), enable reflection and the assessment of alternative decisions (Cousens et al. 2009), present scenarios that require critical skills and are unlikely to be encountered, and allow repetition of similar scenarios which enhance learning (Baker et al. 2005, Salas et al. 2009). It is pertinent to note that the learning style of the new virtual generation (V-gen) is distinct from preceding generations, characterised by higher levels of visuality, interactivity, and problem-solving (Proserpio & Gioia 2007). Despite that, it appears that simulation is largely absent from language studies (see list (1) to follow²), and humanities for that matter, which can be attributed to the fact that their primary focus is on the examination and analysis of existing works instead of the generation of new ideas.

(1)	Game	Area	Computerised/ non-computerised	Type
	[16]	Architecture	Computer based	Drill-based
		Business ethics	Computer based	Problem-based
	[17]	Child development	Non-computerized	Mini-case

²Pharmacy Simulator, Pharmlator, Capitalism II, STEMPilot, Ghost Medical: Animation and VR Surgery, Police Training Simulators: Police 1 by Lexipol are other examples of educational stimulation games.

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[18]	Interviewing skills	Non-computerized	Exercise-based
[19]	Laboratory management	Computer based	Drill and exercise
[20]	Medical	Computer based	Problem-based
[21]	Medical	Computer based	Drill-based
HEART-SENSE game [22]	Physics	Computer based	Drill-based
Spectrum Game [23]	Physics	Computer based	Drill-based
[25]	Physics and calculus	Computer based	Problem-based
Frantic Physics [24]	Mathematics	Computer based	Drill-type
Tic-tac-toe [24]	Computer security	Computer based	Exercise-based
[37]	Software engineering	Computer based	Drill-based
SimSE [39]	Software engineering	Computer based	Mini-case based
SESAM [40]	Software engineering	Computer based	Mini-case based
Project Manager [41]	Information systems	Computer based	Mini-case based
KM Quest [42]	Information systems	Computer based	Problem-based
MagicBlocks [11]	Computer protocols	Non-computerized	Drill-based
Konane [44]	Digital logic	Computer based	Drill-based
Industrial Waste game [70]	Artificial intelligence	Non-computerized	Exercise-based
Educational water quality management game [6]	Environmental engineering	computerized	Problem-based
Learning Sustainable Development [13]	Environmental engineering	Non-computerized	Exercise-based
Internet-based Interactive Construction Management Learning System [32]	Civil engineering	Computer based	drill-based
[33]	Civil engineering	Computer based	Problem-based
[12]	Engineering graphics	Computer based	Drill-based
[4]	Engineering mechanics	Computer based	Drill-based
[54]	System dynamics	Computer based	Problem-based
[50]	Thermodynamics	Computer based	Drill-based
[51]	Thermodynamics	Computer based	
[52]	Thermodynamics	Computer based	Drill-based

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[5]	Enterprise resource planning	Computer based	Drill-based
[8]	Enterprise resource planning	Computer based	Mini-case based
MURSH-Bikes [55]	Enterprise resource planning	Computer based	Mini-case based
Socratic [56]	Production planning and control	Non-computerized	Drill-based
NCTB Game [57]	Production planning and control	Computer based	Mini-case based
SimEnterprise [58]	Production planning and control	Non-computerized	Drill-based
[59]	Production planning and control	Non-computerized	Drill-based
[60]	Production planning and control	Non-computerized	Drill-based
Parade of trade [61]	Production planning and control	Computer-based	Drill-based
[63]	Production planning and control	Computer based	Drill-based
Lean Leap [65]	Supply chain management	Computer based	Exercise based
Logistics Game [66]	Supply chain management	Non computerized	Mini-case based
TAC-SCM [67]	Supply chain management	Computer based	Online multi player simulation competition
SC Puzzle Game [68]	Supply chain management	Non-computerized	Drill-based
[69]	Chemical engineering	Computer based	Problem-based
Virtual Lab [46]	Chemical engineering	Computer based	Exercise based
Vicher [45]	Chemical engineering	Computer based	Drill-based
WinDSK [34]	Electrical engineering	Computer based	Drill-based
Interactive Power Electronics Seminar [35]	Electrical engineering	Computer based	Drill-based

In addition, simulated learning can help learners to achieve higher cognitive learning targets by consolidating the theoretical knowledge they have acquired, telegraphing an opportunity to intervene and guiding actual business processes. Business simulation increases psychological readiness, which is an important factor for novice learners in understanding work. A study conducted on students of law at the University of Fenerbahçe (Doğan et al. 2024) to understand how their immersion in a simulated environment with the help of virtual reality affects their performance in creative tasks, and have reached the conclusion that they

enjoy a learning process that supports creativity. This study suggests that simulated learning can also improve translation or interpretation – in other words, improve a student’s practical preparedness before going to work.

All in all, the concept of simulations in the context of interpreter training and education has significant merits that may prove to be exactly what the field in the region needs to address its different challenges and concerns. Furthermore, the simulations can be nuanced and adapted to be fit-for-purpose in a way that encourages learners to pursue self-study and training on their own accord when we insert the gamification of education into the road map for the creation of the simulations that are utilised in the classroom.

4 Gamification in education

Despite recent growing interest and research in gamification, the concept is not new: it has been utilised in education for many years, predominantly in primary and secondary schools. The use of gamification begins as early as kindergarten, where educators employ tools such as image cards, letter cubes, and colourful abacuses. However, adult students can also benefit significantly from the integration of games into their educational experiences. Given that more people of all ages play games – the average video game player has been involved in gaming for more than twelve years (Kapp 2012, 18) – gamification may be even more advantageous for today’s students, as they are more accustomed to non-traditional learning methods. It provides a way for educators to adapt the learning process to meet student learning styles and new requirements, as contemporary learners are also viewed as “digital natives”, having grown up with technologies (Kiryakova et al. 2014: 679). This can make the combination of gamification and technology a highly effective approach to enhance learning outcomes.

Games have been utilised in the past to create a stimulating environment, helping individuals overcome different types of turmoil or to come to terms with certain adversaries that arise in different fields. For example, there is a game called *Re-mission* (2008) which has been designed to help young patients fight cancer. The game has been localised to French and Spanish and has, according to Tiemann (2008), proved to help patients’ sense of self-efficacy and their attitude towards taking medication.

Another example is a game titled *Revolution 1979*, in which the player takes on the role of a journalist and witnesses the events leading up to the Iranian revolution from a very different perspective or can be perceived as a historical documentary. In 2018, the University of Texas created a game titled ARTE:

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Mecenas, which is the first art history game in the world. Researchers at that university have reported an improvement of 24.7% in recalled knowledge after just two hours of gameplay (Thomas 2018). There are many other examples of games that historically play an important role in providing an educational experience through an interactive medium. In fact, the adoption of games as tools for learning and development has become one of the methods employed by museums to engage with younger audiences (Styx 2022) as it has proven to heighten the benefits gained from visits, and to allow visitors retain information for longer periods of time.

4.1 Why Incorporate Gamification into Education?

As learning and gaming share many fundamental mechanisms, more educators are exploring the possibility of gamifying the learning process to make it as automatic and enjoyable as gaming (Wang 2021). Gamified learning approaches, often based on digitalisation, increase accessibility for students, promote individualized learning, sustain students' attention, motivate them, and enable the free flow of information, leading to various educational possibilities and a positively developing educational environment. These approaches positively change students' behavior and attitudes towards learning, enhancing their engagement and motivation. Consequently, they potentially improve academic outcomes, increase subject mastery, level of commitment, and understanding of the educational content (Balalle 2024: 4, Kiryakova et al. 2014: 679). Wang (2021) further argues that incorporating a gamified learning process makes learning less laborious and immerses students in the curricular content. Similarly, a study by Smiderle et al. (2020: 7) shows a significant improvement in the quality of submitted solutions by participants in a gamified group, who achieved higher accuracy. The same study reveals that introverted participants scored higher in terms of points, medals, and logins than extroverted ones, demonstrating a statistically significant difference in the accuracy improvement of the introverted participants. This indicates how gamification can potentially contribute to creating a more inclusive and safer environment that accommodates students' diverse needs and personalities.

Kapp (2012: 9) emphasizes the importance of placing individuals in a safe learning environment, allowing them to gain experience through trial and error, and introducing an allowable element of failure into the learning process. The integration of gamification can also improve accessibility to educational material remotely, thereby increasing both reach and engagement. Furthermore, gamification has been successfully applied in a wide range of disciplines such as engi-

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neering, finance, marketing, and healthcare. As [Dichev & Dicheva \(2017: 1\)](#) state, effective collaboration in different contexts, such as customer engagement and employee performance, is among the reasons for gamification's growing popularity and its applicability across various domains. An example of gamified learning can be found in almost every industry, as many organizations employ gamification to train employees and educate students. These organizations range from schools to government institutions. For instance, INNOV8 created by IBM, is an interactive first-person thinker game used in business and information technology programs in hundreds of schools internationally. It teaches complex ideas of business process management through players' decision-making ([Kapp 2012: 20](#)).

4.2 The need for comprehensive research in gamification

Although it is suggested that gamification can enhance students' motivation, engagement, knowledge attainment, and understanding of educational content, [Dichev & Dicheva \(2017: 12\)](#) assert that more studies are needed to support this claim, as there is still insufficient empirical work that thoroughly examines gamification's educational potential. They argue that understanding the target population of a gamified system is vital to effectively gamify a learning activity. Continued theoretical research and systematic empirical studies across diverse gamification settings and contexts are necessary to establish a practical, comprehensive, and methodical understanding of the benefits of applying gamification in educational contexts. In the same vein, recognizing the significance of understanding the needs and preferences of each group of learners, as well as the need for a more accessible and safe learning environment, led to the idea of project [Al]phra, an interpreting simulator.

5 Project [Al]phra

Creating a situated learning simulation that puts interpreters in real-life situations can foster a three-dimensional comprehension of basic interpreting strategies. It can enable interpreters to develop self-awareness and reflection skills, enhancing their capacities even after entering the professional market. Additionally, it assists in obtaining a better understanding of the cognitive and emotional processes that take place during interpreting, especially in public service contexts. Therefore, this chapter introduces Project [Al]phra, a new interdisciplinary approach to interpreter education and training using gamification as underlying notions to support capacity building in interpreting. [Al]phra addresses the

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lack of adequate training for interpreters, with a particular emphasis on sustainable innovation for community interpreters. It seeks to establish an interpreter-centered use of technology to provide access to real-life scenario-based situations, creating a stimulating and engaging environment to increase motivation and engagement among novice and professional interpreters. Moreover, as an interdisciplinary endeavor, the project aims to create long-lasting connections with other fields such as IT, the sciences, and others from which lessons can be learned.

5.1 Methodology

5.1.1 Phase 1: Developing [Al]phra

Phase one of the project involves two main steps: diagnostic research and initiation of execution. Research begins by gathering preliminary data about simulated learning and studying lessons learned from other studies, acting like a comprehensive linguistic and cognitive mapping process. The simulation conducts a multi-dimensional evaluation of the learner's linguistic capabilities, going beyond traditional language proficiency tests. As an initial step, this research aims to create an adaptive testing algorithm to precisely measure not just vocabulary and grammar, but also interpretation-specific skills like cognitive processing speed, context switching ability, and cross-linguistic transfer competence. Through fed data, a set of learner profiles that captures linguistic strengths and weaknesses, learning preferences, and potential cognitive challenges specific to Arabic-English interpretation are built in real time.

This forms the basis for the scientific research that will enable step two. Step two involves hiring software developers and animators and adapting existing simulators from other fields to create a prototype of the interpreting simulator, [Al]phra. Upon agreeing on the necessary modules and alpha scripts, the testing extends to functionality and usability with only a small group of testers that is mostly comprised of educators and trainers.

5.1.2 Phase 2: Testing and modification of [Al]phra

This phase moves the learning profile of a trainee interpreter into a structured, gamified learning experience by breaking down the intricate process into manageable, interconnected skill modules. Each of the skills is integrated within the interface as an interactive target. The cognitive and linguistic competencies required for high-level interpretation translate into levels, questions, and scripted interactions. The cognitive processing module, for instance, challenges learners

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with rapid-fire exercises that simulate the intense mental demands of community interpretation. The linguistic conversion aspect entails exposure to specialized terminology within a realistic context which focuses on cultural and contextual adaptation, presenting learners with scenarios that require not just linguistic knowledge but deep cultural understanding. By gamifying these skill components, the simulation transforms what could be a dry, challenging learning process into an engaging, motivational experience.

In this phase, the simulation creates an environment that simulates real-world interpretation scenarios with intentional complexity. Unlike traditional training methods, these scenarios are not linear. They are responses to the created profiles and the feed of assessment results provided within the span of the testing phase. The simulation uses generative AI to adapt the scripts created for learners based on their linguistic proficiency, their cultural awareness, their cognitive agility, and their adaptability to change. Since feedback will be provided in real time, the learner will be able to receive immediate insight into their real time feedback mechanisms and provide immediate, constructive insights into performance, highlighting linguistic errors, suggesting improvement strategies, and tracking cognitive processing metrics. The branching narrative structure ensures that each scenario feels unique and unpredictable, mirroring the complex, high-stakes environments where professional interpreters operate.

Since this phase involves testing [AI]phra with student focus groups and collecting data, a call for volunteers will be created and invitations sent to universities and selected professionals. The focus group will test the beta version, extracting data that will then be analysed and used to modify [AI]phra as necessary.

5.1.3 Phase 3: Final modifications and release of [AI]phra

In this phase, the simulation creates a rich, dynamic environment that simulates real-world interpretation scenarios with unprecedented depth and complexity. Unlike traditional training methods, these scenarios are not static or linear, but dynamically responsive to the learner's performance, creating a truly adaptive learning experience. Virtual environments range from international conferences and diplomatic negotiations to medical consultations and technical briefings, each designed to challenge different aspects of interpretation skills. The simulation uses advanced AI to generate contextually nuanced scenarios that test not just linguistic accuracy, but also cognitive agility, emotional intelligence, and professional adaptation. Real-time feedback mechanisms provide immediate, constructive insights into performance, highlighting linguistic errors, suggesting improvement strategies, and tracking cognitive processing metrics.

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Phase three includes final modifications made to [Al]phra based on the results of the first two phases, which involves thorough analysis of the data collected during the earlier phases. This may include user performance metrics, error patterns, and participant feedback. Adjustments at this point will address weaknesses, such as difficulties in managing shifts in scenarios or issues with cognitive load. During this phase, a final test run of [Al]phra should be conducted before it is officially released. Interpreting students, novice interpreters, and professionals will be invited to participate in a controlled pilot study through which they will perform tasks using [Al]phra, along with other traditional training methods. The goal is to measure improvements in target language quality, speed, accuracy, and confidence. Quantitative metrics such as error rates and scenario completion times, combined with qualitative insights from participants and supported by pre- and post-test evaluations will provide empirical evidence of [Al]phra's efficiency. The outcomes of the data analysis will be incorporated into version Alpha of the project.

5.1.4 Phase 4: Evaluation

The fourth and final phase entails evaluating [Al]phra post-release. This evaluation will involve obtaining user feedback and surveys to determine if [Al]phra is achieving its desired objectives. User engagement and performance, focusing on leaderboard activity, feedback, and completion rates, will be tracked using post-release data analysis combined with surveys conducted months after the release. This will contribute to assessing the long-term impact of [Al]phra on users' professional skills, such as handling high-pressure scenarios and managing cultural nuances. This will also be reflected in the research paper accompanying the pitch to the industry for real-life testing and distribution.

5.1.5 Maintenance

Ongoing maintenance and support for [Al]phra will be provided throughout the project and after release. This will include bug fixes, security updates, and any other necessary adjustments or updates. It is important to ensure that scenarios and features integrated into [Al]phra remain relevant to emerging interpreting practices, regular updates will be informed by users' and industry partners' feedback.

The estimated timeframe for completing [Al]phra is three to five years, underscoring the need for comprehensive research on gamification.

5.1.6 Expected experimental setup and results

Project [Al]phra will use a simulated environment and a gamified approach to interpreting, involving a virtual reality environment with various scenarios where interpreters can practice in realistic situations. It will provide a range of different settings such as medical, legal, and humanitarian. To gamify this environment, there will be rewards for completing tasks or achieving goals, as well as a leaderboard to track progress. This will be combined with a tutorial and assessment system to ensure that interpreters are making progress and advancing. The project will also develop an online platform to link interpreters and researchers, sharing additional resources, including videos, articles, and other materials related to interpreting.

The expected results of [Al]phra on interpreting students, novice, and professional interpreters include:

- Increased understanding of the interpreting process, as [Al]phra will provide an interactive and engaging platform where interpreters can practice their skills in an accessible and safe environment and gain insights on how to recognize and appropriately convey meaning across languages and effectively manage time and resources.
- Increased efficiency in the interpreting process.
- Increased accuracy and speed of interpreting.
- Improved communication between interpreters and clients.
- Increased understanding of cultural nuances.
- Enhanced confidence in interpreters' abilities.
- Increased understanding of how to approach various interpreting tasks.
- Sharpened cognitive and receptive skills.
- Increased motivation and engagement among interpreters.
- Better handling of unexpected and challenging scenarios.

6 Conclusion

It remains crucial to continuously collect and coordinate data that demonstrate the successes and failures of gamification to establish an empirical knowledge base, compile best practices, extract guidelines, and ultimately develop predictive theories (Dichev & Dicheva 2017). Building on this foundation, this chapter outlines Project [Al]phra's potential to bridge the divide between academic preparation and market demands and address gaps in interpreting training through the application of gamification and simulation-based learning. These gaps include the lack of practical experience with lifelike high-pressure scenarios, insufficient attention to emotional and psychological challenges and the limited integration of technological competence despite its integral role in remote interpreting services and the management of translation memories and specialized terminology. The interdisciplinary approach of [Al]phra tackles pedagogical limitations while aligning with recent developments and emerging trends in education and professional training. A gamified, simulation-based training could effectively transform interpreting education especially where interpreting training programs and regulatory frameworks are underdeveloped. [Al]phra seeks to provide a learning space to address key challenges such as linguistic accuracy, cultural competence and cognitive agility, which are essential for interpreters, especially those working in complex, demanding and high-risk settings. It serves as a basis for further research and development focusing on the validation of the feasibility and efficacy of gamified and simulation-based training in interpreting education.

Although the project's reliance on data-driven modifications and agility may ensure its relevance, its scope in the initial pilot phase will be limited by the focus on the use of standard English and classical Arabic. This is because the sample of participants will consist only of learners who speak English and Arabic. In subsequent phases, inclusion of more languages and dialects will be prioritized to increase accessibility and inclusivity. Furthermore, there is the potential to develop a mobile application for [Al]phra, enabling learners to access it at any time and on their preferred devices. This can be particularly beneficial for younger generations, who are heavily reliant on their mobile phones. Project [Al]phra represents a major step forward in incorporating simulation and gamification into interpreter education. It has the potential to revolutionize the field and embrace the use of technology and AI and employing them in ways that collectively serve interpreters and other end users.

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Chapter 12

Embracing machine translation in L2 education: Bridging theory and practice in the AI Age

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This chapter examines the evolving role of machine translation (MT) in second language (L2) education. It first summarizes recent research on MT's pedagogical applications, synthesizing findings from systematic reviews that highlight MT's effectiveness in L2 writing when used appropriately. The chapter explores learner and teacher perceptions, noting a generally positive attitude among students but a more divided stance among educators. It presents a comprehensive framework for understanding factors influencing MT use in L2 writing, encompassing linguistic, personal, contextual, and ideological dimensions. The chapter introduces the concept of "MT as Augmented L2 Competence," offering a new paradigm for integrating MT into language learning. It discusses various models of MT instruction, including the Guided Use of MT (GUMT) model. Finally, the chapter proposes the Metacognitive Resource Use (MRU) framework, which positions learners as metacognitive agents capable of strategically utilizing a wide range of language resources, including MT and generative AI (GenAI) tools. This integrated approach aims to foster autonomous learners who can effectively navigate the complex landscape of digital language learning resources.

1 Current Status of MT Use in L2 Education

1.1 Pedagogical applications

As evidenced by the recent increase in papers on MT use for L2 learning and teaching, a wealth of research reports various pedagogical applications for MT in

Atsushi Mizumoto. 2025. Embracing machine translation in L2 education: Bridging theory and practice in the AI Age. In JC Penet, Joss Moorkens & Masaru Yamada (eds.), *Teaching translation in the age of generative AI: New paradigm, new learning?*, 237–252. Berlin: Language Science Press. DOI: ??

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L2 education. These applications include using MT as a resource for vocabulary acquisition (Lo 2023, 2024) and grammar practice (Lee & Kang 2024), fostering metalinguistic awareness through comparative analysis of original and machine-translated texts (Tsai 2019), and enhancing learner autonomy by promoting self-directed learning (Lee 2020).

Indeed, the proliferation of MT-related research over the past two decades has led to the emergence of several systematic reviews (Jolley & Maimone 2022, Klimova et al. 2022, Lee 2023, Ohashi 2024). These comprehensive analyses provide valuable insights into the current state of MT use in L2 education and highlight key trends and findings across multiple studies. Jolley & Maimone (2022) reviewed MT studies from 2000 to 2019, focusing on MT accuracy, educational applications, and perceptions of learners and teachers. They noted the improvement in MT quality over time and its effectiveness in L2 writing, while also highlighting uncertainties about its long-term impact on language learning. Klimova et al. (2022) analyzed 13 studies published between 2018 and 2021, specifically examining the impact of neural machine translation (NMT) in foreign language education. Their review found that NMT can be beneficial for language learning, particularly for advanced learners, but also identified a lack of teacher training in this area.

Lee (2023) conducted a systematic review of 87 studies from 2000 to 2019, investigating the effectiveness of MT in foreign language education. The review highlighted MT's usefulness in L2 writing, mixed perceptions among learners, and discrepancies between teacher and student perceptions. It also emphasized the need for further research on long-term learning effects and appropriate usage methods. Lee also conducted a meta-analysis with 12 studies that satisfied predefined criteria. It revealed that utilizing MT had a positive impact on L2 learning with a small effect size ($g = 0.345$, 95% CI [0.201, 0.489]). The analysis of subcategories showed that using MT was effective in various linguistic areas, including lexical accuracy ($g = 0.616$, 95% CI [0.243, 0.676]), syntactic accuracy ($g = 0.562$, 95% CI [0.371, 0.754]), syntactic complexity ($g = 0.453$, 95% CI [0.181, 0.726]), orthography ($g = 0.741$, 95% CI [0.424, 1.059]), and overall writing quality ($g = 0.768$, 95% CI [0.547, 0.990]). However, the effect on lexical complexity was not statistically significant ($g = 0.253$, 95% CI [-0.058, 0.564]). These findings provide quantitative evidence for the positive impact of MT on various aspects of language learning, particularly in writing skills. This additional information strengthens the overall narrative by providing specific quantitative evidence of MT's effectiveness in various aspects of language learning. It supports the general consensus among the reviews that MT can be beneficial for L2 writing when

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used appropriately, while also highlighting areas where its impact may be less pronounced (such as lexical complexity).

Ohashi (2024) examined 14 studies published between 2020 and 2022, reporting on the latest trends and key findings in MT research. This review emphasized that MT can be a good starting point for language learning, but its effectiveness varies depending on learner proficiency and target language. It also stressed the importance of communication between teachers and learners and the need for teacher training in MT use.

The systematic reviews collectively indicate that MT can be an effective tool for language learning when used appropriately. They emphasize that MT is particularly beneficial for L2 writing, helping students reduce lexico-grammatical errors and focus more on content (Klimova et al. 2022, Lee 2023). However, the reviews also point out that the effectiveness of MT use may vary depending on factors such as students' proficiency levels, target language pairs, and specific learning tasks (Ohashi 2024). Furthermore, these reviews underscore the importance of proper guidance and training for both students and teachers in effectively using MT for language learning. They highlight a gap between teachers' perceptions and students' actual practices, emphasizing the need for better communication and mutual understanding (Jolley & Maimone 2022, Lee 2023). The reviews also call for more research on the long-term effects of MT use on language acquisition and the development of appropriate pedagogical strategies to integrate MT into language curricula effectively.

1.2 Learner and teacher perceptions

As reported above, MT use is widespread among language learners, who utilize these tools for a variety of purposes. Studies have found that a significant percentage of students, particularly those in higher education settings, report frequently using MT tools like Google Translate for language learning tasks, including writing and translation (Lee 2020, Tsai 2019). This widespread use persists despite learners often being aware of MT's limitations, particularly regarding accuracy and the potential for grammatical errors (Jolley & Maimone 2022).

Learners primarily view MT as a convenient and beneficial aid for language learning, using it for tasks such as looking up words and phrases, facilitating translation, and enhancing the revision process. While some learners express concerns about over-reliance on MT and its potential impact on learning, research indicates a generally positive attitude towards MT's role in supporting language acquisition (Chung & Ahn 2022).

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In contrast to learner perspectives, the literature highlights a more divided stance among teachers regarding the use of MT in L2 education. While many teachers continue to express concerns about MT's potential to hinder language acquisition and facilitate academic dishonesty, others recognize its pedagogical potential as a valuable learning resource (Ducar & Schocket 2018).

The reluctance among teachers to embrace MT stems from several factors. Firstly, the historical association of MT with inaccurate translations, often referred to as a “bad model,” and its perceived threat to traditional teaching methods contributes to skepticism (Lo 2023). Concerns about academic integrity, with MT being viewed as a form of cheating, also contribute to this reluctance. This often leads teachers to focus on detecting and preventing MT use rather than exploring its potential benefits (Jolley & Maimone 2022).

However, there is a growing movement among researchers and practitioners to integrate MT into language classrooms in a meaningful and responsible manner (Hellmich 2023, Lee 2020). This shift is driven by the understanding that MT use is now ubiquitous and cannot be effectively banned (Ducar & Schocket 2018). Instead, advocates of MT integration call for instructing learners about its strengths and weaknesses, equipping them to use it critically and ethically as a tool for language learning (Klimova et al. 2022).

1.3 Factors and views influencing MT use

The scoping review by Jiang et al. (2024), which synthesizes 29 MT studies in L2 writing from 2009 to 2023, examines the factors influencing MT use in L2 writing, focusing on four main categories: linguistic, person/individual, contextual, and ideological factors, framed through cognitive, sociocultural, and critical theoretical perspectives.

- Linguistic factors (cognitive perspective): The review analyzes MT's role as a linguistic processor, exploring how the cognitive processing of MT impacts L2 writing. It explores the effects of different writing tasks on MT utilization and assesses how the quality of input and output from MT systems affects the resulting text complexity and structure. These considerations are crucial in determining how both teachers and students perceive and employ MT, shaping its integration into educational practices.
- Person/individual factors (sociocultural perspective): MT is viewed as a mediational artifact/tool within social contexts, incorporating factors such as students' beliefs about MT's effectiveness, their writing strategies, L2

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proficiency, and digital literacy. Additionally, teacher-related factors include their attitudes towards MT, pedagogical integration of MT, and ability to detect its use. These individual factors critically influence educational approaches and the effectiveness of MT.

- Contextual factors (sociocultural perspective): The role of MT is influenced by the dynamics within educational environments, such as peer interactions, the presence of MT training or scaffolding in instructional settings, and institutional policies that may either support or restrict MT usage. These contextual factors are pivotal in shaping the operational and situational environments for MT, impacting its perception and efficacy in educational settings.
- Ideological factors (critical perspective): Ideological considerations focus on language-related ideologies, including monolingual biases, standard language ideologies, and translanguaging stances, and how these influence the use and perception of MT. Such ideologies challenge conventional language education practices and influence how MT is integrated and impacts linguistic hierarchies within educational contexts.

Jiang et al. (2024) propose a layered conceptual framework to understand the factors influencing MT use in L2 writing. With ideological factors as the outermost layer, contextual factors encompass both person/individual and linguistic factors, with the latter being positioned at the innermost core. These layers interact to shape the core activities involved in the cognitive processing of linguistic factors. The model effectively illustrates the complex interplay from broad societal and ideological influences to specific individual and cognitive interactions, highlighting the necessity of considering a range of factors from societal ideologies to individual cognitive processes when integrating MT into L2 education.

Building on the comprehensive framework proposed by Jiang et al. (2024), we can see how ideological factors form the outermost layer influencing MT use in L2 education. This ideological dimension is particularly evident in two recent studies that highlight contrasting perspectives on MT use in academic settings.

Grieve et al. (2024) conducted a qualitative study exploring the ethical views of nursing and midwifery students on using AI machine translation software for university assignments. Their findings reveal a complex interplay of factors influencing students' ethical decision-making, including ownership of ideas, fairness and respect, and personal growth. Importantly, the study identified a tension between deficit-oriented and translanguaging perspectives among students, with

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the latter referring to an approach that views learners' use of multiple languages as a resource rather than a problem.

The deficit-oriented perspective, often aligned with traditional language teaching methodologies, perceives L1 or MT use as detrimental to L2 acquisition. This perspective typically leads to policies discouraging or prohibiting MT use, viewing it as an impediment to authentic language learning. With this perspective, it is believed that that MT use may hinder the development of critical thinking skills in the target language and interfere with immersive language experiences.

Conversely, the translanguaging view adopts a more inclusive approach to language learning. This perspective values learners' entire linguistic repertoire as a resource (Wei 2018). Advocates of translanguaging argue that MT can be a valuable tool for accessing and leveraging learners' full range of linguistic knowledge, potentially enhancing both language awareness and learning outcomes. This view aligns with contemporary understandings of bilingualism and multilingualism, which conceptualize languages as part of an integrated communication system rather than as separate entities.

These contrasting perspectives on MT use in language education are further exemplified in the approaches educators and institutions take when addressing student use of MT. Jolley and Maimone's (2022) comprehensive review of three decades of MT research in language teaching and learning highlights two distinct approaches: the "MT as Cheating" approach, which leads to a Detect-React-Prevent Response, and the "MT as Resource" approach, which encourages an Integrate-Educate-Model strategy.

The "MT as Cheating" perspective, aligned with the deficit-oriented view, treats MT use as a form of academic dishonesty. This approach focuses on strategies to detect unauthorized MT use, react punitively, and prevent future occurrences. Proponents of this view recommend implementing clear syllabus policies against MT use, designing assignments resistant to MT use, and educating students about the pitfalls of relying on MT. This perspective often leads to policies that ban MT use outright, viewing it as incompatible with language learning goals.

In contrast, the "MT as Resource" approach, more closely aligned with the translanguaging view, sees MT as a potential tool for language learning. This perspective advocates integrating MT into the curriculum, educating students on its appropriate use, and modeling effective strategies for leveraging MT in language learning. Researchers like Stapleton & Kin (2019) and Niño (2020) argue for accepting the reality of MT use and finding ways to incorporate it meaningfully into language education. This approach acknowledges the ubiquity of MT in modern life and seeks to prepare students to use it critically and effectively.

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The shift from the Detect-React-Prevent mindset to the Integrate-Educate-Model approach reflects a growing recognition of the inevitability of MT use in language learning contexts. As [Ducar & Schocket \(2018\)](#) note, the key question is no longer whether teachers can prevent learners from using MT, but rather how to help them use it ethically and effectively as part of their language learning journey.

These contrasting approaches to MT use in language education exemplify the broader ideological tensions identified in [Grieve et al. \(2024\)](#) and reflect the outermost layer of ideological factors in Jiang et al.'s (2024) framework. They demonstrate how deeply held beliefs about language acquisition and the role of technology can shape educational policies, pedagogical practices, and ultimately, students' engagement with and perceptions of MT in their L2 development process. This interplay between ideological stances and practical approaches underscores the complexity of integrating MT into language education and highlights the need for context-sensitive strategies that consider both the potential benefits and challenges of MT use in L2 learning and teaching.

2 Integrating MT into L2 education: A new paradigm

2.1 MT as augmented L2 competence

The translanguaging perspective and “MT as Resource” approach, implemented through the Integrate-Educate-Model strategy, provide a theoretical and practical foundation for incorporating MT into L2 education. Building upon these concepts, we can further conceptualize MT use in language learning through the lens of “MT as Augmented L2 Competence.” This model offers a visual representation of how MT can enhance learners' language abilities, particularly in bridging the gap between receptive and productive skills. By viewing MT as a tool for augmenting competence rather than replacing language learning, we align with the translanguaging idea of fluid language practices and the “MT as Resource” approach. The Integrate-Educate-Model strategy can then be applied to help learners effectively utilize MT to expand their augmented competence zone, while simultaneously developing their own language skills. This integrated perspective not only justifies the use of MT in language learning but also provides a framework for understanding its role in enhancing overall L2 proficiency.

The concept of MT as augmented L2 competence is illustrated in [Figure 1](#), which provides a visual representation of how MT, and also GenAI such as ChatGPT, can enhance language learners' abilities.

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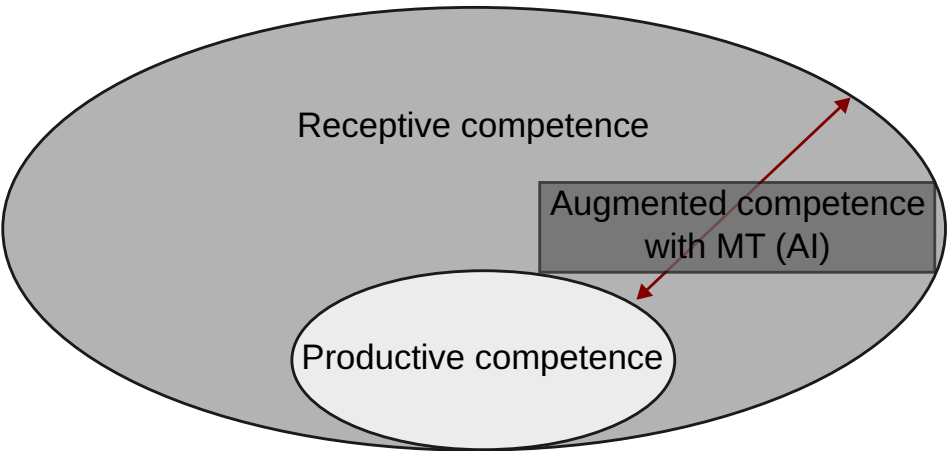


Figure 1: The concept of MT as augmented L2 competence

The figure demonstrates the relationship between receptive competence, productive competence, and the potential for augmented competence through MT use. Here is a breakdown of the key elements:

- **Receptive Competence:** This is represented by the larger, outer oval. It refers to the ability to understand the target language (L2), which is typically more developed than productive skills. For most L2 English language learners, their capacity to comprehend English exceeds their ability to produce it.
- **Productive Competence:** Shown as the smaller, inner oval, this represents the learner’s ability to actively use the language. It is generally more limited than receptive competence, which aligns with theories like Swain’s output hypothesis (1985), emphasizing the importance of language production in second language acquisition.
- **Augmented Competence with MT (AI):** This is depicted by the dark gray area extending beyond the productive competence oval. It illustrates how MT can bridge the gap between what learners can recognize as correct (receptive knowledge) and what they can produce on their own.

Figure 1 suggests that MT can serve as a tool to augment learners’ competence, particularly in areas where they can recognize correctness by sight but struggle to produce it accurately. This augmentation is especially beneficial for more proficient learners, as supported by previous studies (Klimova et al. 2022, Ohashi

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2024). Higher proficiency learners tend to have a larger gap between their receptive and productive skills, providing more room for MT to assist in bridging this divide.

Importantly, this model underscores that there remains a strong rationale for studying English (or any L2). The augmented competence provided by MT is built upon the foundation of the learner's own language skills. Without developing one's own receptive and productive competencies, the benefits of MT augmentation would be limited. Furthermore, as learners' proficiency increases, they become better equipped to effectively utilize MT, maximizing its potential as a learning tool.

This conceptualization of MT as augmented L2 competence aligns with the findings from systematic reviews (Jolley & Maimone 2022, Lee 2023) that highlight MT's effectiveness when used appropriately, particularly for more advanced learners. It also supports the need for proper guidance and training in MT use, as the tool's effectiveness is contingent upon the learner's ability to critically evaluate and apply its output.

In sum, the model presented here provides a framework for understanding how MT can be integrated into language learning processes. It emphasizes that MT is not a replacement for language study, but rather a tool that can enhance and extend learners' existing competencies, potentially accelerating their progress towards higher levels of language proficiency.

2.2 MT instruction for L2 learning

Niño (2009) proposed four models of MT use in L2 education: a "bad model," a "good model," vocational applications (particularly in translation-related fields), and as a computer-assisted language learning (CALL) tool. Initially, MT was employed as a "bad model," where students identified and corrected errors through post-editing, a process necessitated by the limited accuracy of early systems. In contrast, the "good model" involved using MT outputs as exemplars for students. These models, reflecting the evolution of MT technology and its pedagogical applications, illustrate a significant shift in focus. As MT technology has advanced, its primary use has transitioned to serving as a CALL tool, where it facilitates student engagement in solving language problems independently, as evidenced in recent studies (Lee 2020, Stapleton & Kin 2019, Tsai 2019).

Lee (2023) recommends that teachers should provide guidelines for using MT and explicitly teach effective strategies to students prior to using it, which leads to enhancing student performance, as supported by O'Neill (2016).

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While many researchers discuss the need for explicit MT instruction in language classrooms, only a limited number of studies provide concrete examples (Cancino & Panes 2021, Chang et al. 2022, Mirzaeian 2021, O'Neill 2016, Ryu et al. 2022), all of which report positive results. Among these, Ryu et al. (2022) implemented a model called the Guided Use of MT (GUMT). The GUMT model activities were developed and implemented in an upper-elementary Korean as a foreign language course at a U.S. university. The GUMT model has five steps:

- Instructional Session on MT Tools: As Activity 1, Students review strengths and weaknesses of popular MT resources like Google Translate and Naver Papago, discussing differences in speech styles and politeness levels. As Activity 2, students engage in pragmatic and grammar evaluation exercises, assessing the appropriateness of translations by considering contexts such as formality and speaker-listener relationships.
- Practice with Writing Assignments: Throughout the semester, students apply learned concepts in five writing tasks, using MT to aid their translation from Korean (L2) to English (L1) and vice versa.
- Reflection on MT Use: After each writing assignment, students reflect on their use of MT, assessing its effectiveness and making adjustments based on their experiences.
- Instructor Feedback: Instructors provide written feedback on students' drafts, focusing on grammatical and pragmatic accuracy, and highlight areas needing correction.
- In-Class Review Sessions: Students refine their drafts in class based on instructors' feedback, discussing and revising highlighted issues to improve their understanding and application of MT in writing.

Ryu et al. (2022) reported that implementing the GUMT model significantly contributed to the development of MT use strategies and enhanced students' confidence and self-assessed fluency in L2 writing, as evidenced by analyses of both pre- and post-surveys and student reflections.

2.3 Learners as metacognitive agents

The role of learners as metacognitive agents in using MT and other language resources is crucial for effective language learning. Mizumoto (2023) proposed a

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framework called Metacognitive Resource Use (MRU), which provides a comprehensive approach to understanding how learners can strategically utilize various language resources, including MT and GenAI tools like ChatGPT (Figure 2).

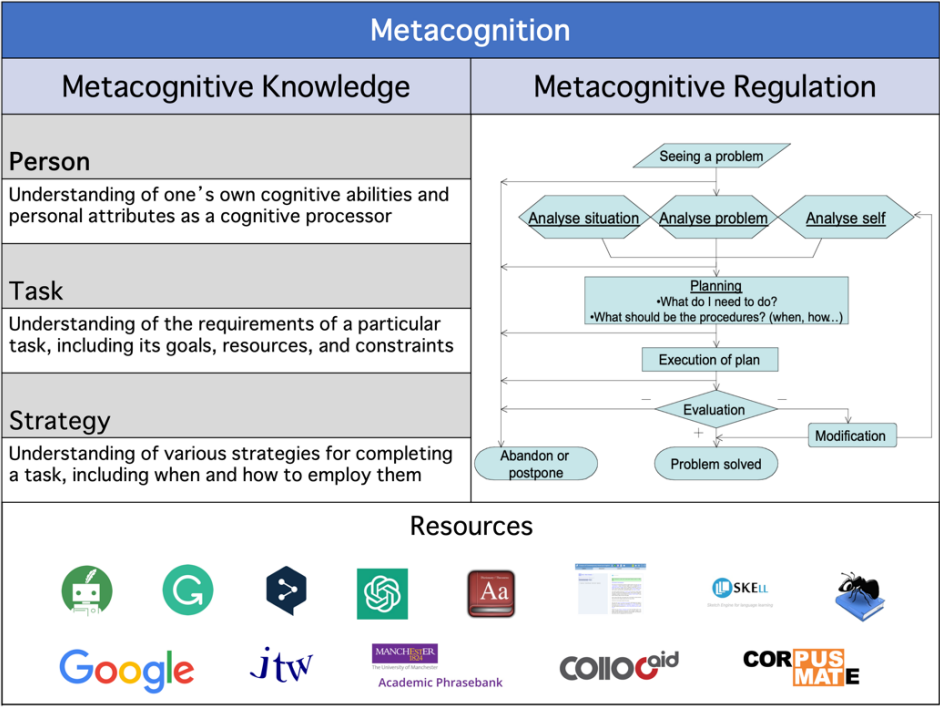


Figure 2: Framework of metacognitive resource use (adapted from Mizumoto 2023).

The MRU framework is grounded in metacognition theory and consists of two main components: metacognitive knowledge and metacognitive regulation. Metacognitive knowledge encompasses understanding of the person (self-knowledge), task (requirements and constraints), and strategy (available approaches). Metacognitive regulation involves the active management of this knowledge through planning, monitoring, and evaluating one's use of resources through employing learning strategies.

The framework visualizes a range of resources that learners can utilize, including online dictionaries, corpus-based data-driven learning (DDL) tools like concordancers, web apps, search engines, GenAI systems like ChatGPT, MT tools, and grammar correction tools such as Grammarly. This reflects the real-world usage of tools by learners when engaging with language tasks.

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For instruction on metacognitive resource use, Mizumoto (2023) suggests adopting principles from learning strategy instruction (Chamot & Harris 2019). This approach involves enhancing learners' metacognitive knowledge and imparting specific metacognitive and related cognitive strategies. The instructional model focuses on exposing students to a diverse array of tools and guiding them through the processes of planning, executing, evaluating, and adapting their strategic resource use.

It is important to note that the GUMT model employed by Ryu et al. (2022) aligns well with this approach. The GUMT model does not rely on a single tool but incorporates various resources, including different MT platforms and reflective practices. Furthermore, the instruction model in Ryu et al. (2022) shares similarities with both DDL and strategy instruction approaches, demonstrating that these methodologies can be discussed within the same theoretical framework that Mizumoto (2023) advocates for integrating DDL and GenAI.

This integrated perspective allows for a more comprehensive approach to language resource use. Instead of focusing solely on MT, learners should be encouraged to become metacognitive agents capable of effectively utilizing a wide range of resources, including GenAI tools like ChatGPT. By adopting this approach, we can foster the development of autonomous learners who are adept at selecting and using the most appropriate tools for their language learning tasks.

The MRU framework not only enhances learners' awareness and use of MT but also extends to their conscious utilization of GenAI tools. This holistic approach has the potential to cultivate truly autonomous learners who can navigate the complex landscape of language learning resources effectively.

Moreover, the research findings accumulated in the field of MT should not be disregarded but rather re-examined and validated within this integrated and extended framework. By doing so, we can ensure that the valuable insights gained from MT research continue to inform and enhance our understanding of language learning processes in the era of diverse digital resources.

In conclusion, by positioning learners as metacognitive agents within the MRU framework, we can create a more comprehensive and effective approach to language learning that embraces the full spectrum of available resources, from traditional tools to cutting-edge AI technologies.

2.4 Distinguishing MT from GenAI

While this chapter has primarily focused on MT, it is important to acknowledge the emerging role of GenAI tools like ChatGPT in L2 education. Although both MT and GenAI can assist with language tasks, they differ fundamentally

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in their capabilities and potential applications. Traditional MT systems are designed specifically for translation between languages, focusing on maintaining semantic equivalence while adapting to target language conventions. In contrast, GenAI systems can engage in open-ended language generation, including not only translation but also extending to explanation, summarization, paraphrasing, and interactive dialogue.

The differences between these technologies extend to their ability to handle context and maintain coherent interactions. While MT primarily processes text at the sentence or paragraph level, GenAI can maintain longer conversations, take broader context into account, and engage with complex prompts across multiple exchanges. This enhanced contextual awareness allows GenAI to serve not just as a tool for language conversion, but as an interactive partner in the language learning process.

These fundamental differences lead to distinct educational applications. MT serves primarily as a translation tool, though it can be used pedagogically for language comparison and error analysis. GenAI, however, can potentially function as an interactive tutor, providing explanations, examples, practice exercises, and feedback on language use. This broader functionality suggests that while MT typically serves as a standalone tool for translation tasks, GenAI has the potential to integrate and potentially replace multiple language learning tools, including dictionaries, grammar checkers, and corpus resources.

3 Concluding remarks

Looking ahead, the integration of MT, GenAI, and other AI-powered language tools in L2 education is likely to continue evolving rapidly. Future research should focus on developing and empirically testing comprehensive frameworks like the MRU, which embrace a wide range of digital resources beyond just MT. As the capabilities of GenAI continue to expand, there is a particular need to understand how these tools can complement or potentially transform traditional MT use in language learning contexts. There is a need for longitudinal studies to assess the long-term impact of these tools on language acquisition and learner autonomy, specifically examining how the interactive capabilities of GenAI might differ from traditional MT in supporting language development. Additionally, as AI language models become more sophisticated, practitioners and researchers will need to continually adapt their approaches, balancing the benefits of these tools with the core objectives of language learning. The field may see a shift towards more personalized, AI-assisted language learning experiences, necessitating new pedagogical strategies and ethical guidelines for their implementation.

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in L2 learning and teaching. Understanding the distinct roles and potential synergies between MT and GenAI will be crucial for developing effective educational practices in this rapidly transforming field.

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